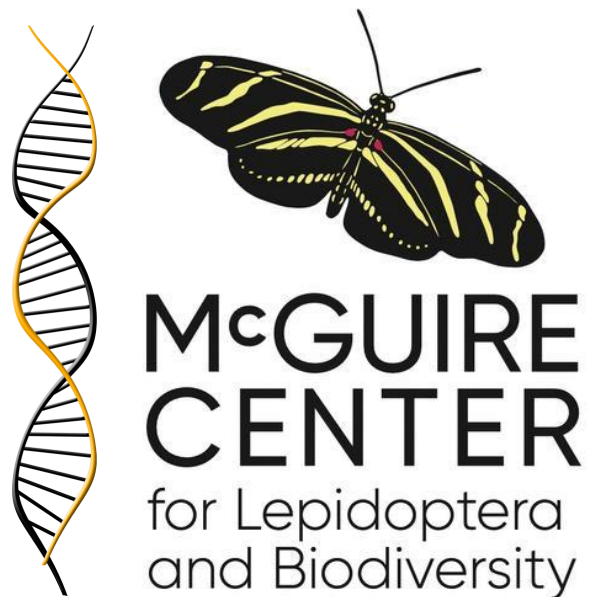


Module II: Genome Assembly

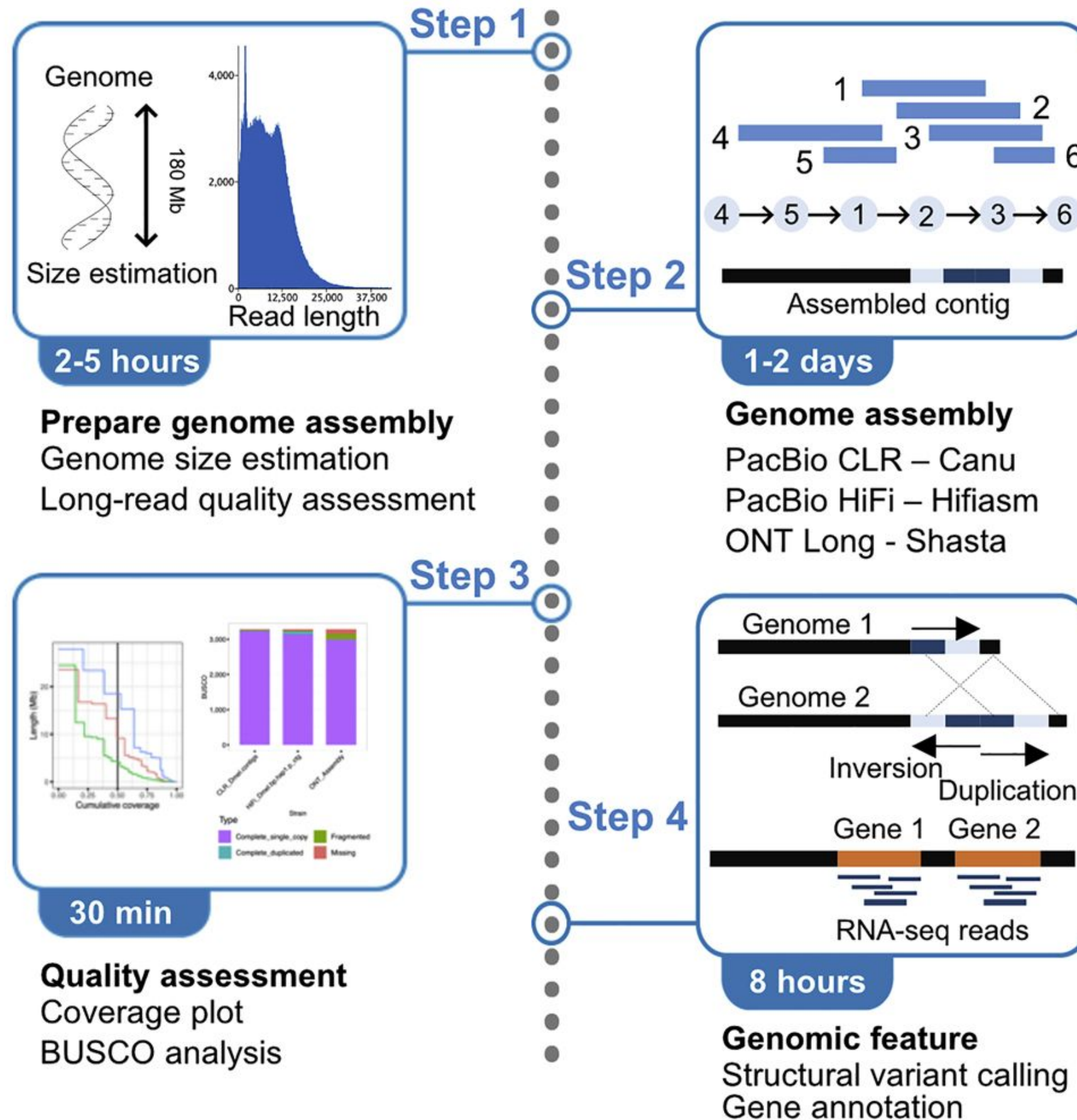
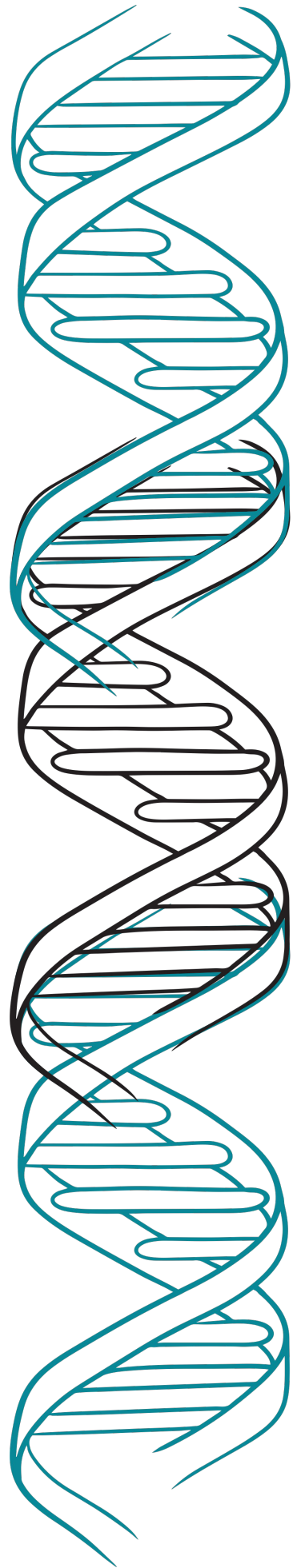
Saturday, January 17th 2026

R. Keating Godfrey, PhD
Assistant Professor,
Neuroscience Program
Middlebury College

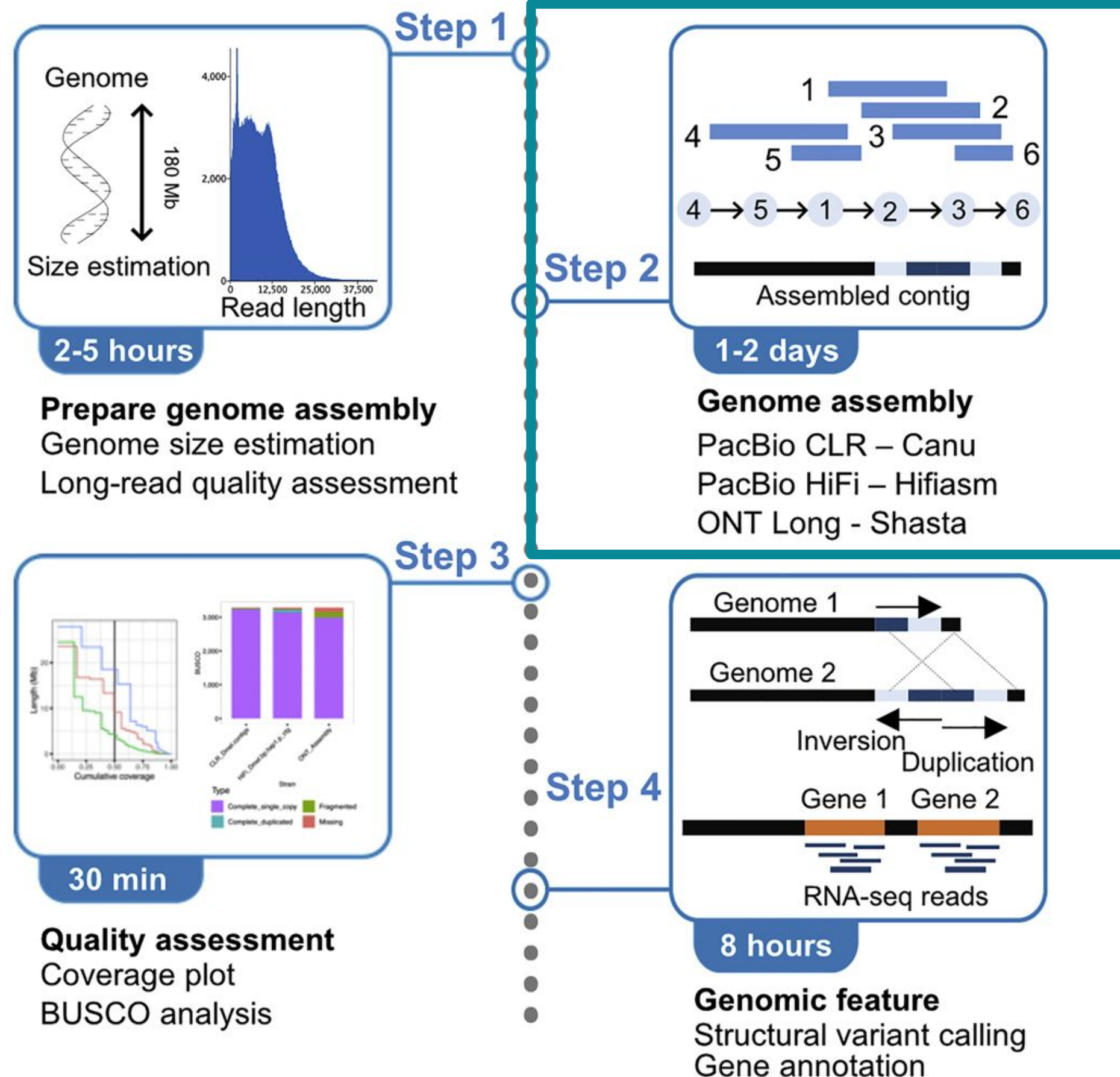
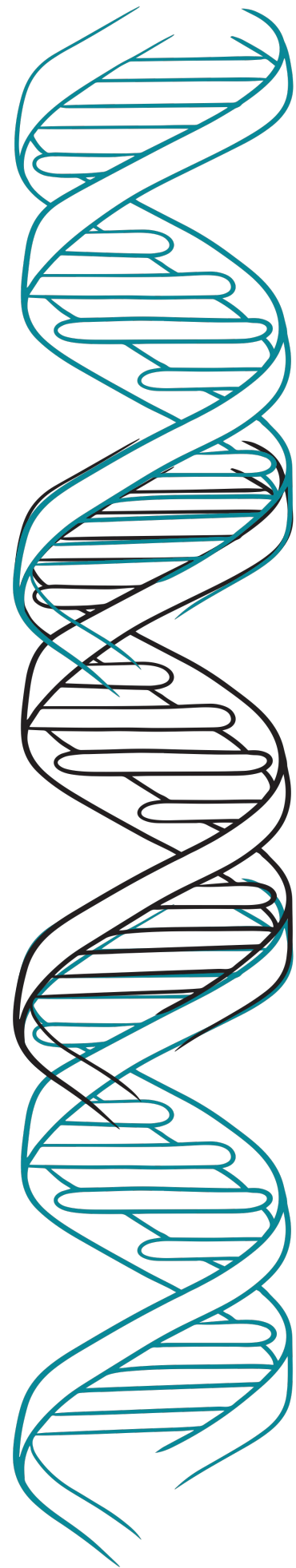
rkeatinggodfrey@middlebury.edu



Overview of Genome Assembly & Annotation



Module II Genome Assembly **Objectives**



- Understand differences between short- and long-read sequencing
- Describe how genome assemblers work, including challenges to assembly
- Interpret GitHub and HiperGator instructions for bioinformatic software
- Construct shell scripts that communicate specific instructions to an HPC (HiPerGator) about how to run a genome assembler (Hifiasm).

*look over example submission
scripts & discuss outputs*

Our Assembly Pipeline

SRA toolkit

download publicly available
long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size,
heterozygosity

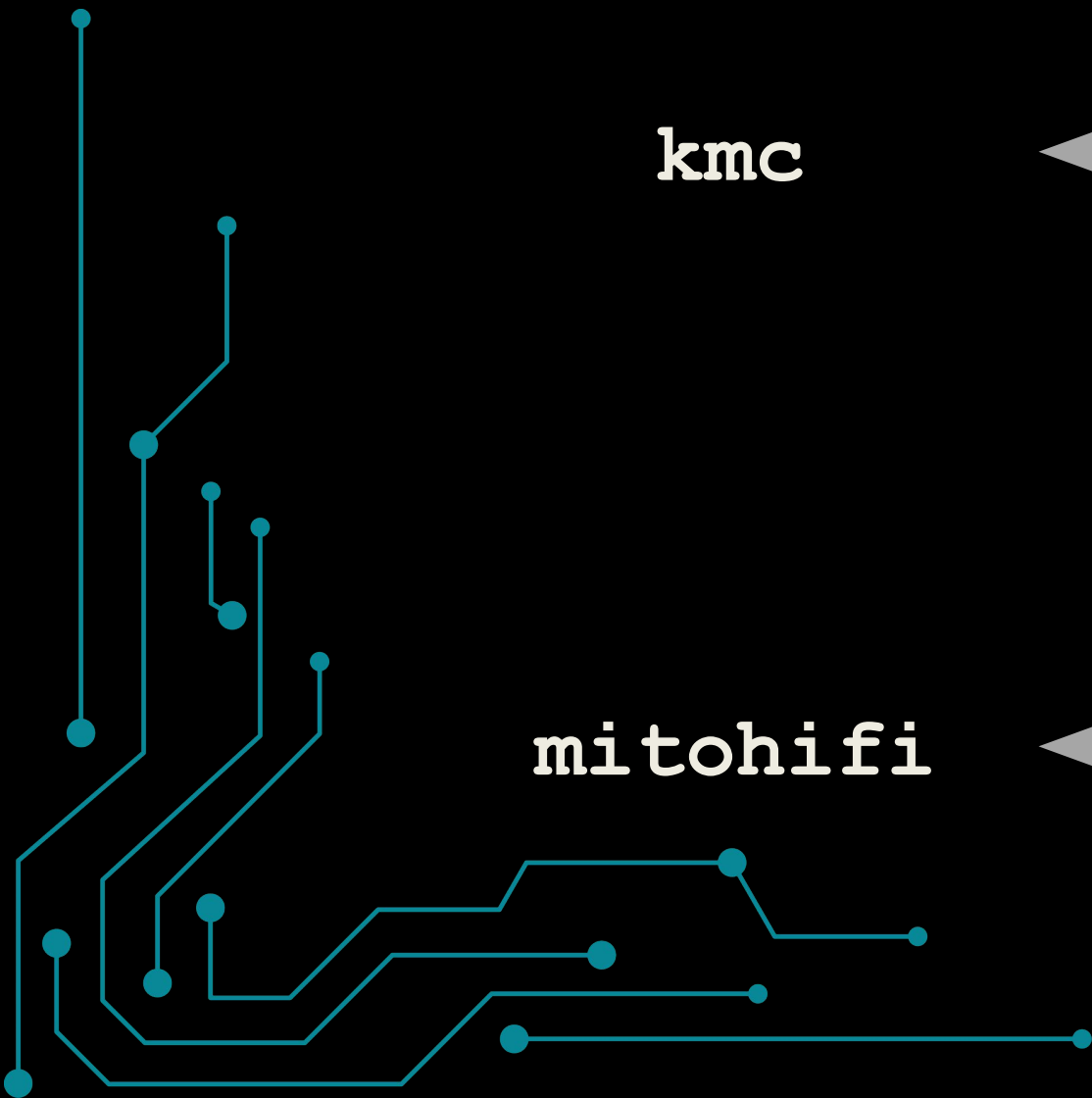
build assembly de novo using
long-read data

Hifiasm

mitohifi

pull mitochondrial genome
sequences from assembly

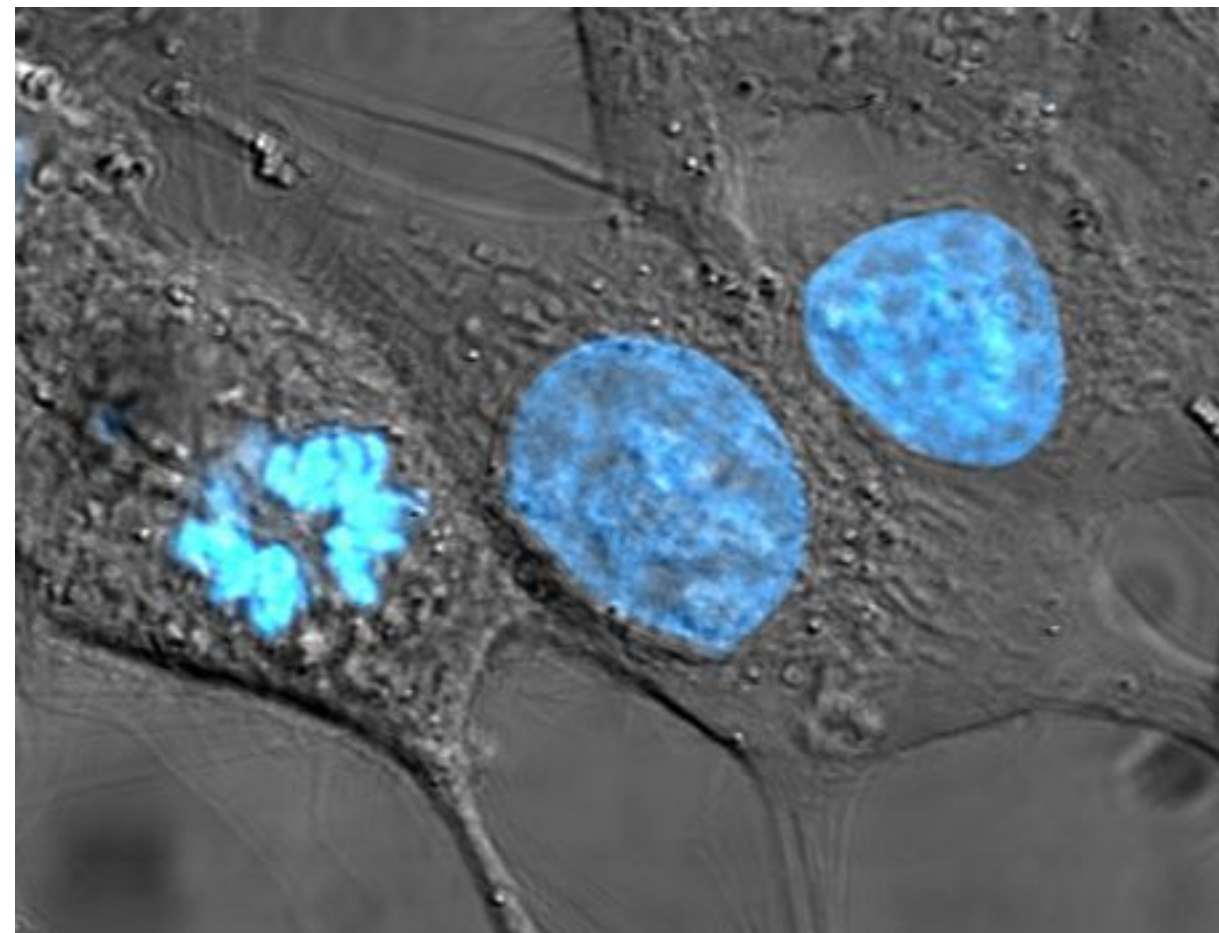
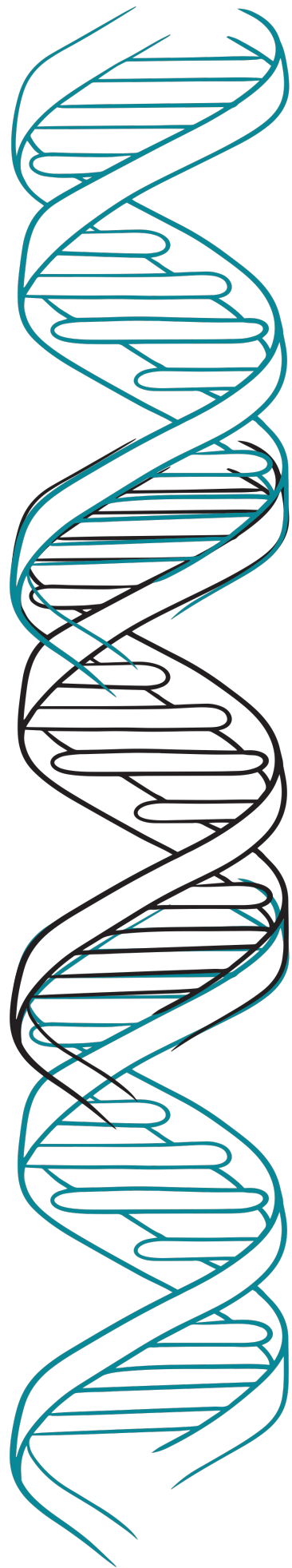
*build submission script, submit
job, and assess output*



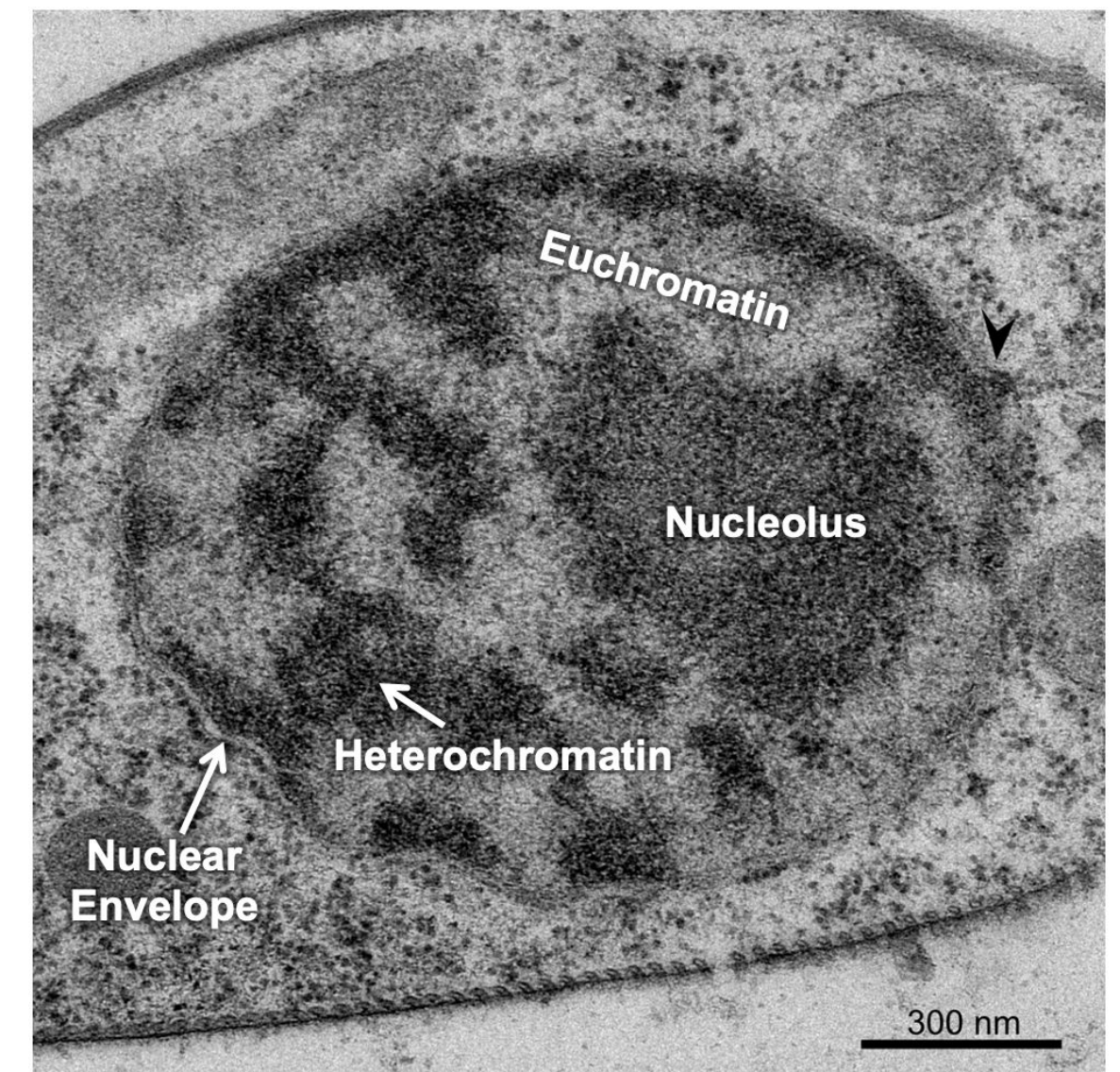
Genome Assembly: What is a Genome?

DNA instructions for the construction and maintenance of a specific biological being

- including protein-coding and non-coding regions
- comprised of long strings of four nucleic acids
- packaged around protein (histones) into discrete packages (chromosomes)

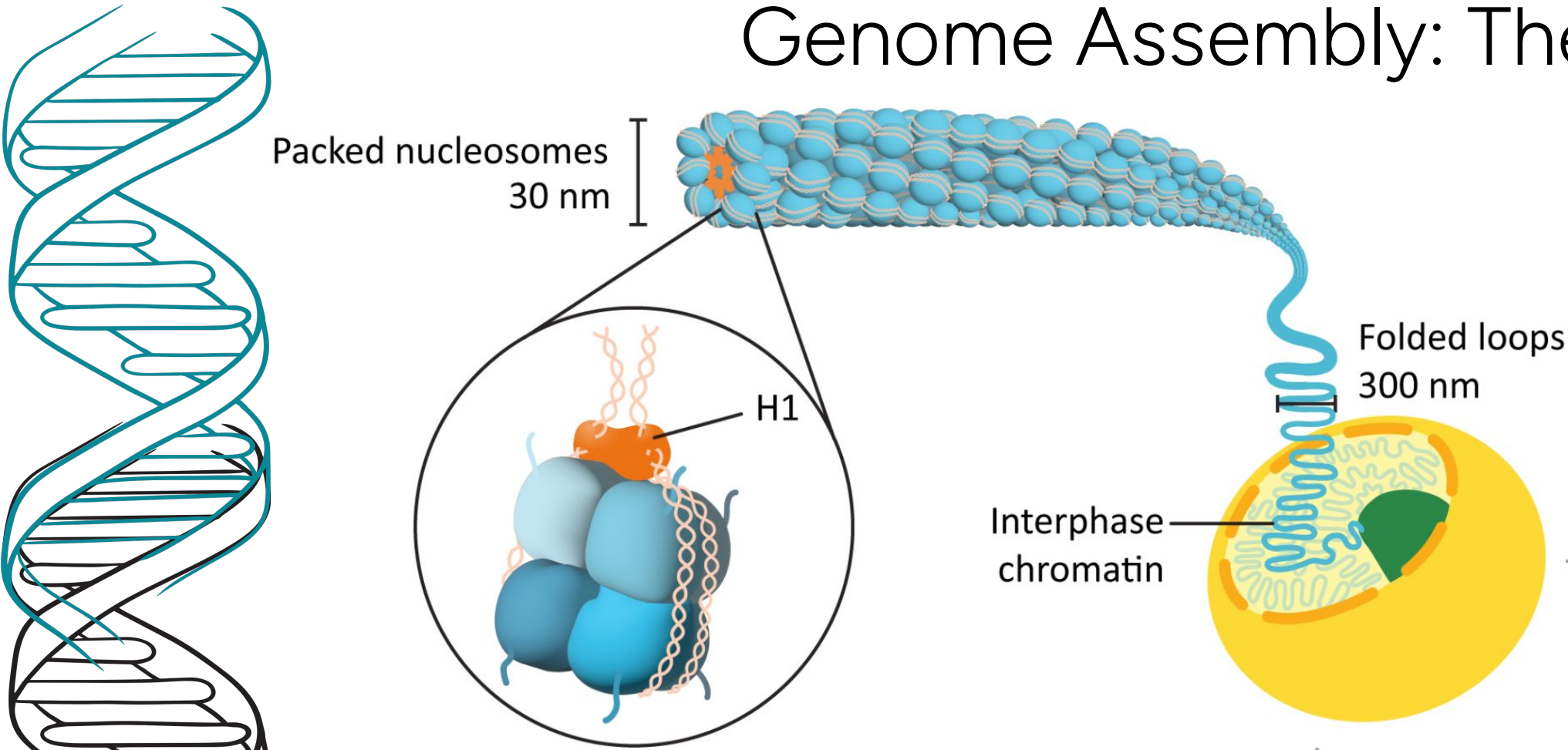


wikimedia: TenofAllTrades

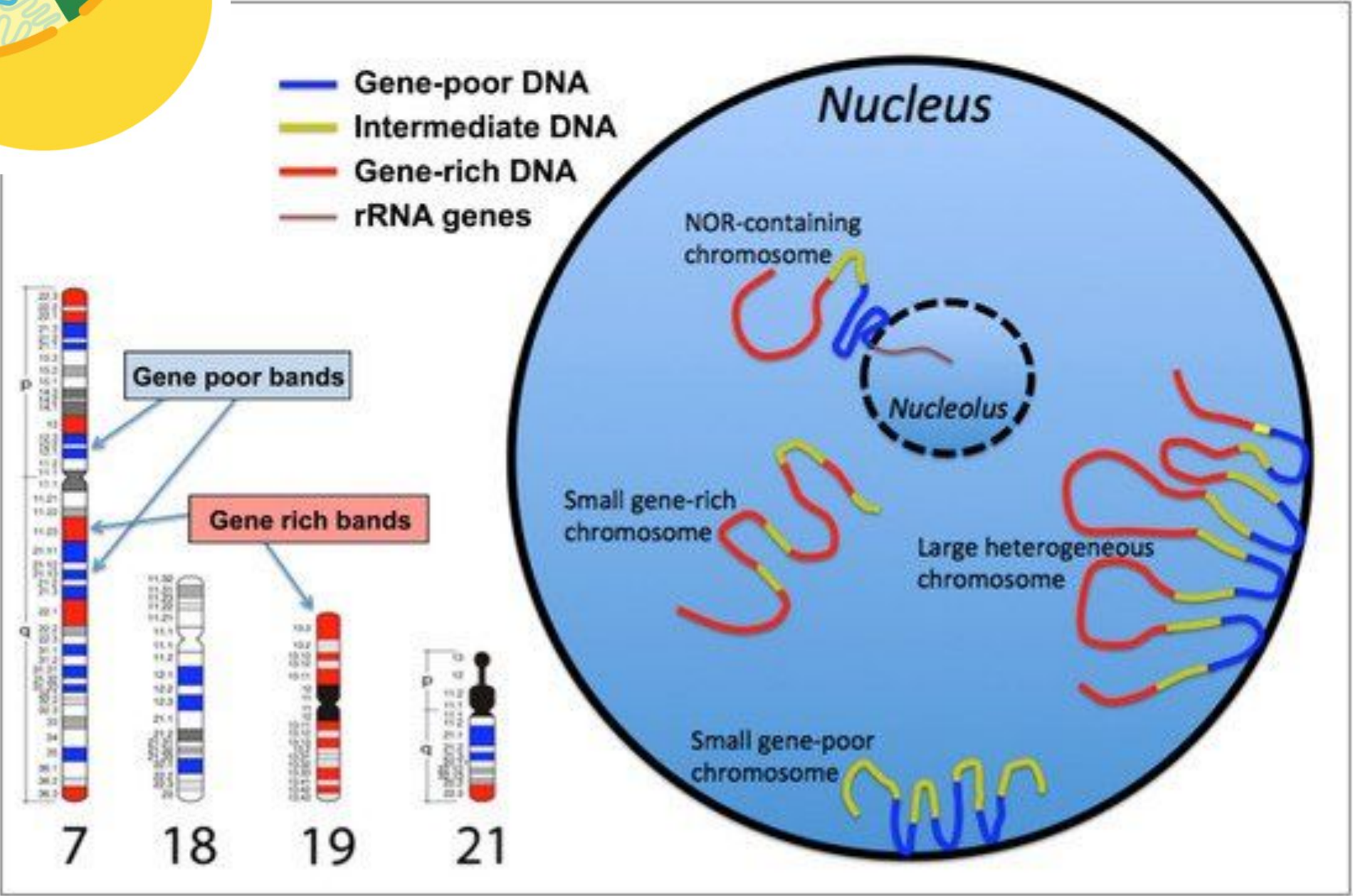


Oregon State University

Genome Assembly: The Organization of Genetic Material



Oregon State University

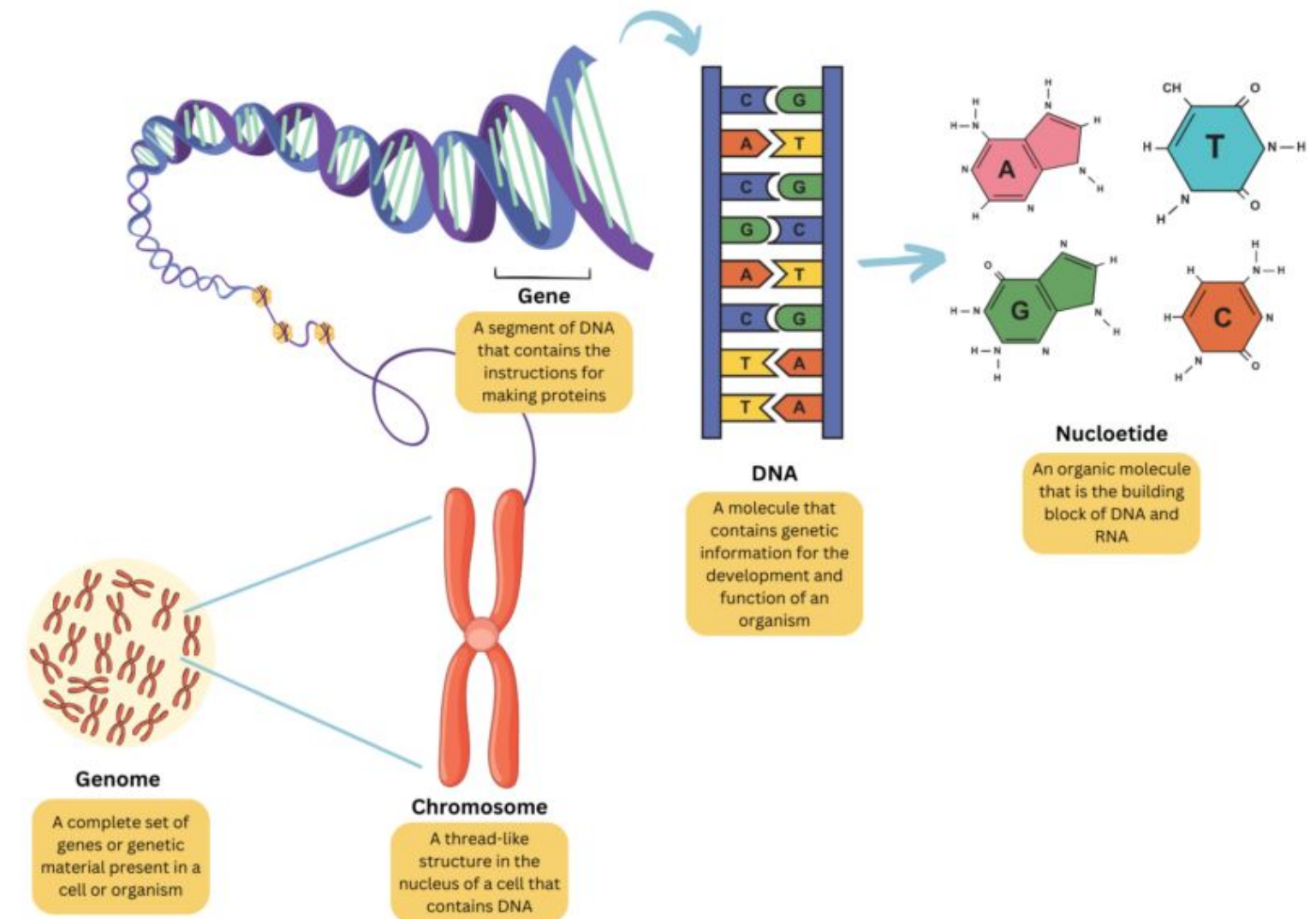


Genome Assembly: What are the goals?

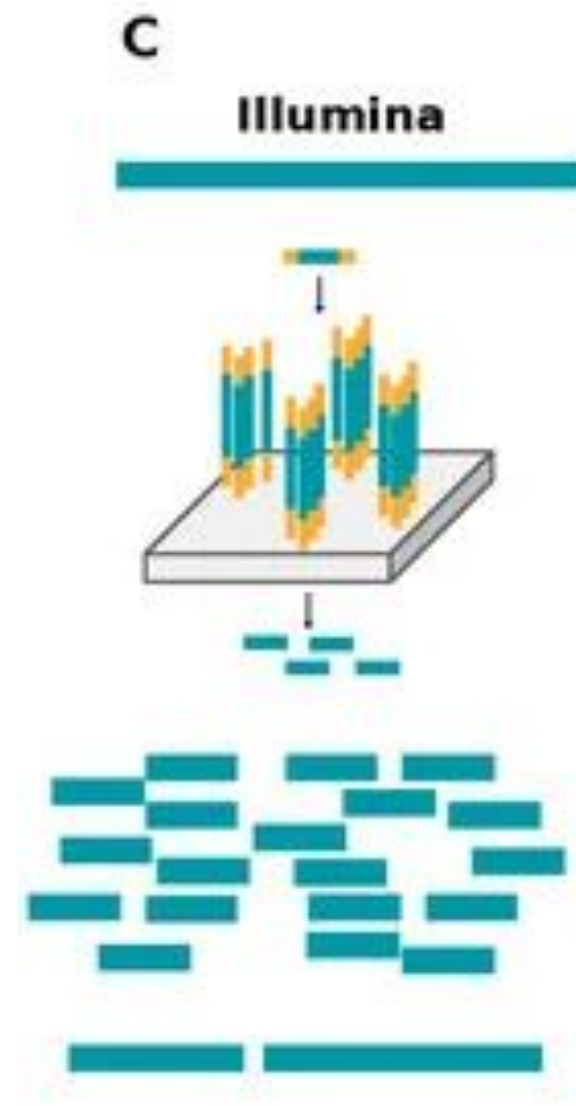
1. Determine the complete nucleotide sequences of an organism
2. Describe genome organization and structure (e.g., chromosomes)
3. Understand the functional consequences of the genome (e.g., annotate protein-coding genes and regulatory regions)

For what purpose?

- Evolutionary biology
- Conservation biology
- Pest science
- Ecological interactions
- Behavior



Sequencing Techniques



**Library and
Sequencing**

Reads

**Assembled
Genome**

- High throughput and speed
- Accuracy and reliability
- Scalable and versatile
- User-friendly and well established technology
- Low cost per gigabase

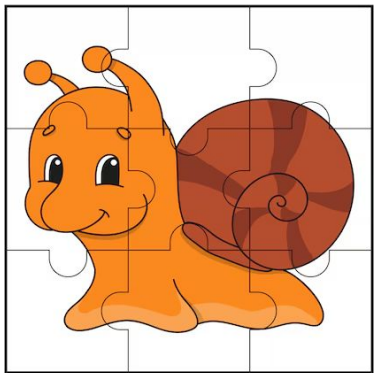
Short-read
50 - 250 bp



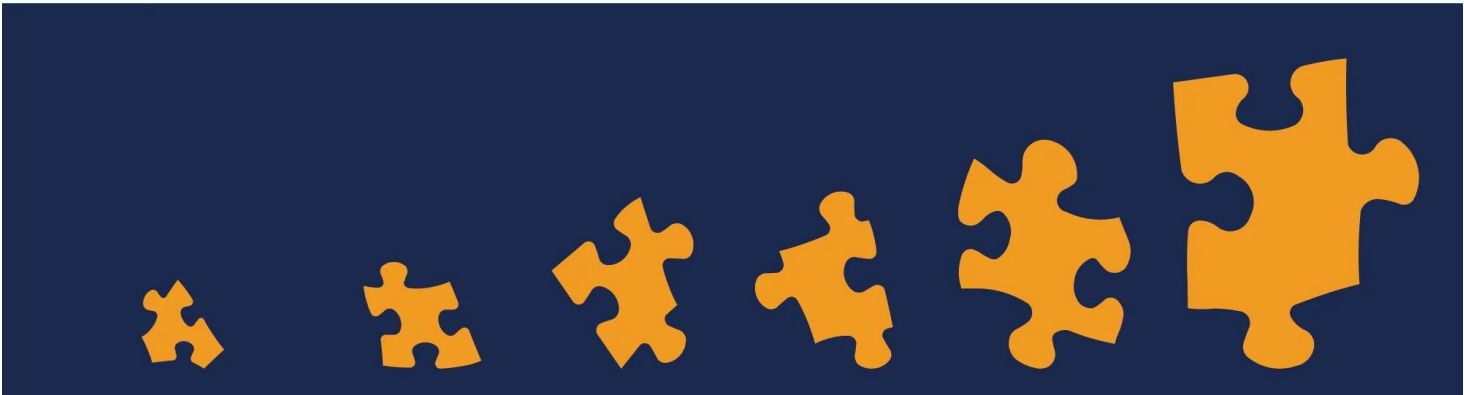
The Puzzle Analogy

Long read sequencing & genome assembly

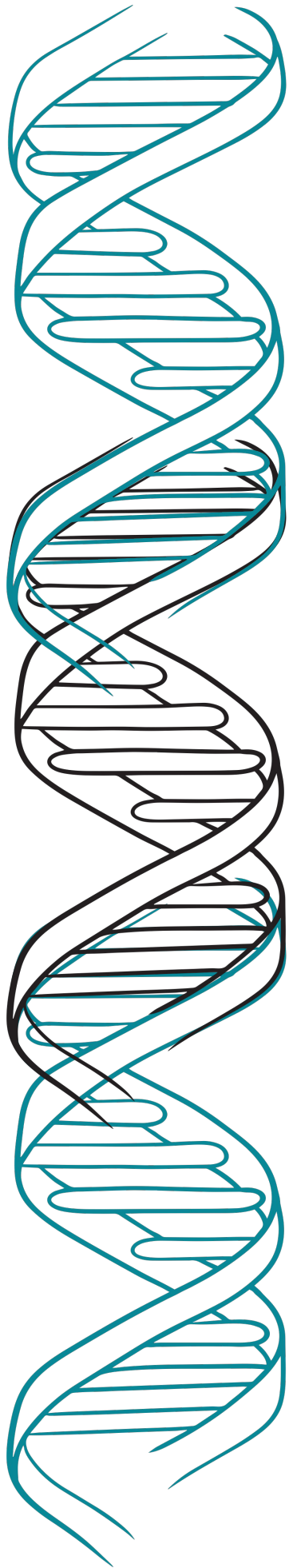
Complexity



Size



Genome Assembly: *Tineola bisselliella*



"This moth has evolved to be an efficient, dynamic, super-survival machine."

-Isabel Novick, PhD candidate, BU

<https://entomologytoday.org/2023/03/21/defense-webbing-clothes-moths-marvels-evolution/>

- digests keratin (notoriously hard to digest) along with plastics and toxic metals
- metabolizes (creates) its own water
- survives temperature range from 5 F to 95 F
- is light averse!? (this is just weird for a moth)

look over example submission scripts & discuss outputs

Our Assembly Pipeline: getting raw reads

SRA toolkit

download publicly available long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size, heterozygosity

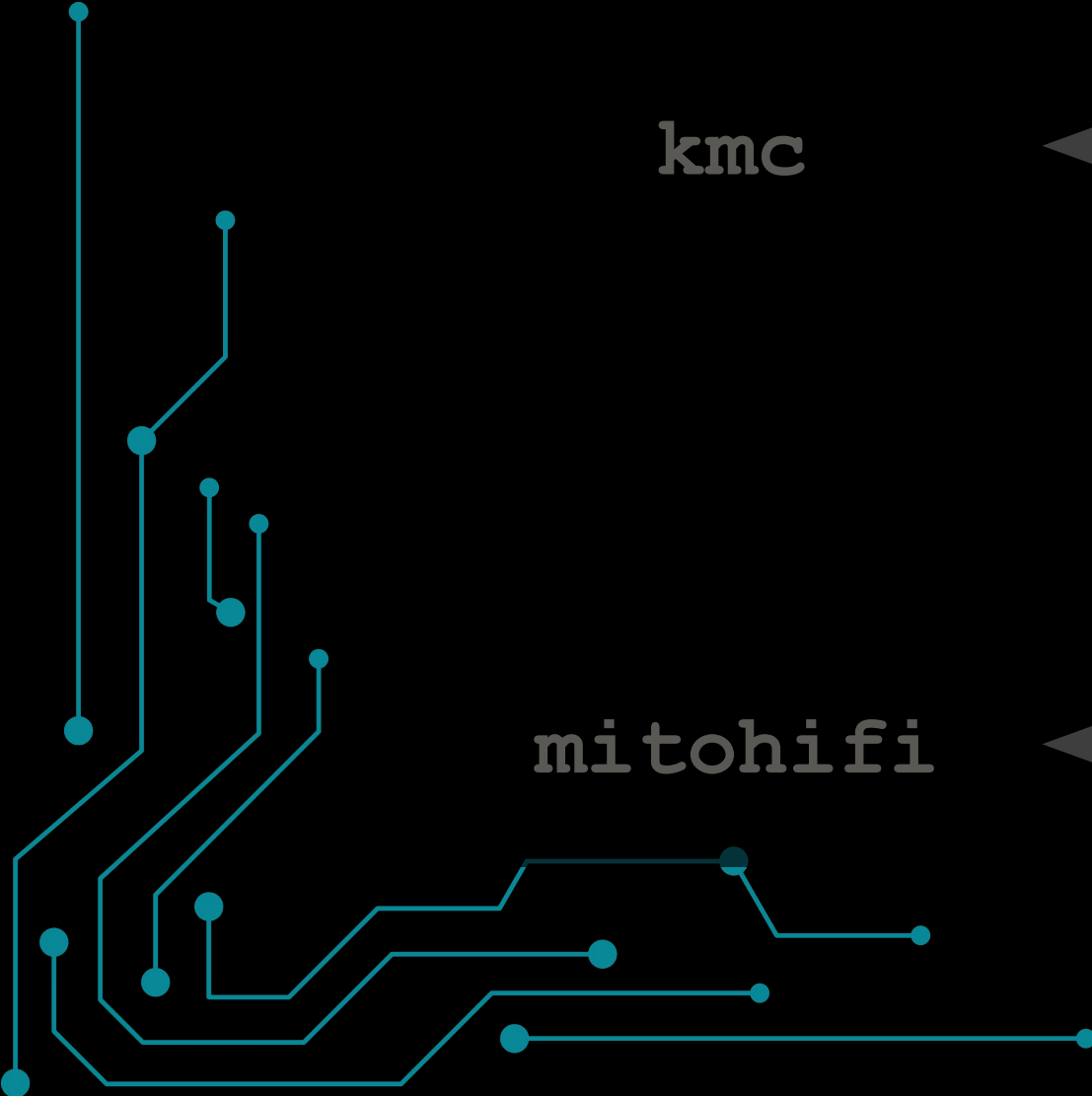
build assembly de novo using long-read data

Hifiasm

mitohifi

pull mitochondrial genome sequences from assembly

build submission script, submit job, and assess output



Steps to getting raw reads

Goal: To download raw reads from the National Center for Biotechnology Information

Tasks:

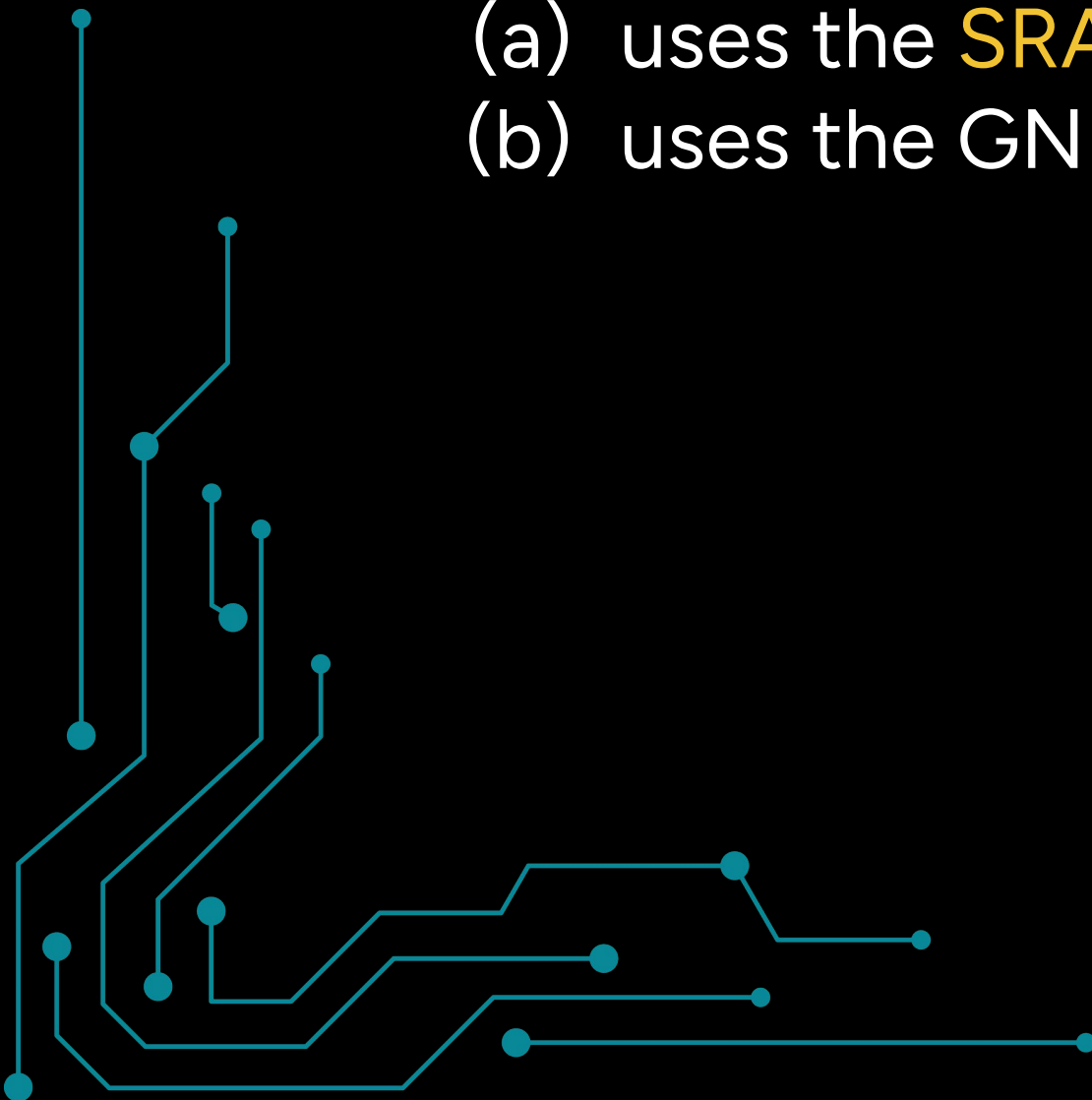
- (1) Locate the reads on the NCBI website and make note of the file name
- (2) Construct a Slurm script that
 - (a) uses the SRA toolkit to download the file to HiperGator
 - (b) uses the GNU utility, gzip, to compress the fastq file



Simple Linux Utility for Resource Management

Sequence Read Archive: repository for high-throughput sequencing data

Widely adopted GNU utility used on Unix-like systems (Linux, macOS) for file compression



SRA toolkit bash script

```
#!/bin/bash
#SBATCH --job-name=SRA
#SBATCH -o SRA_%j.out
#SBATCH --mail-user=rkeatinggodfrey@middlebury.edu
#SBATCH --mail-type=ALL
#SBATCH --cpus-per-task=1
#SBATCH --mem-per-cpu=8gb
#SBATCH --time=00:30:00
#SBATCH --account=kawahara

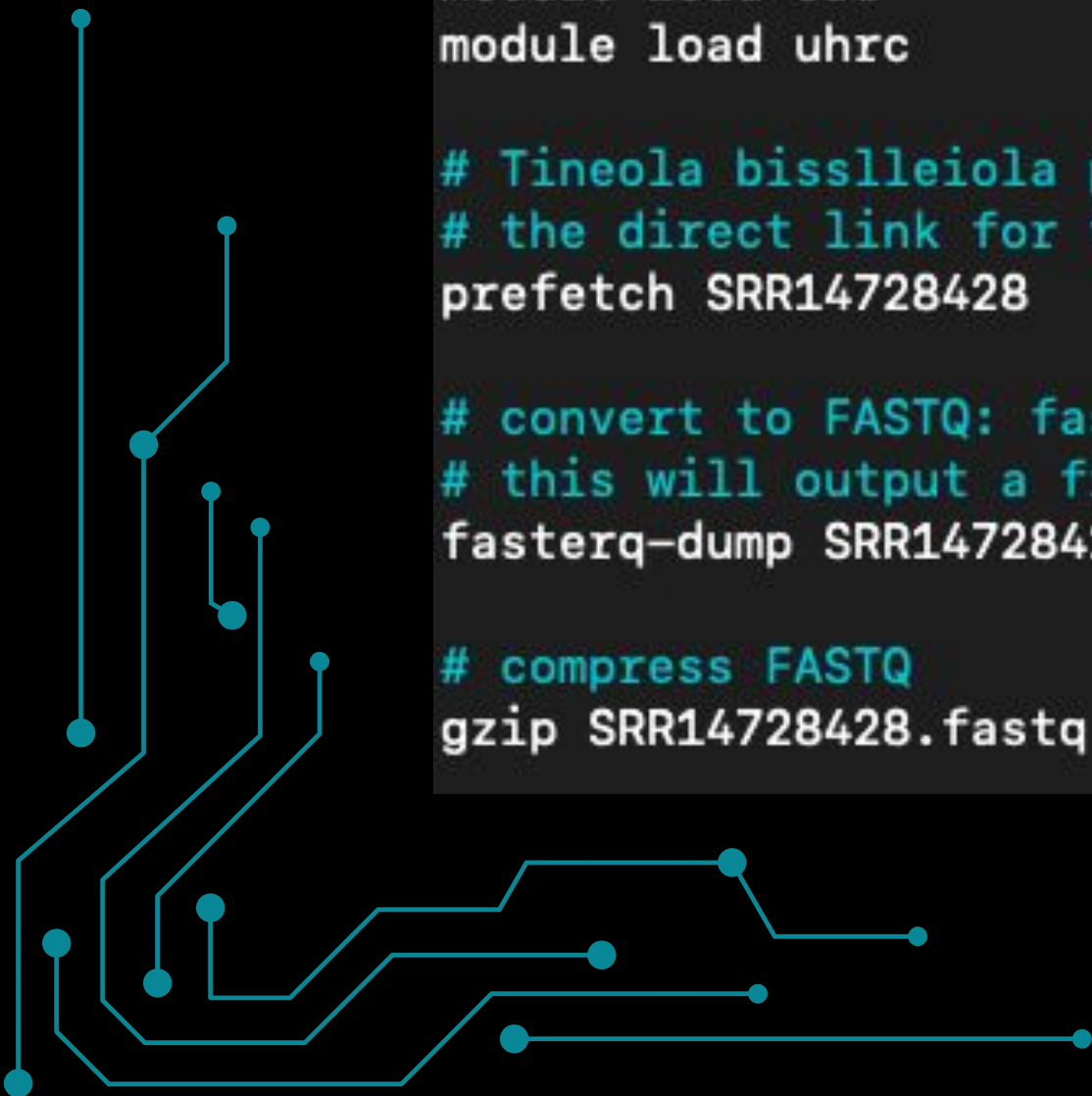
# Job name
# Log file name with %j as job number
# Email address
# Types of notifications
# Number of CPU cores per task
# Max memory allotted to each cpu
# Time limit hrs:min:sec
# Account where job will be run

module load sra
module load uhrc

# Tineola bisslleiola pac bio hifi reads
# the direct link for this sequence read archive: https://www.ncbi.nlm.nih.gov/sra/SRX24886948\[acn\]
prefetch SRR14728428

# convert to FASTQ: fastq-dump will convert .sra to .fastq
# this will output a file of the same name with a fastq extension
fasterq-dump SRR14728428

# compress FASTQ
gzip SRR14728428.fastq
```



SRA toolkit bash script

```
#!/bin/bash
#SBATCH --job-name=SRA
#SBATCH -o SRA_%j.out
#SBATCH --mail-user=rkeatinggodfrey@middlebury.edu
#SBATCH --mail-type=ALL
#SBATCH --cpus-per-task=1
#SBATCH --mem-per-cpu=8gb
#SBATCH --time=00:30:00
#SBATCH --account=kawa

# Job name
# Log file name with %j as job number
# Email address
# Types of notifications
# Number of CPU cores per task
# Max memory allotted to each cpu
# Time limit hrs:min:sec
# Account name where job will be run

module load sra
module load uhrc

# Tineola bisslleiola pac bio hifi reads
# the direct link for this sequence read archive: https://www.ncbi.nlm.nih.gov/sra/SRX24886948[accn]
prefetch SRR14728428

# convert to FASTQ: fastq-dump will convert .sra to .fastq
# this will output a file of the same name with a fastq extension
fasterq-dump SRR14728428

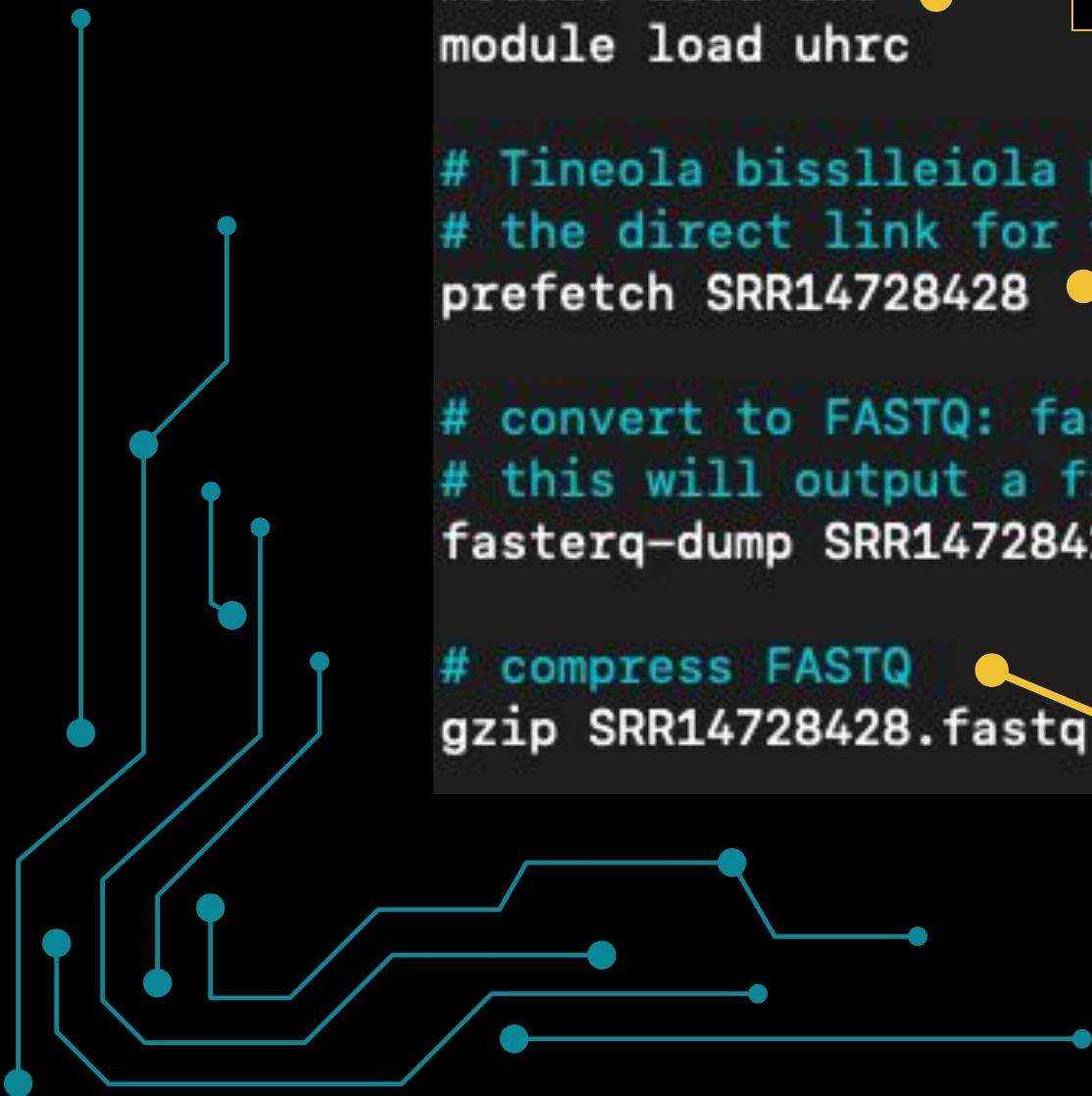
# compress FASTQ
gzip SRR14728428.fastq
```

initializing programs so they are loaded into your environment

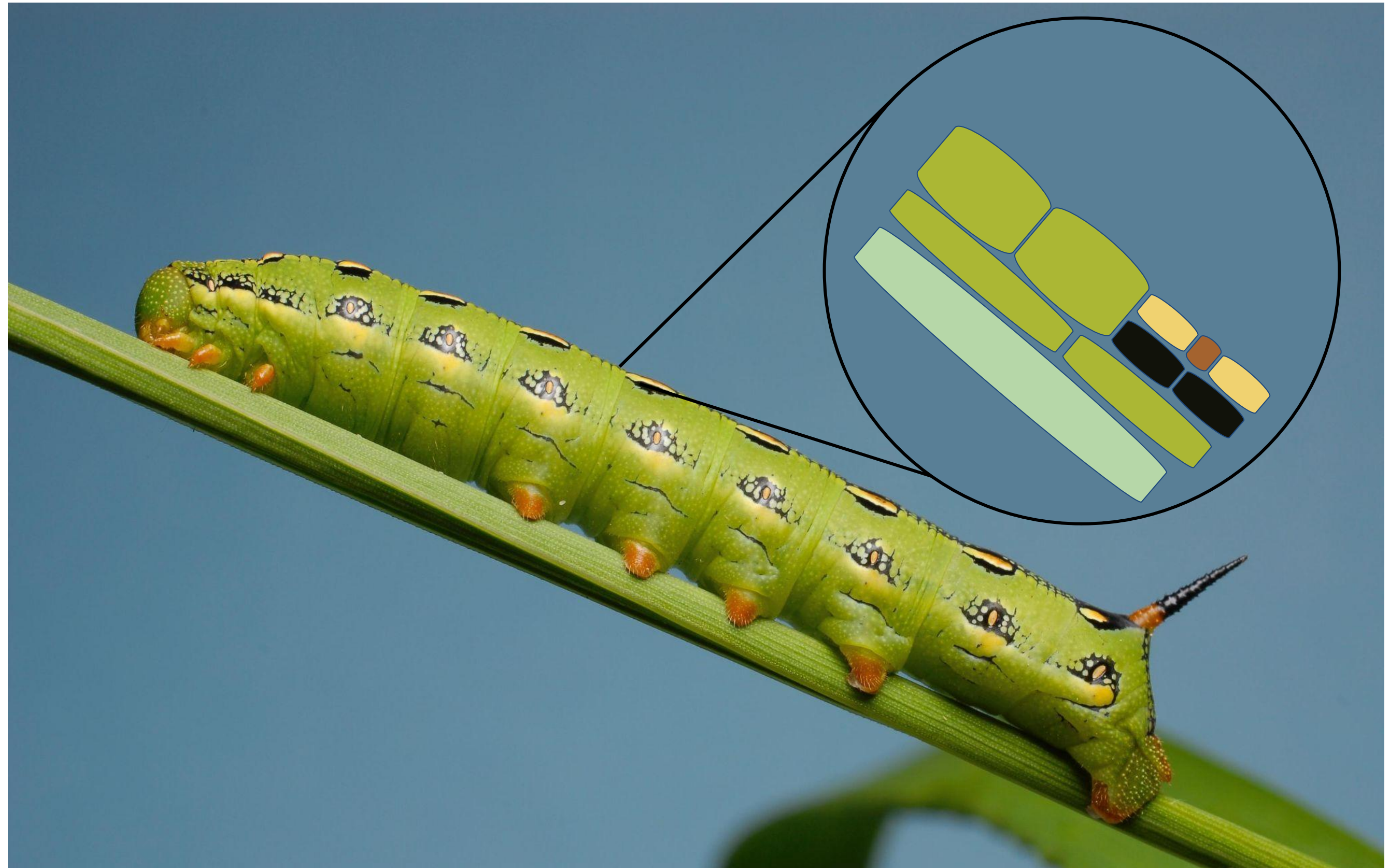
using "prefetch" command to get the reads from NCBI

"dump" those reads into FASTQ file format

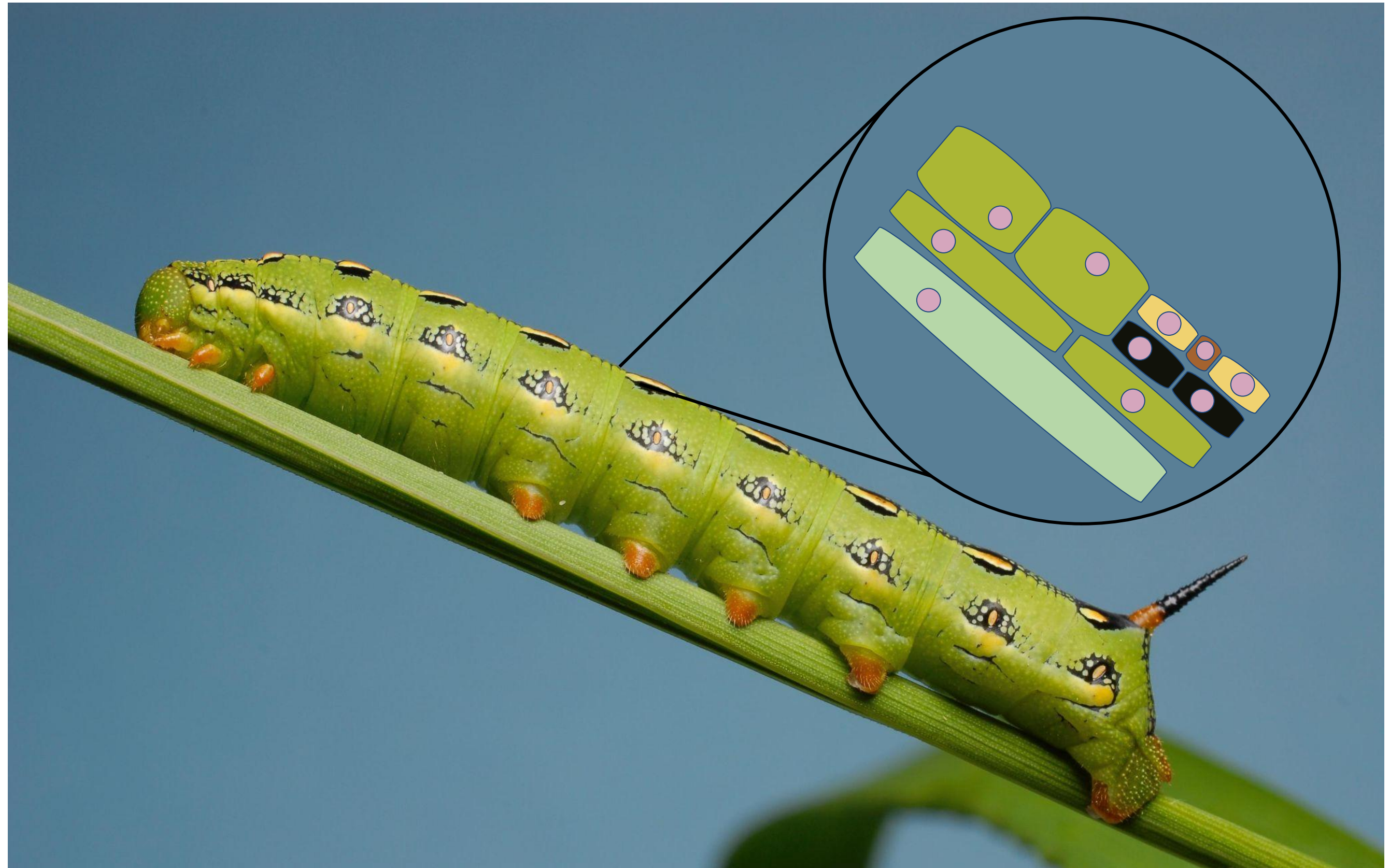
Compress the FASTQ into fastq.gz



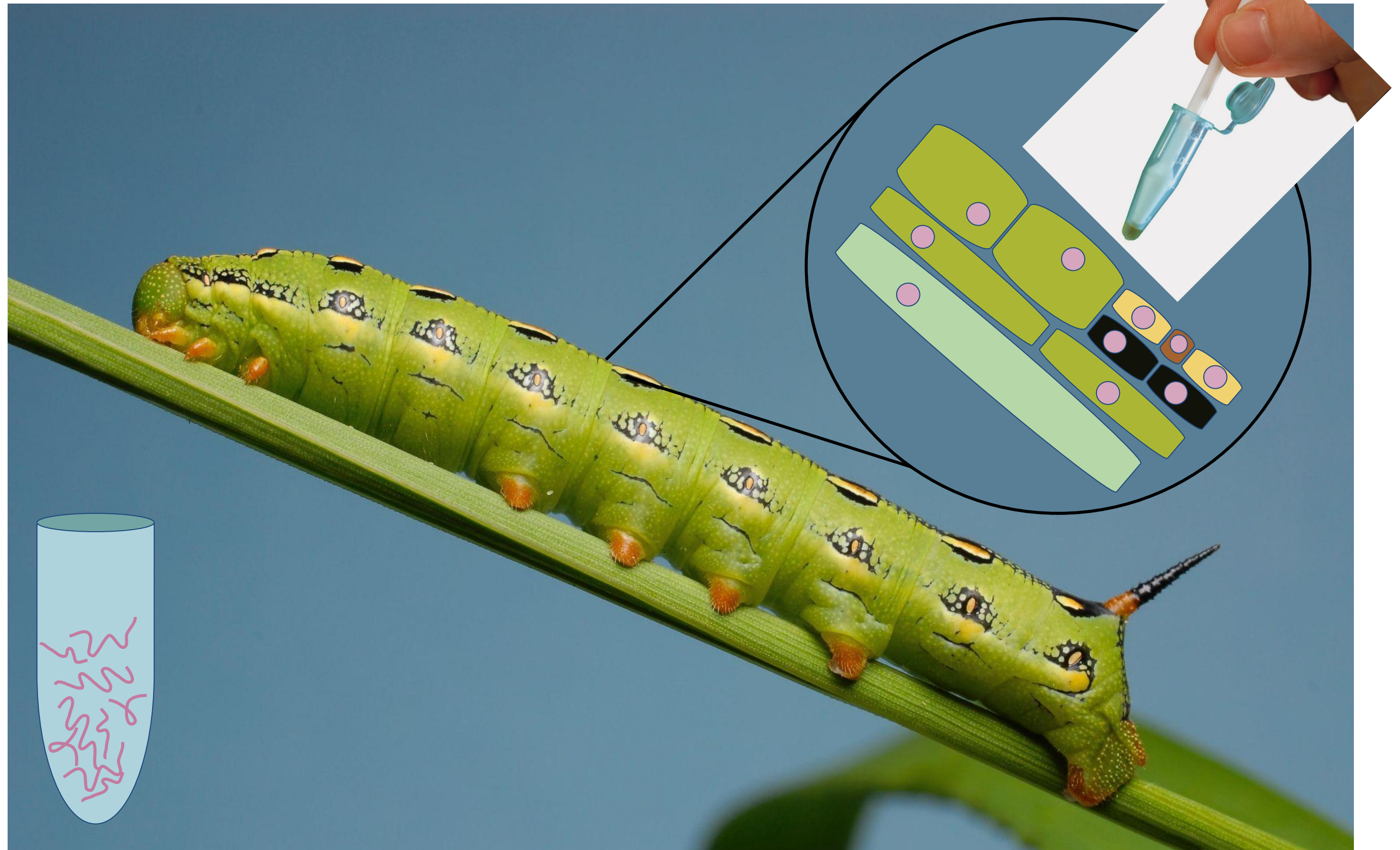
Genome Assembly: So what are these reads?



Genome Assembly: So what are these reads?

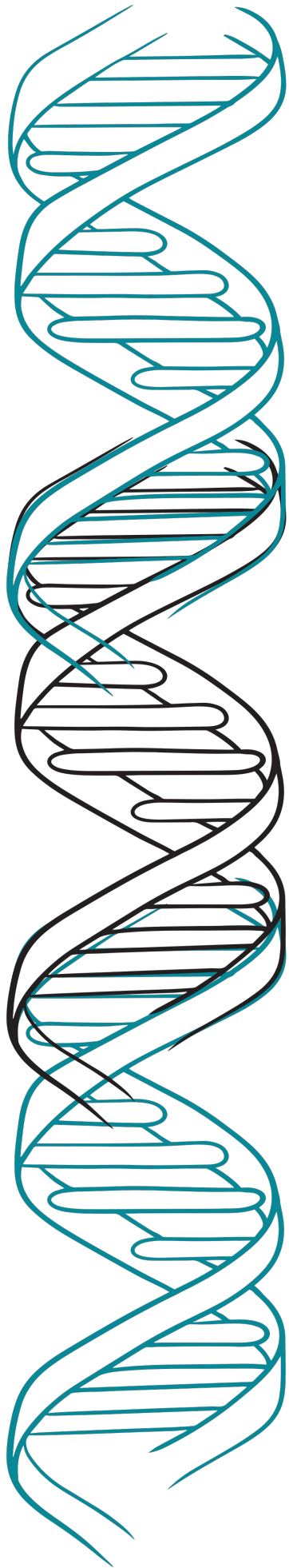


Genome Assembly: So what are these reads?



Genome Assembly: The Newspaper Analogy

Bundle of NYTimes from the
same date



Genome Assembly: The Newspaper Analogy



How can we recreate the newspaper from this date?



Genome Assembly: The Newspaper Analogy



How can we recreate the newspaper from this date?

he ybab ythe reai ntno ea sywa yout iwo ntb
ackdo wnhe yiwil lsta ndm ygro unda ndiw ontb
ackd own wel likno wwha tsri ghti go tju sto
nel ife ina wor ld tha tke epso npus hin mea
roun dbut ill sta ndmy gro und andi wont bac
kdo wn

hey bab yt here ain tnoea sy wayo utiwo ntb
ack dow nhe yi will sta ndm ygro unda ndiw
ontb ack do wnwe lli kno wwha tsri ghti go tju
sto ne life inaw orld tha tke epso npus hin
mea roun dbut il lsta ndm ygro und an diw ont
bac kdo wn



Genome Assembly: The Newspaper Analogy



How can we recreate the newspaper from this date?

heybab ythereaintno easywayout iwontbackdo
wnheyiwillsta ndmygro undandiw ontbackd own
wellikno wwhatsri ghtigotju stonelifaina
worldtha tkeepso npushin mearoundbut illsta
ndmygro undandi wontbackdown

heybabythere aintnoeasy wayoutiwontb ackdow
nheyiwill standmygro undandiwontb ackdownwe
lli knowwha tsrighti gotju stone lifeinaw orld
thatkeepso npushinmea roundbut illsta ndm
yground andiw ontbac kdown

Genome Assembly: The Newspaper Analogy



How can we recreate the newspaper from this date?

What affects the difficulty of the reconstruction (assembly) problem?

- Sequence length
- Repeat content
- Reference or template
- (errors and real variation, too)



look over example submission scripts & discuss outputs

Assessing read quality

SRA toolkit

download publicly available long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size, heterozygosity

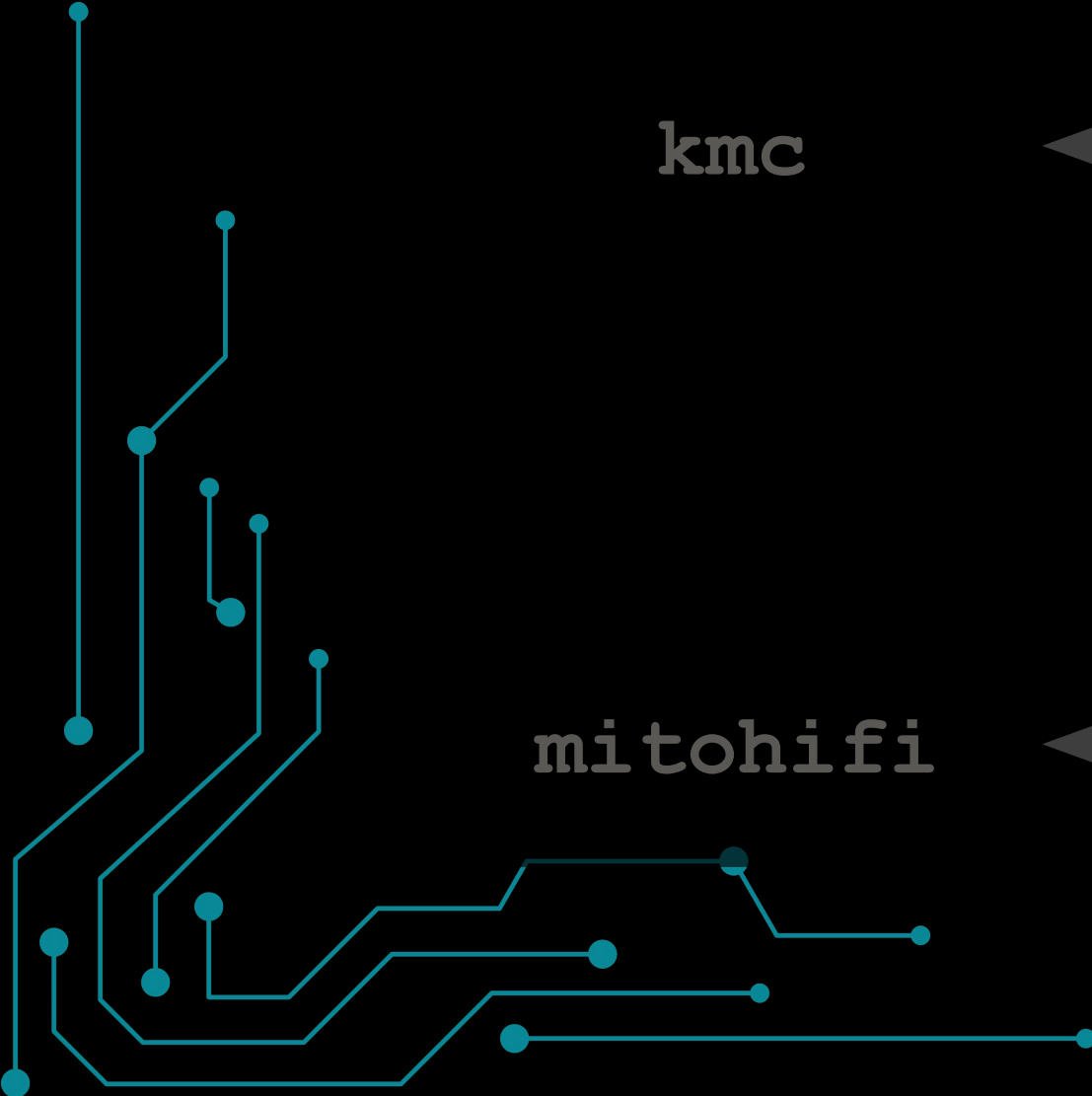
build assembly de novo using long-read data

Hifiasm

mitohifi

pull mitochondrial genome sequences from assembly

build submission script, submit job, and assess output



longqc bash script

```
#!/bin/bash
#SBATCH --job-name=longqc           #this is the name that will show up in the queue
#SBATCH --output=longqc_260113.out  #this is the outfile
#SBATCH --mail-user=rkeatinggodfrey@middlebury.edu #your email address
#SBATCH --mail-type=FAIL,END        # email if job fails or when it ends
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem-per-cpu=8gb
#SBATCH --time=24:00:00
#SBATCH --account=kawahara

## This script runs a longc on the fastq file output from PacBio HiFi Sequel IIe

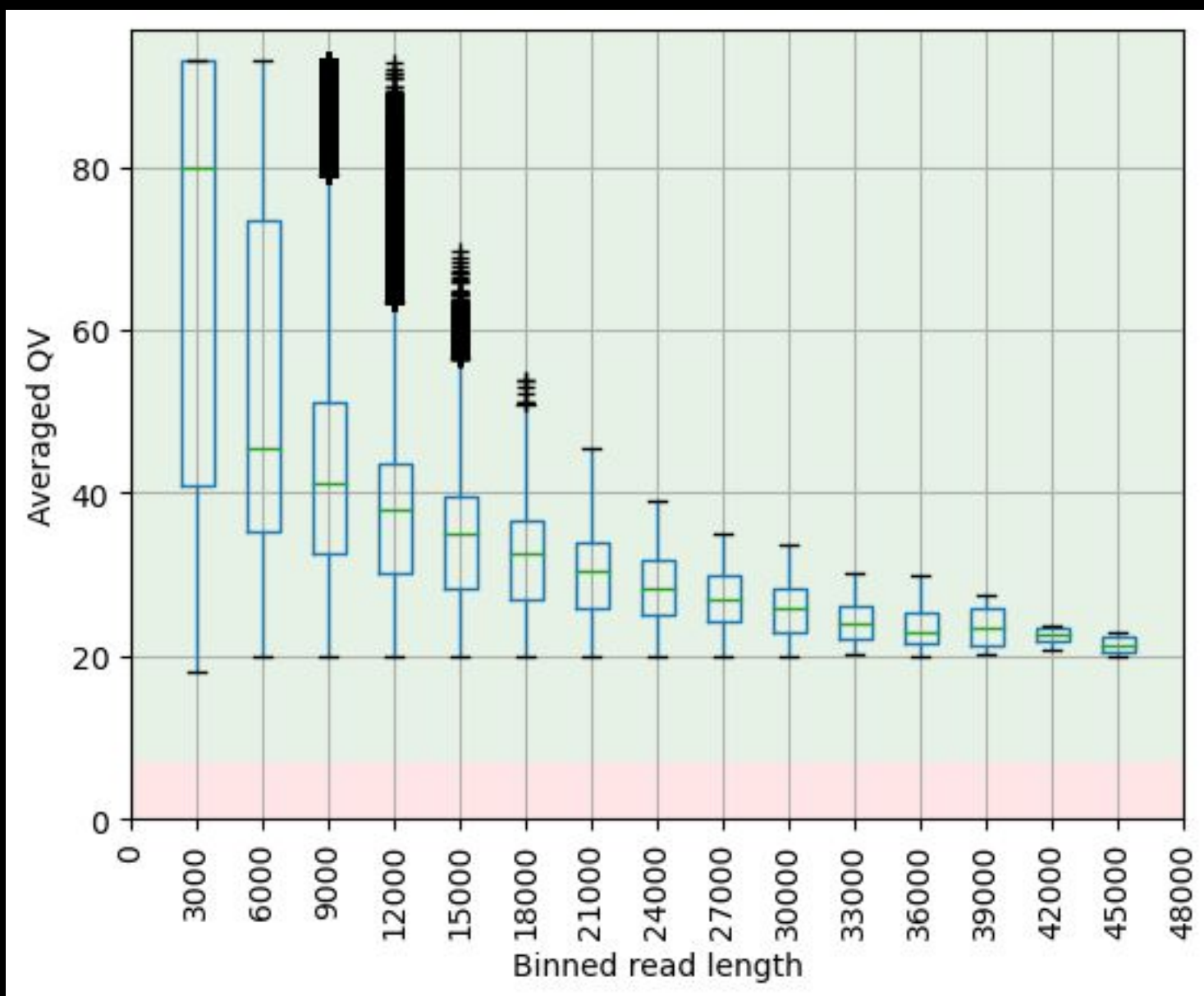
## load longqc module
module load ufrc
module load longqc

## run longqc (a phyon script) with the function sampleqc
## -x indicates the type of sequence and -o is the output directory
## -f indicates fast mode
## the path to the fastq.gz file with reads to be assessed
longQC.py sampleqc -x pb-sequel -o longqc_tbis -f /blue/kawahara/rkeating.godfrey/genome_assembly/01_rawreads/SRR29372512.fastq.gz
```

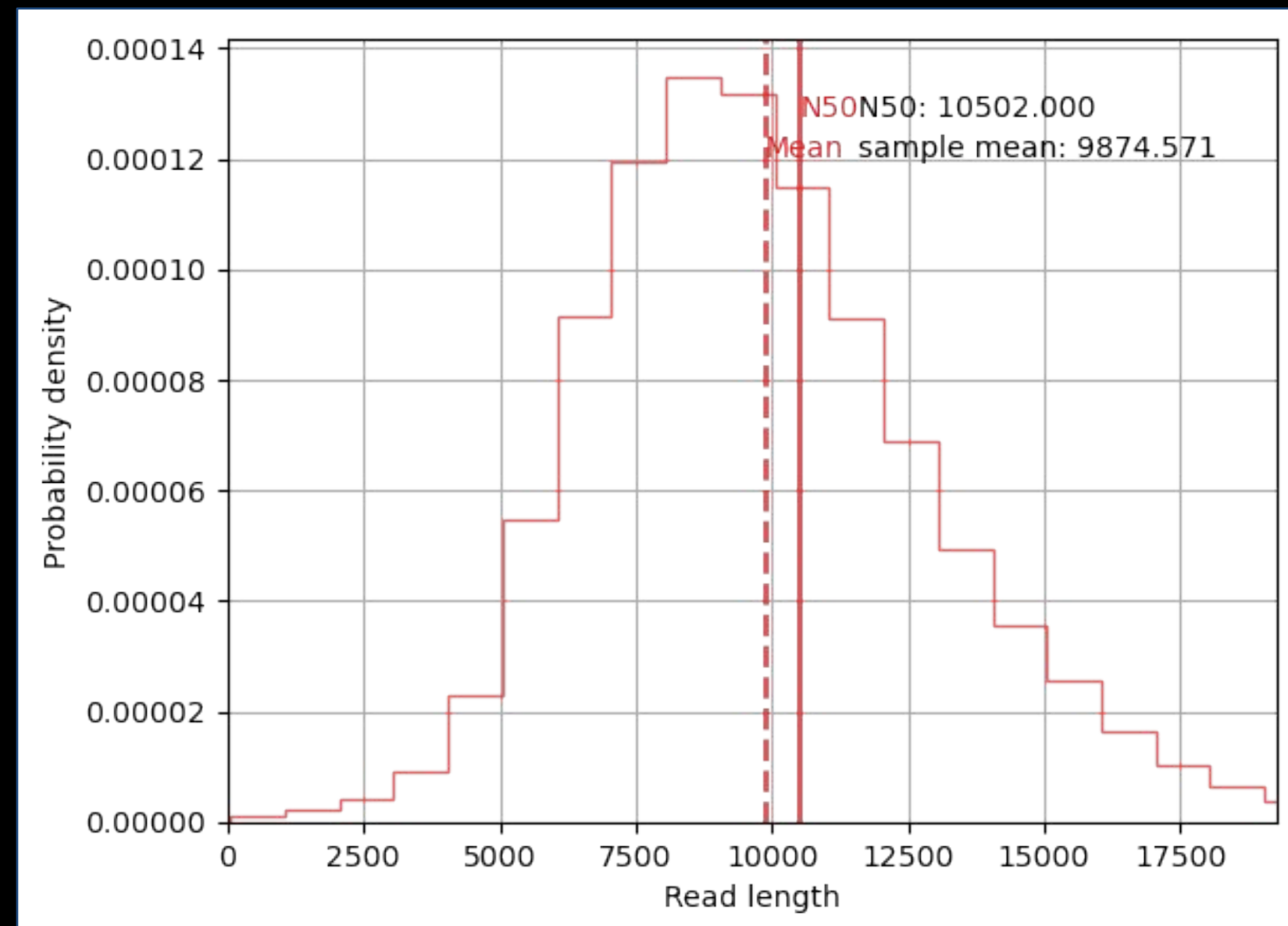
flags, indicated by a dash or two dashes set parameters or "settings" for running a program

longqc output

Quality Scoring
(higher QV = higher confidence in base call)



Higher N50 value generally means better assembly continuity with fewer, longer pieces



N50 = the length of the shortest read where all reads equal to or longer than that length comprise 50% of the total length of all reads summed.

Wait. What is a genome again?

DNA instructions for the construction and maintenance
of a specific biological being



Most sexually reproducing organisms have two or more
slightly non-identical copies of their genome in each cell

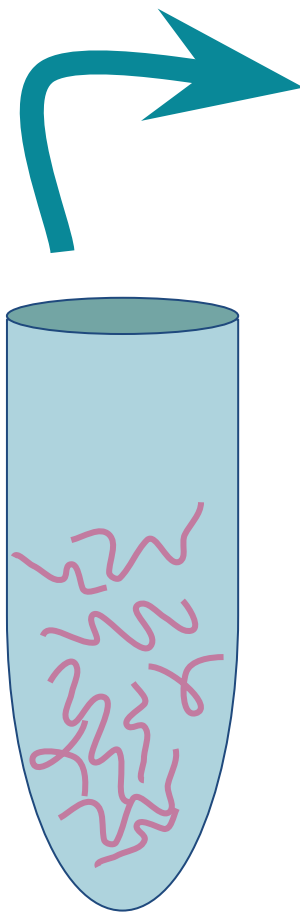
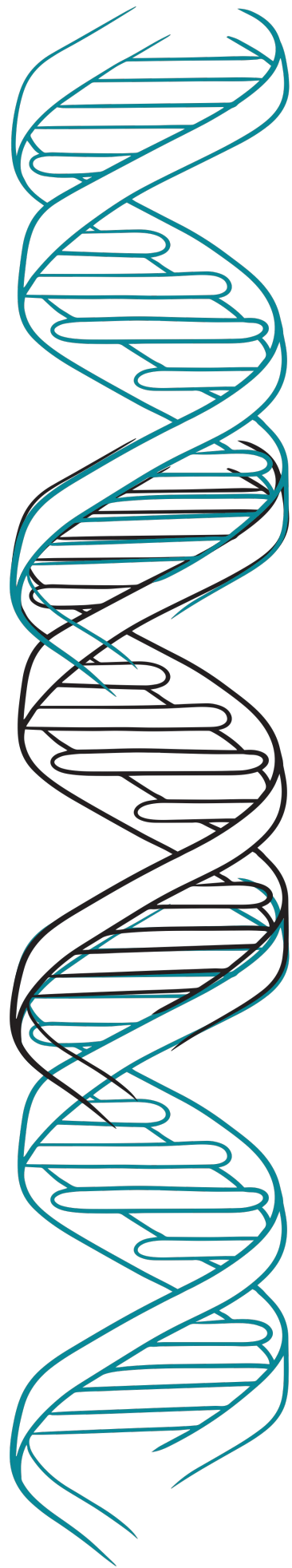


And what is a genome again?

DNA

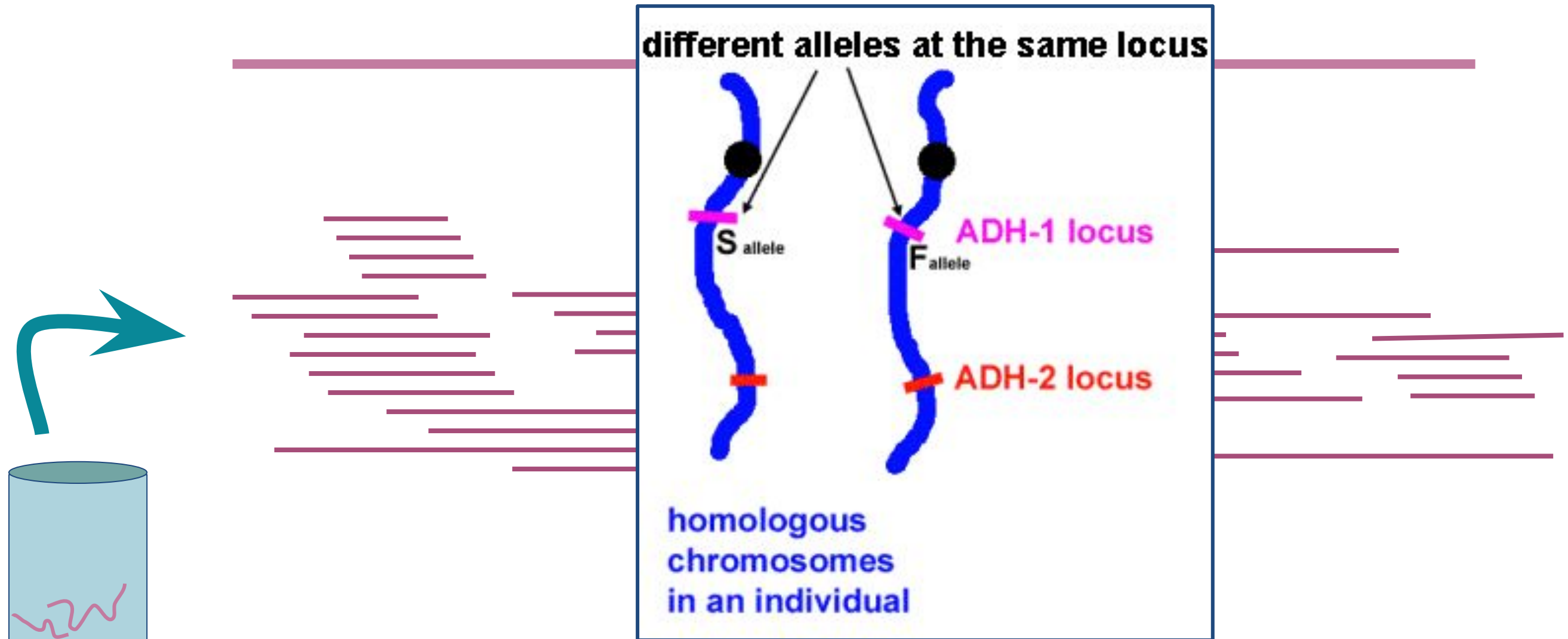


Most sexually reproducing organisms have two or more slightly non-identical copies of their genome in each cell



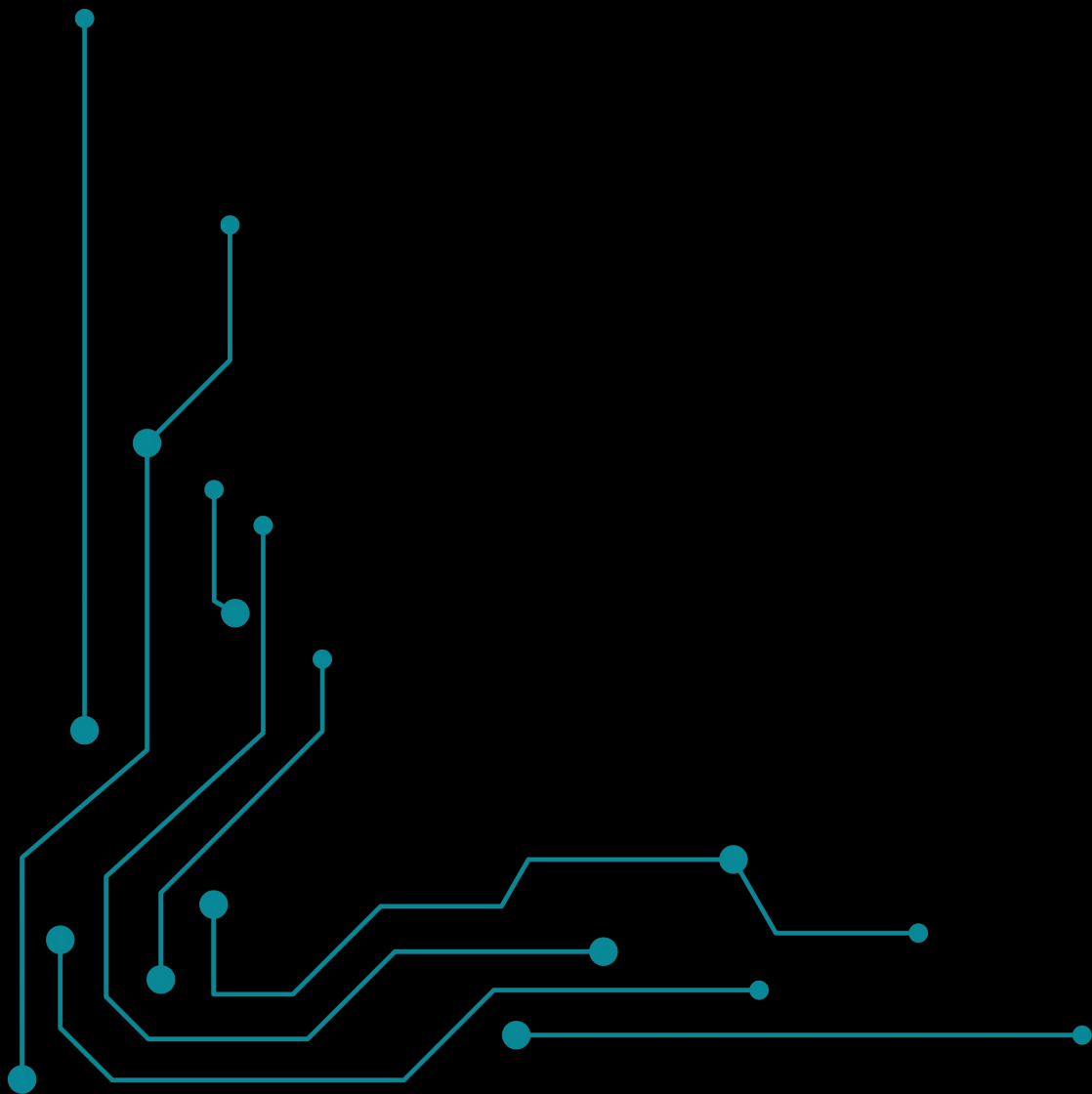
Ploidy in genome assembly

DNA instructions for the construction and maintenance
of a specific biological being



Most sexually reproducing organisms have two or more
slightly non-identical copies of their genome in each cell

How do we assess heterozygosity and coverage?



look over example submission scripts & discuss outputs

Assessing coverage and heterozygosity

SRA toolkit

download publicly available long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size, heterozygosity

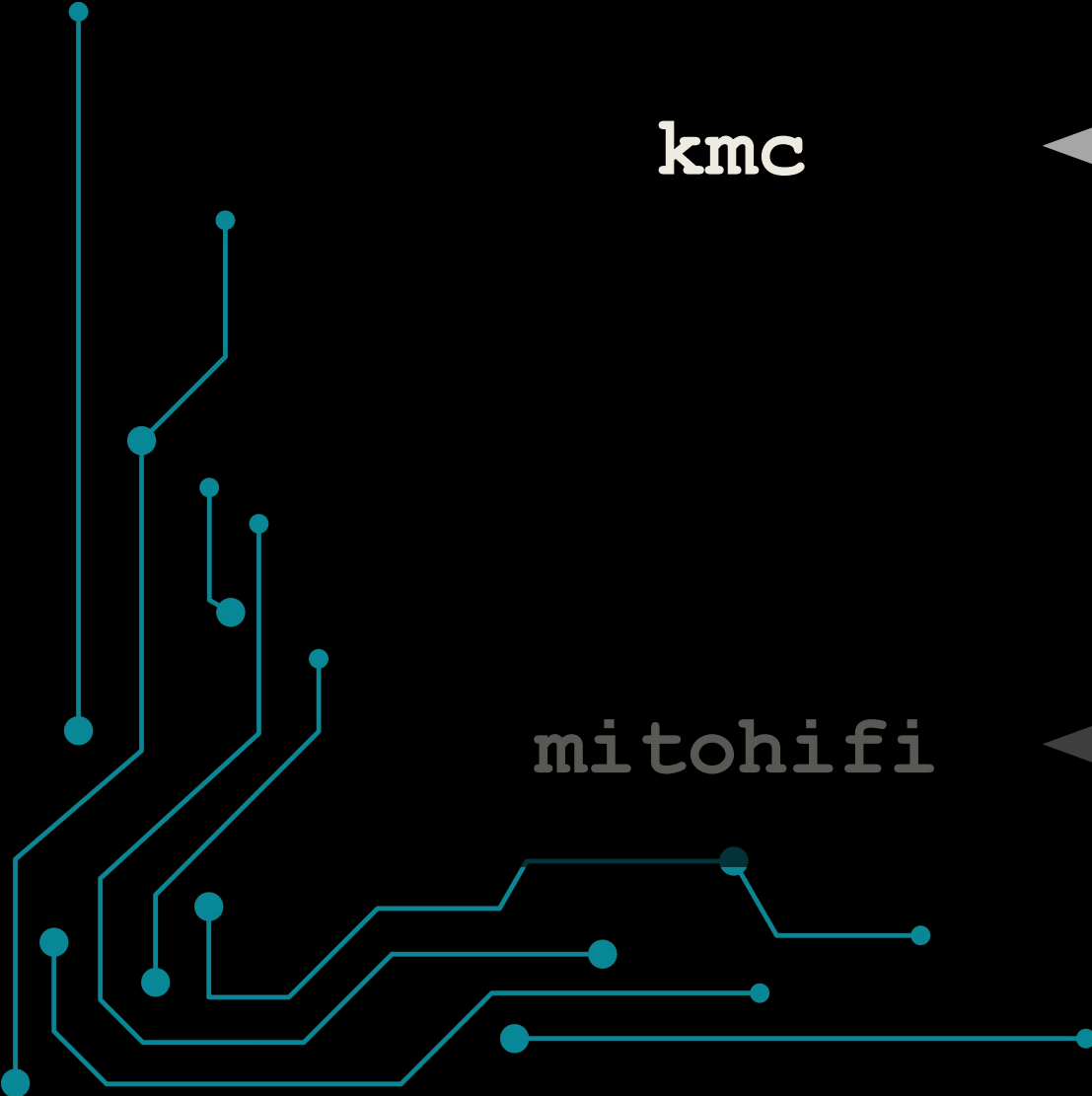
build assembly de novo using long-read data

Hifiasm

mitohifi

pull mitochondrial genome sequences from assembly

build submission script, submit job, and assess output



k-mers: definitions & concepts

Contiguous substring of length k from a longer sequence

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA

k-mers: definitions & concepts

Contiguous substring of length k from a longer sequence

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA

AAATCGT

AATCGTA

ATCGTAT

TCGTATG

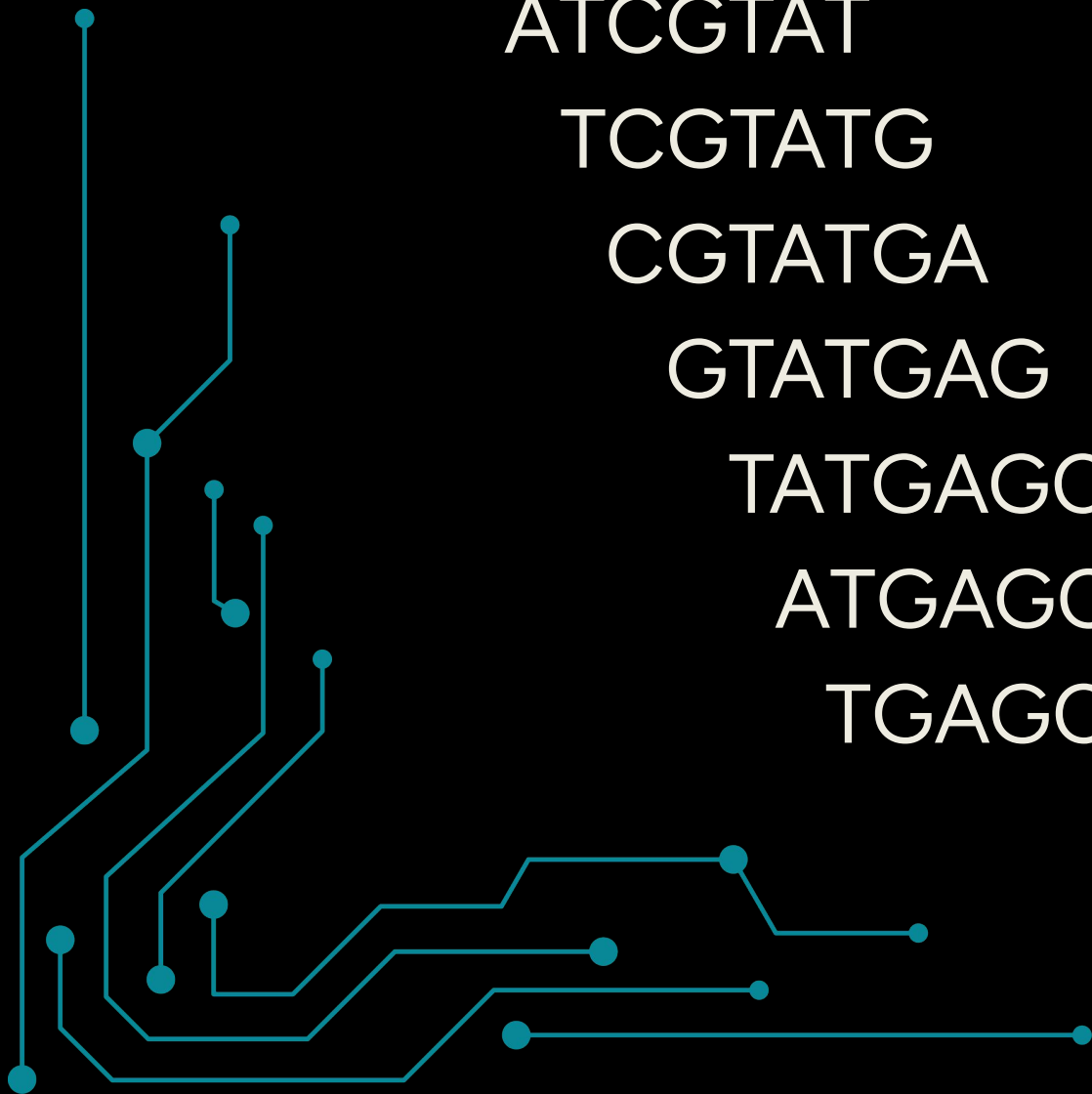
CGTATGA

GTATGAG

TATGAGC

ATGAGCA

TGAGCAA



k-mers: definitions & concepts

Contiguous substring of length k from a longer sequence

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA

7-mers

AAATCGT

AATCGTA

ATCGTAT

TCGTATG

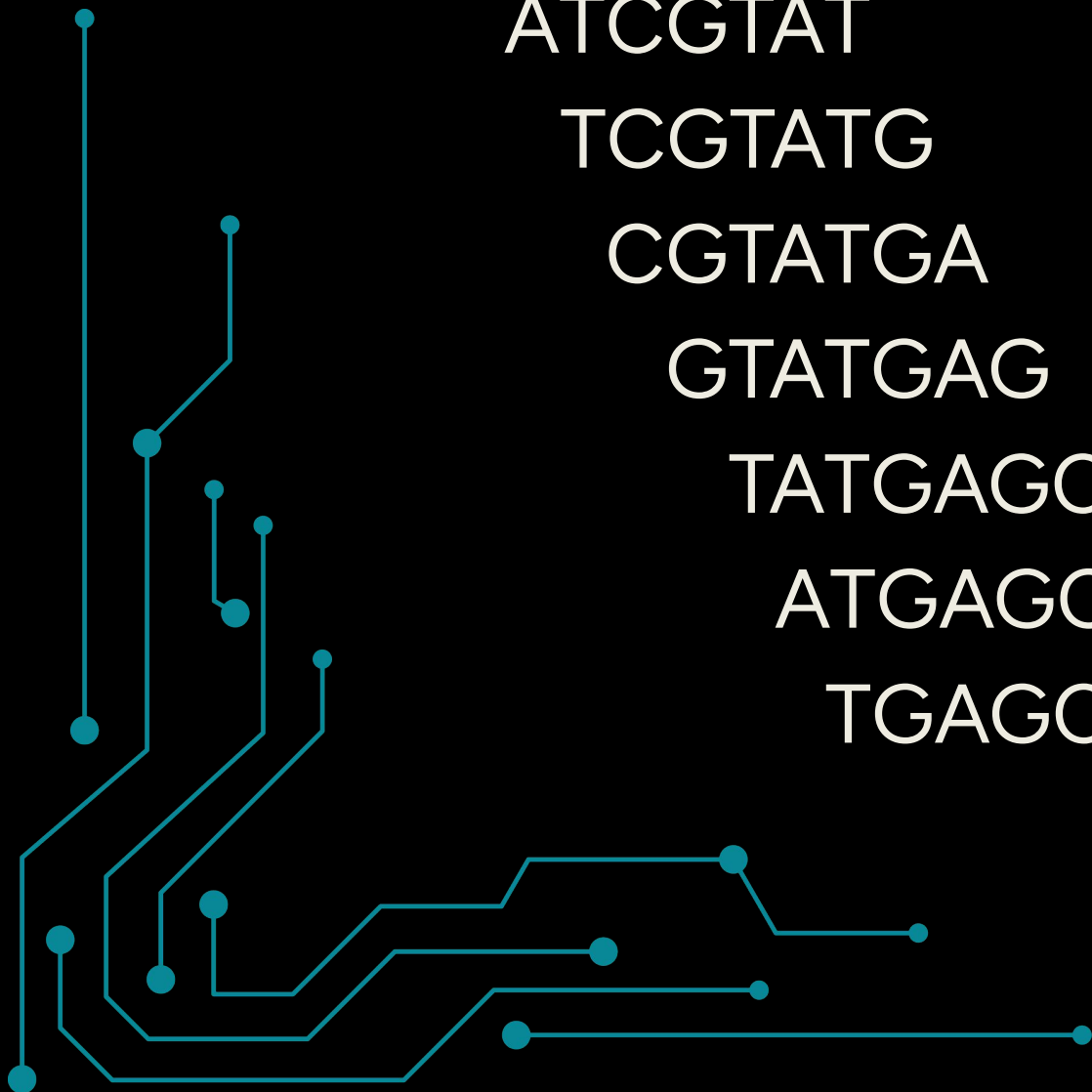
CGTATGA

GTATGAG

TATGAGC

ATGAGCA

TGAGCAA



k-mers: definitions & concepts

We count the number of times each k-mer shows up in our sequences

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA

7-mers

AAATCGT

1

AATCGTA

2

ATCGTAT

TCGTATG

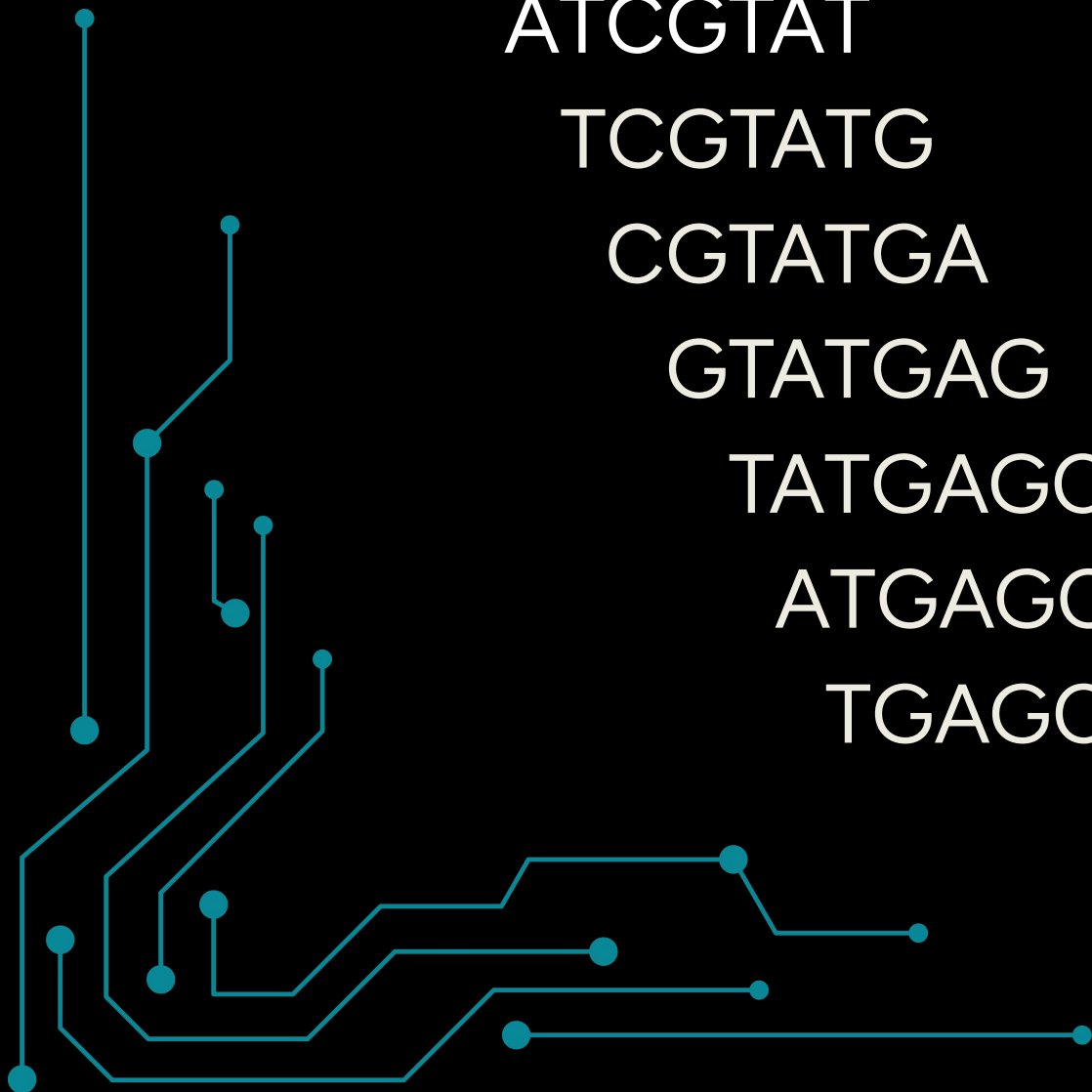
CGTATGA

GTATGAG

TATGAGC

ATGAGCA

TGAGCAA



k-mers: definitions & concepts

We count the number of times each k-mer shows up in our sequences

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA 7-mers

AAATCGT 1

AATCGTA 1

ATCGTAT 3

TCGTATG

CGTATGA

GTATGAG

TATGAGC

ATGAGCA

TGAGCAA



k-mers: definitions & concepts

We count the number of times each k-mer shows up in our sequences

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA	7-mers
AAATCGT	1
AATCGTA	1
ATCGTAT	3
TCGTATG	1
CGTATGA	2
GTATGAG	2
TATGAGC	2
ATGAGCA	1
TGAGCAA	



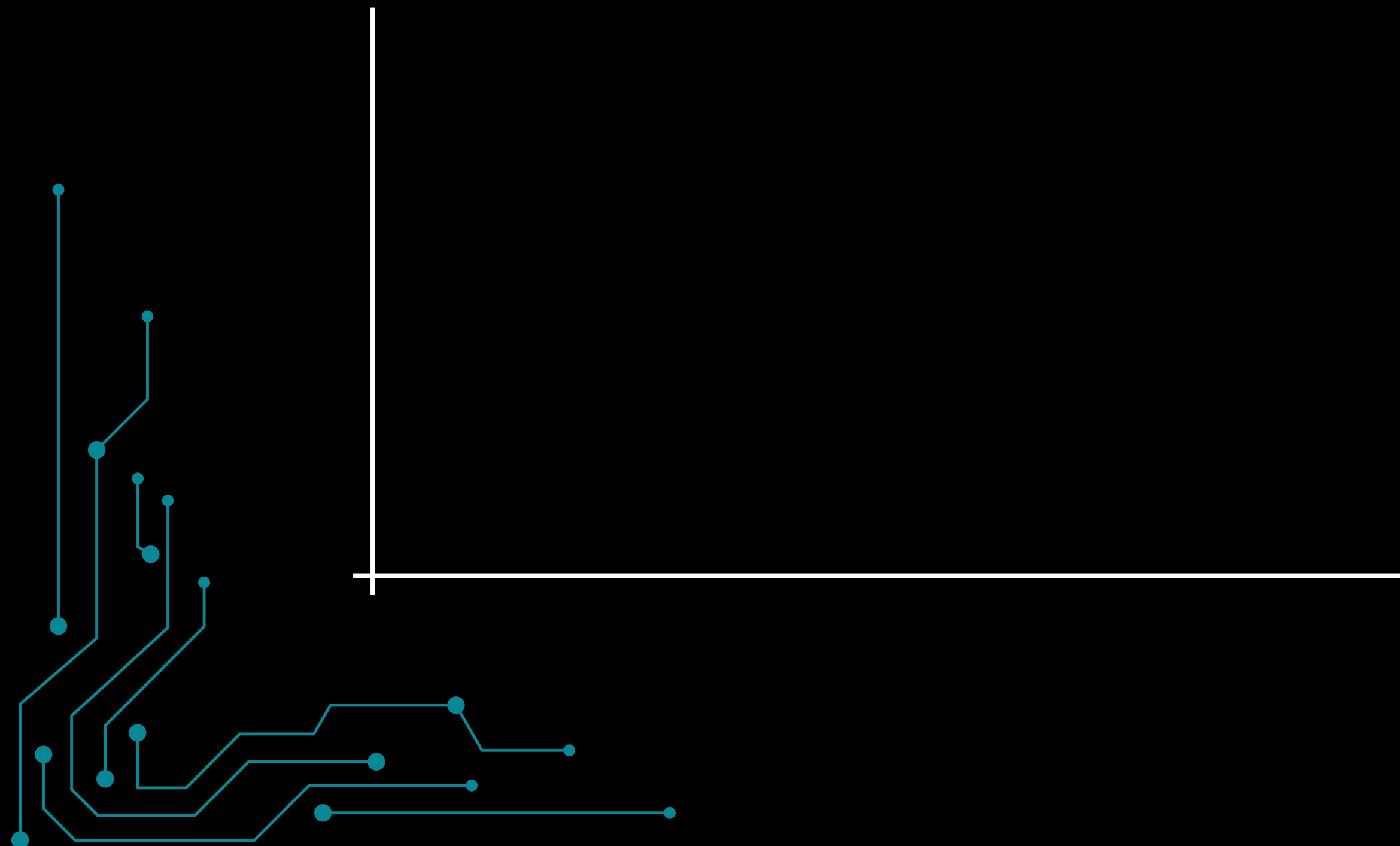
k-mers: definitions & concepts

The number of times a particular k-length sequence
occurs in your read data

AAATCGTATGAGCAATCGTATGAGCTAGGTATCGTATAA	7-mers
AAATCGT	1
AATCGTA	1
ATCGTAT	3
TCGTATG	1
CGTATGA	2
GTATGAG	2
TATGAGC	2
ATGAGCA	2
TGAGCAA	1



The number of times a particular k-length sequence
occurs in your read data



7-mers

1

1

3

1

2

2

2

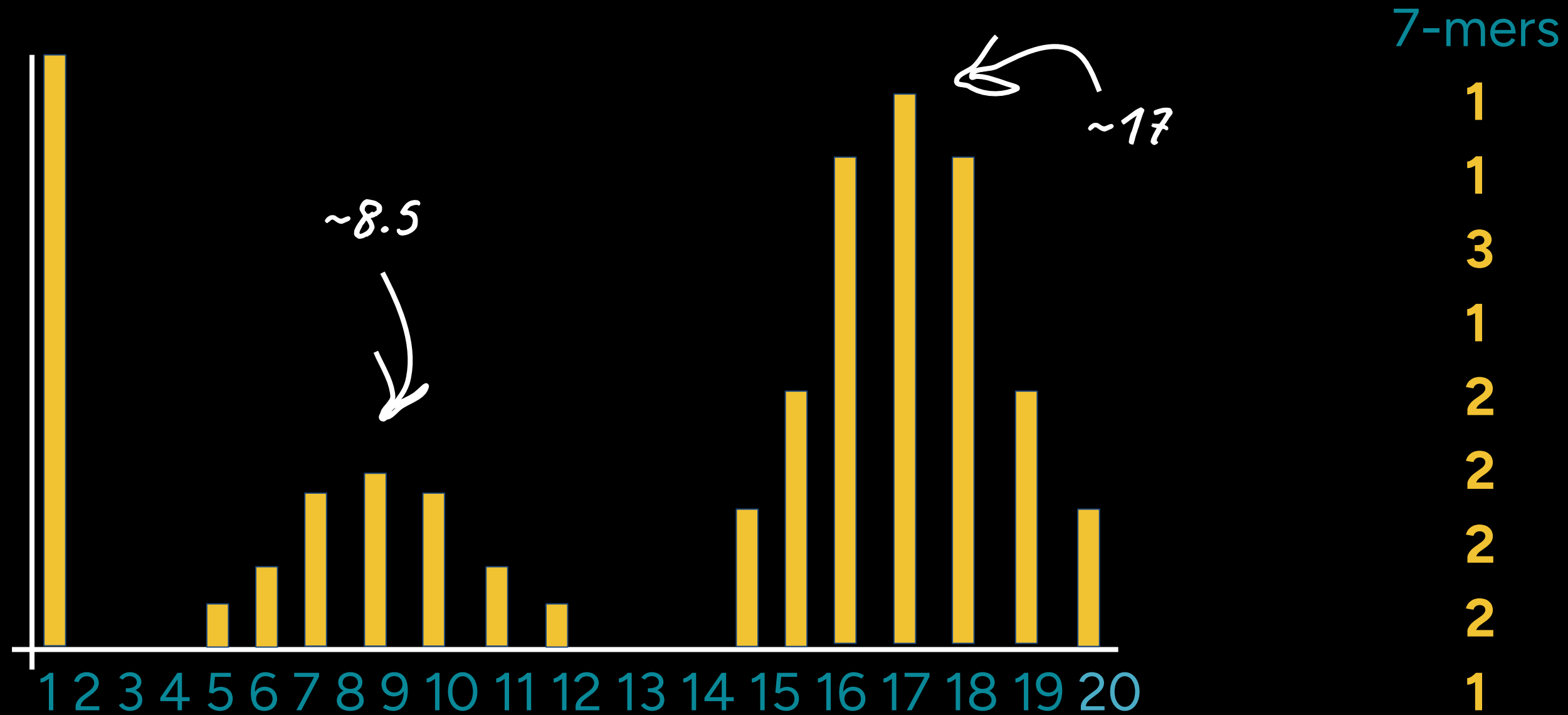
2

1

The number of times a particular k-length sequence occurs in your read data

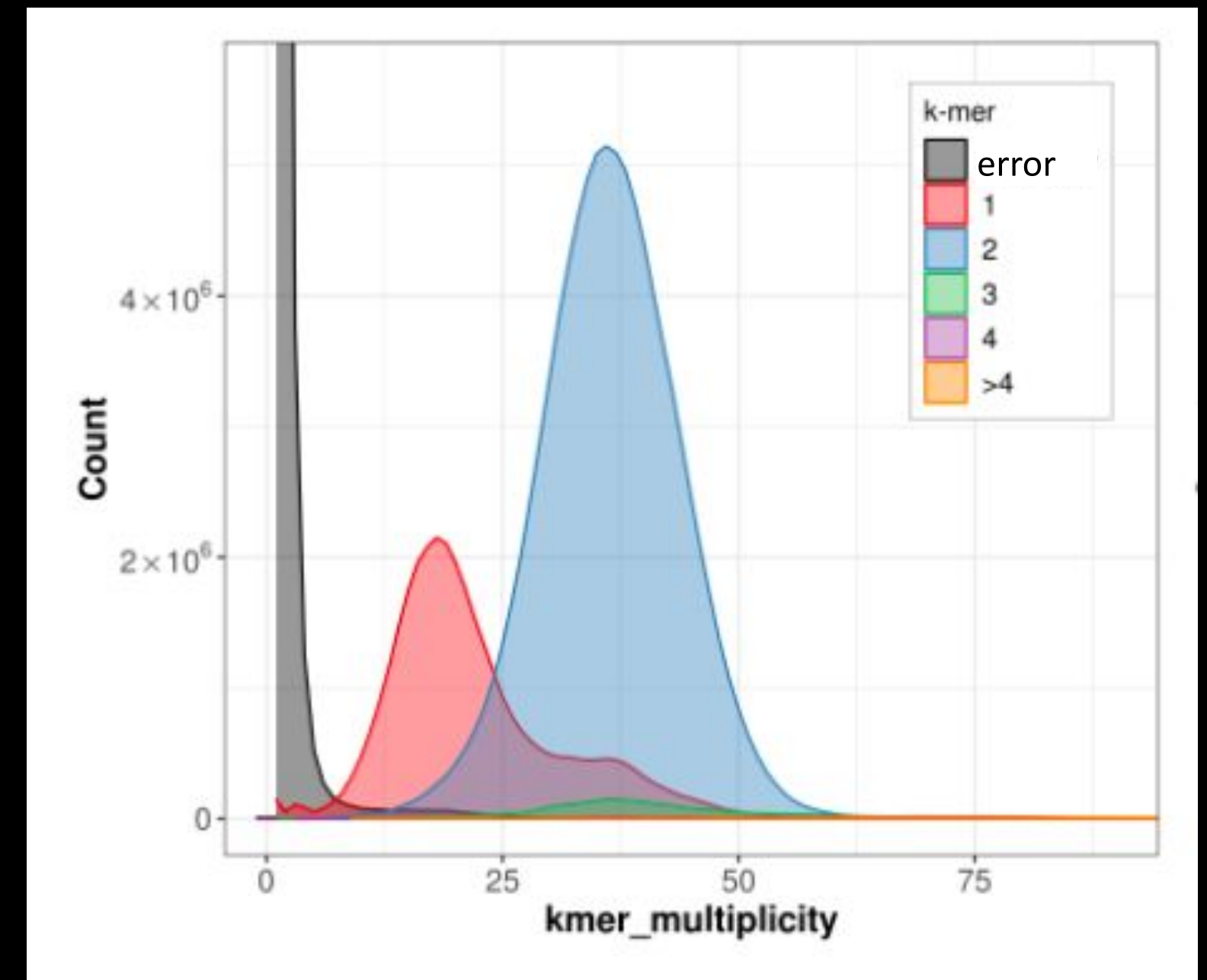
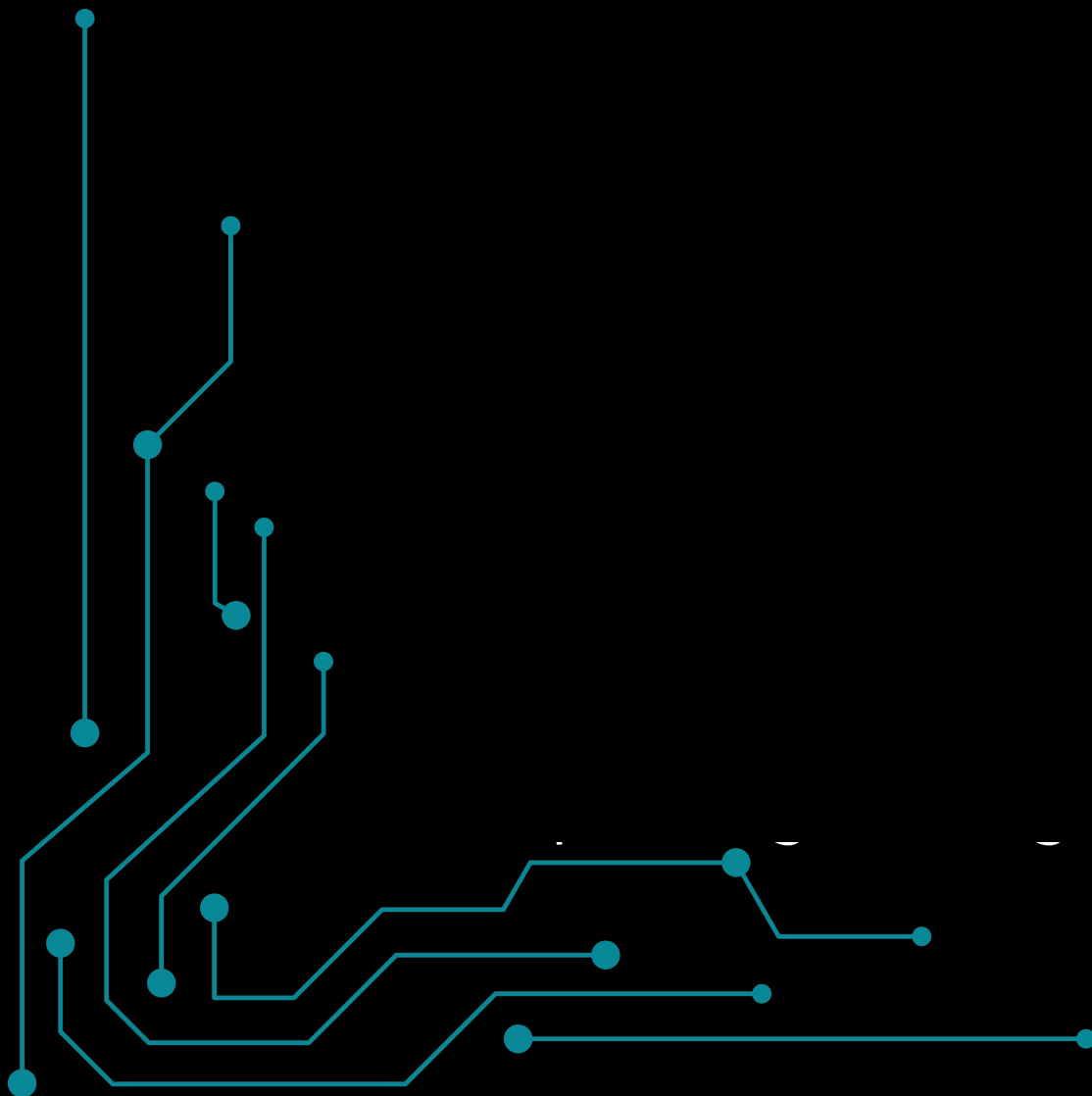


The number of times a particular k-length sequence occurs in your read data



The number of times (multiplicity)
of particular k-length sequences in your read data
We quantify & plot this to estimate coverage and heterozygosity

A k-mer spectrum for a heterozygous, diploid genome consists of two primary peaks:

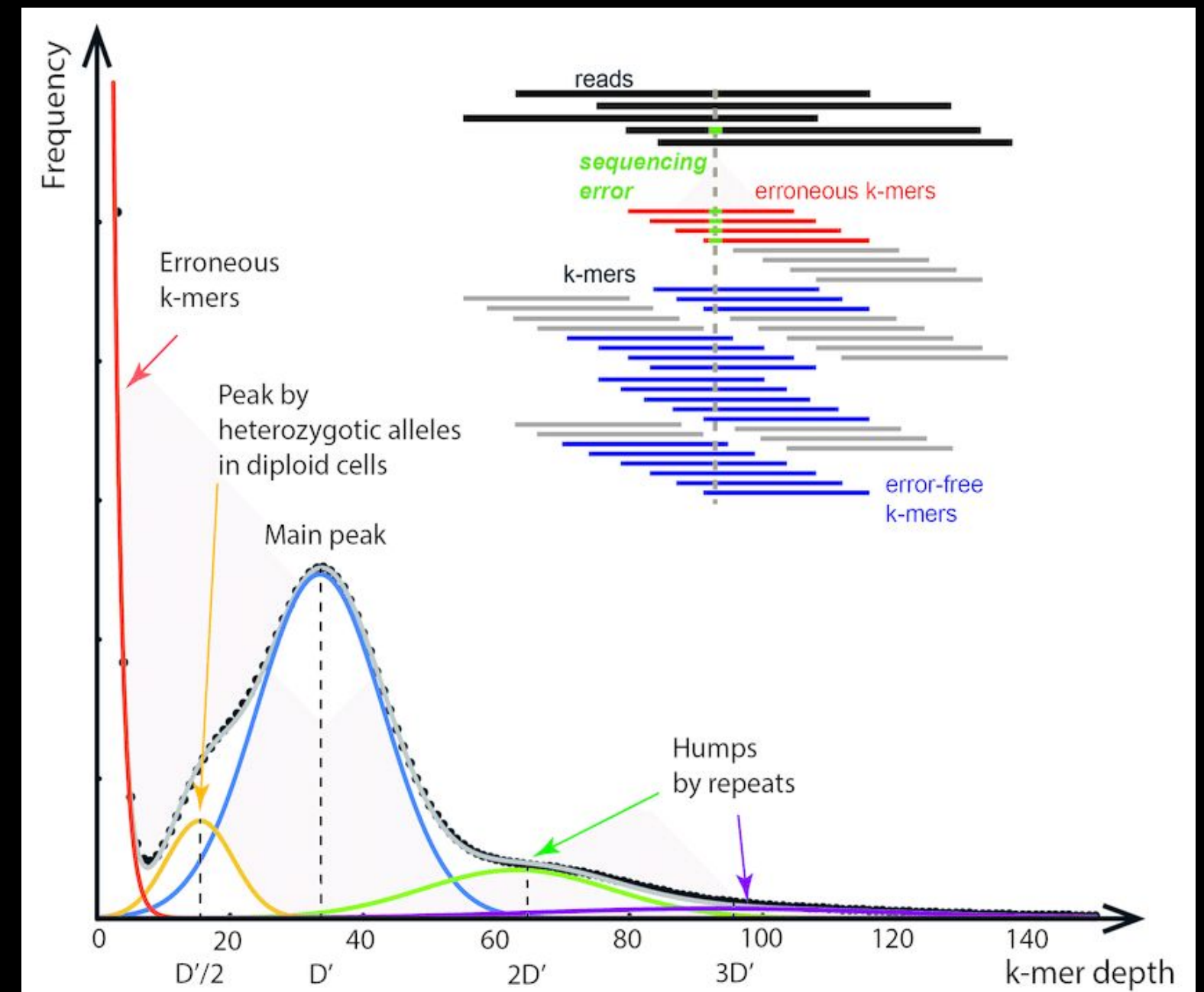


The number of times (multiplicity)
of particular k-length sequences in your read data
We quantify & plot this to estimate coverage and heterozygosity

A k-mer spectrum for a heterozygous, diploid genome consists of two primary peaks:

- (1) one representing k-mers that are 1-copy in the diploid genome (heterozygous, on a single haplotype) and
- (2) one representing those that are 2-copy in the diploid genome (homozygous, on both haplotypes or two copies on one haplotype).

The 2-copy k-mers appear with a frequency approximately equal to the average depth of sequencing coverage



k-mer counter (kmc)

SRA toolkit

download publicly available
long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size,
heterozygosity

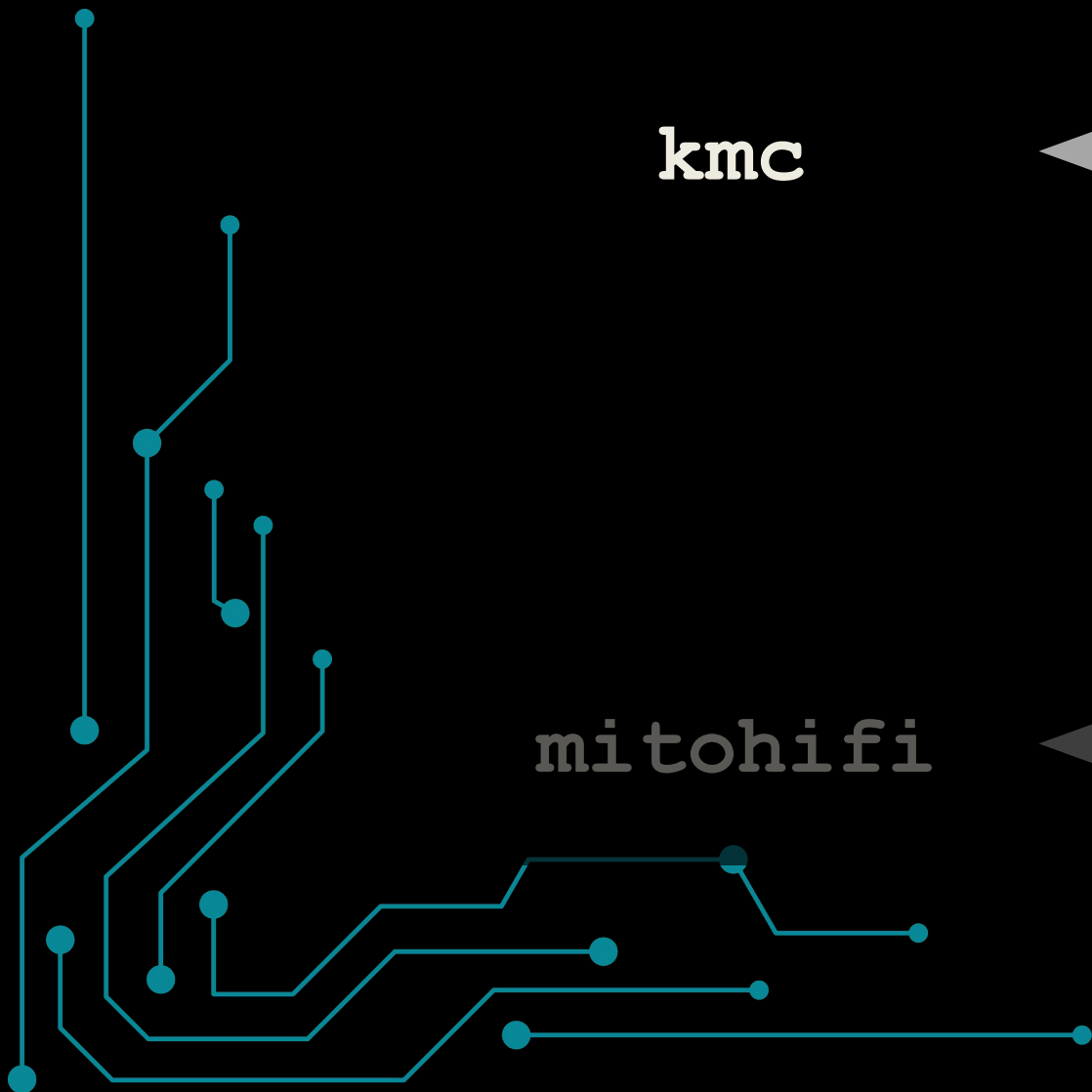
build assembly de novo using
long-read data

Hifiasm

mitohifi

pull mitochondrial genome
sequences from assembly

*build submission script, submit
job, and assess output*



k-mer counter (kmc)

```
#!/bin/bash
#SBATCH --job-name=kmc
#SBATCH --output=kmc_tbis.out
#SBATCH --mail-type=FAIL,END
#SBATCH --mail-user=rkeatinggodfrey@middlebury.edu
#SBATCH --cpus-per-task=4
#SBATCH --mem-per-cpu=8gb
#SBATCH --time=05:00:00
#SBATCH --account=kawahara

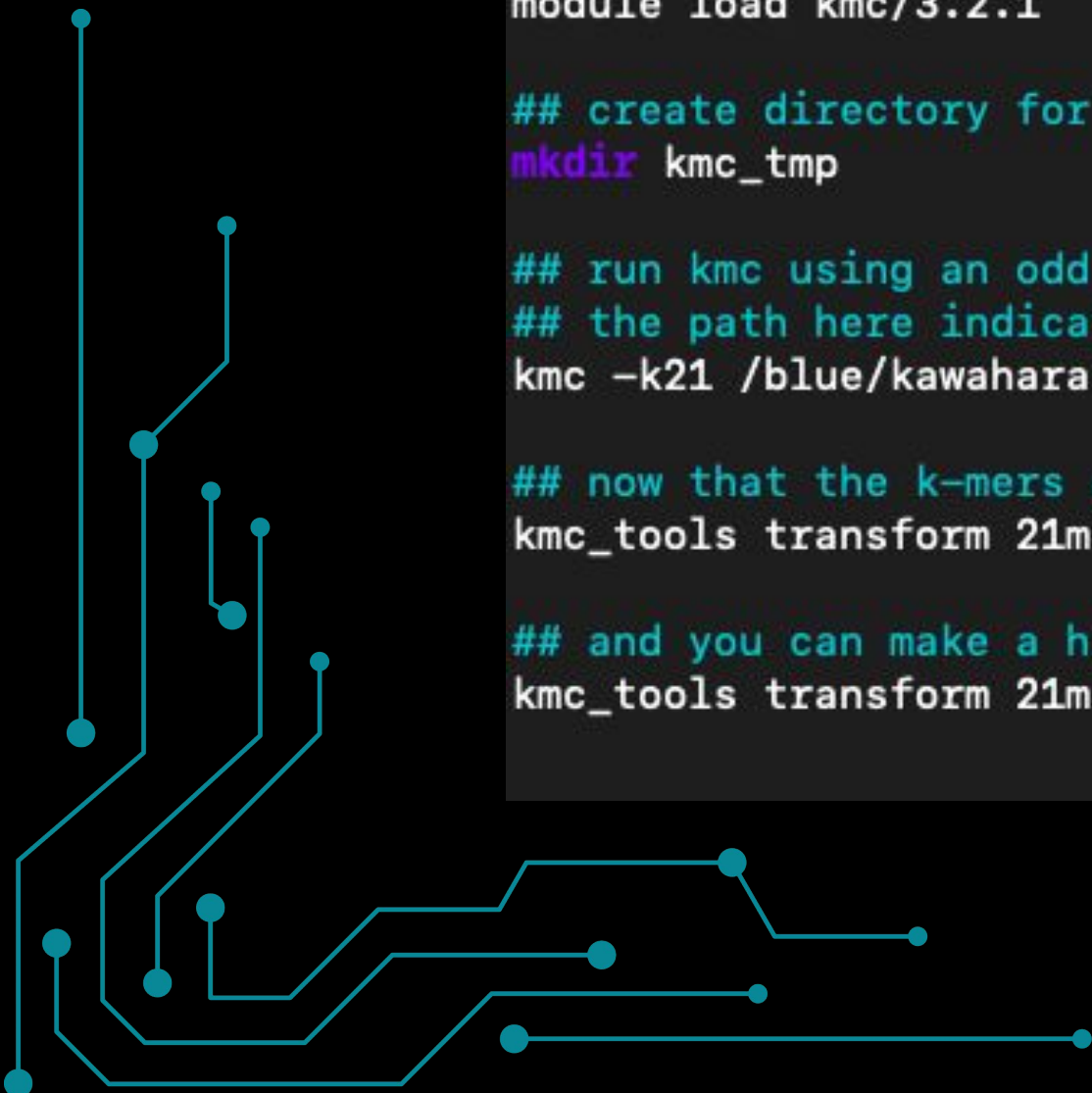
## load the kmc module version 3.2.1
module load kmc/3.2.1

## create directory for kmc temporary files
mkdir kmc_tmp

## run kmc using an odd number kmer, usually starting with k21 or k23
## the path here indicates the path to the fastq file of reads
kmc -k21 /blue/kawahara/rkeating.godfrey/genome_assembly/01_rawreads/SRR29372512.fastq.gz 21mers kmc_tmp

## now that the k-mers are counted you can dump the KMC binary database to textual form with kmc_tools.
kmc_tools transform 21mers dump tbis_21mers.txt

## and you can make a histogram file for visualization using Genomescope 2.0
kmc_tools transform 21mers histogram tbis_21mers.histo
```



k-mer counter (kmc) outputs: kmers.txt

```
rkeating.godfrey kawahara      5242972 Jan 15 15:14 21mers.kmc_pre  
rkeating.godfrey kawahara 1070540123 Jan 15 15:14 21mers.kmc_suf  
rkeating.godfrey kawahara      2404 Jan 15 15:14 tbis_21mers.histo  
rkeating.godfrey kawahara_ 5337795856 Jan 15 15:14 tbis_21mers.txt
```



k-mer counter (kmc) outputs: kmers.txt

```
[rkeating.godfrey@login7 kmer_tbis]$ head tbis_21mers.txt
AAAAAAAAAAAAAAAAACTACATA      3
AAAAAAAAAAAAAAAAAGAAAAAT      26
AAAAAAAAAAAAAAAAATATAAAT      76
AAAAAAAAAAAAAAAAATTAGTAT      65
AAAAAAAAAAAAAAAAATTATAGT      71
AAAAAAAAAAAAAAAAATTATATC      16
AAAAAAAAAAAAAAAAACTACATAA      15
AAAAAAAAAAAAAAAAACTACATAG      2
AAAAAAAAAAAAAAAAACTATAATT      39
AAAAAAAAAAAAAAAAAGAAAAATA      43
[rkeating.godfrey@login7 kmer_tbis]$ tail tbis_21mers.txt
TTTTTTGCAAACCAAAAAAAAAA      35
TTTTTTGGTCAAACCAAAAAAAAAA      2
TTTTTTGTTTGACCAAAAAAAAAA      2
TTTTTTTATAGCAAAAAAAAAA      23
TTTTTTTCGCTGAGAAAAAAAAA      26
TTTTTTTGCAAACCAAAAAAAAAA      34
TTTTTTTGCAAGAAAAAAAAAAA      5
TTTTTTTGCTTCGAAAAAAAAAAA      9
TTTTTTTCTGCAAAAAAAAAAAA      26
TTTTTTTGCAGAAAAAAAAAAA      51
```

head myfile.txt

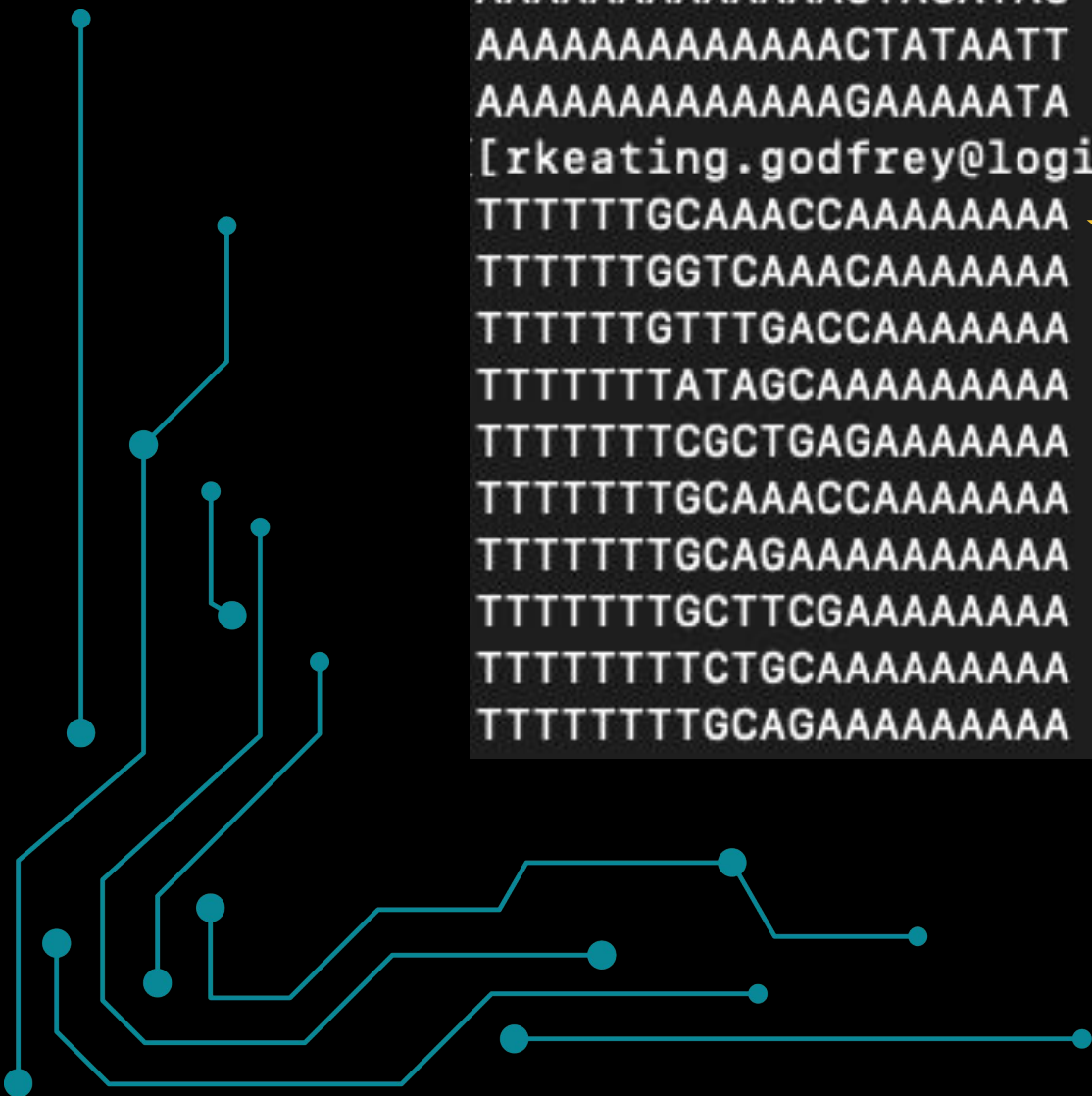
shows first 10 lines of the file

tail myfile.txt

shows last 10 lines of the file

Unique k-mer sequence

Multiplicity or Coverage



Genomescope 2.0 for viewing kmer plot

[GenomeScope](#) [Home](#) [Info](#) [Examples ▾](#)

GenomeScope 2.0

Estimate genome heterozygosity, repeat content, and size from sequencing reads using a kmer-based statistical approach.

Run GenomeScope

Click or drop .histo file here to upload

Description

my sample

Instructions

Upload results from running Jellyfish or KMC. Example: [inputk21.hist](#)

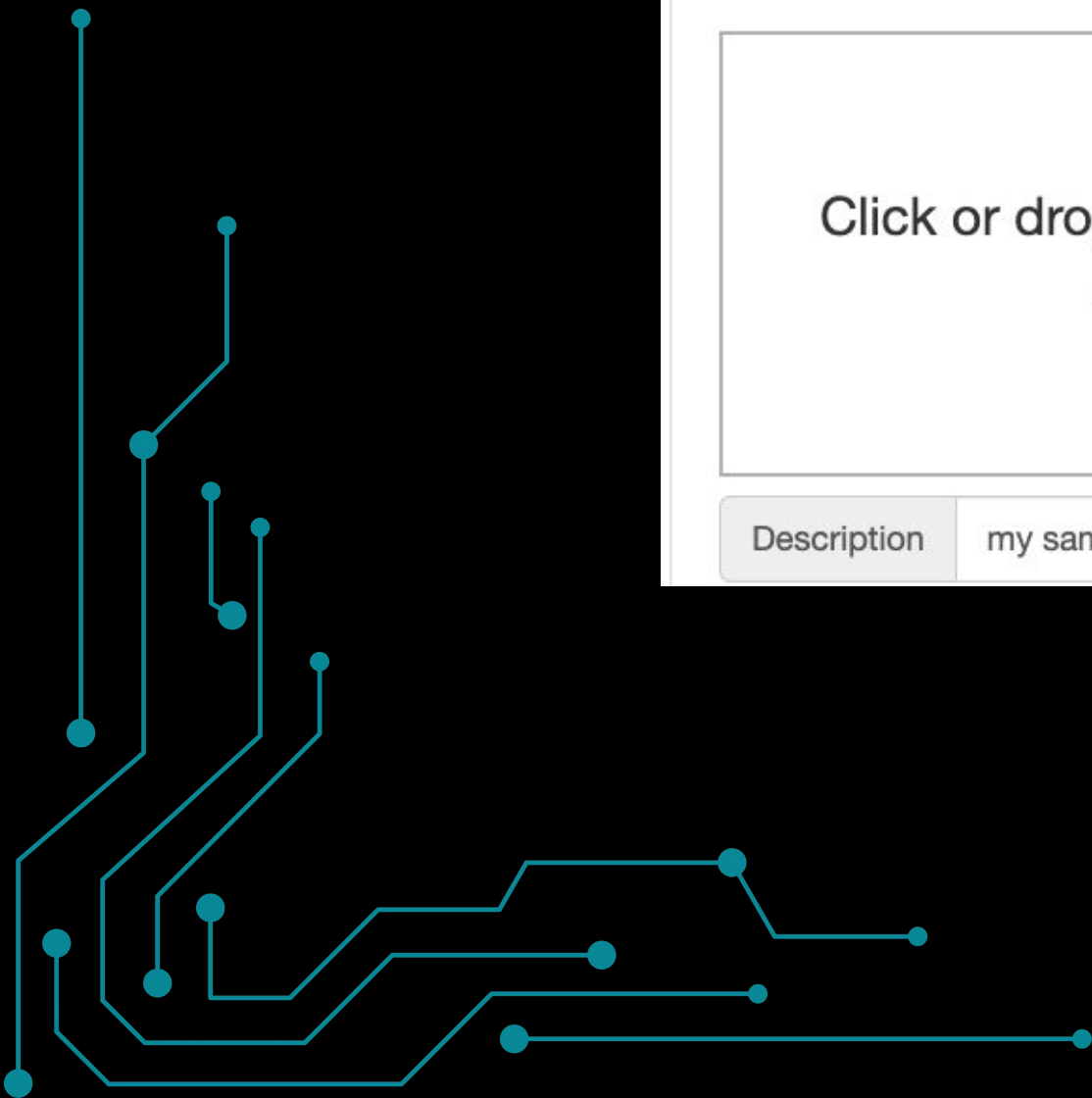
Instructions for running Jellyfish:

1. Download and install jellyfish from: <http://www.genome.umd.edu/jellyfish.html#Release>
2. Count k-mers using jellyfish:

```
$ jellyfish count -C -m 21 -s 1000000000 -t 10 *.fastq -o reads.jf
```

Note you should adjust the memory (-s) and threads (-t) parameters according to your server. This example will use 10 threads and 1GB of RAM. The k-mer length (-m) may need to be scaled if you have low coverage or a high error rate. You should always use "canonical k-mers" (-C).
3. Export the k-mer count histogram

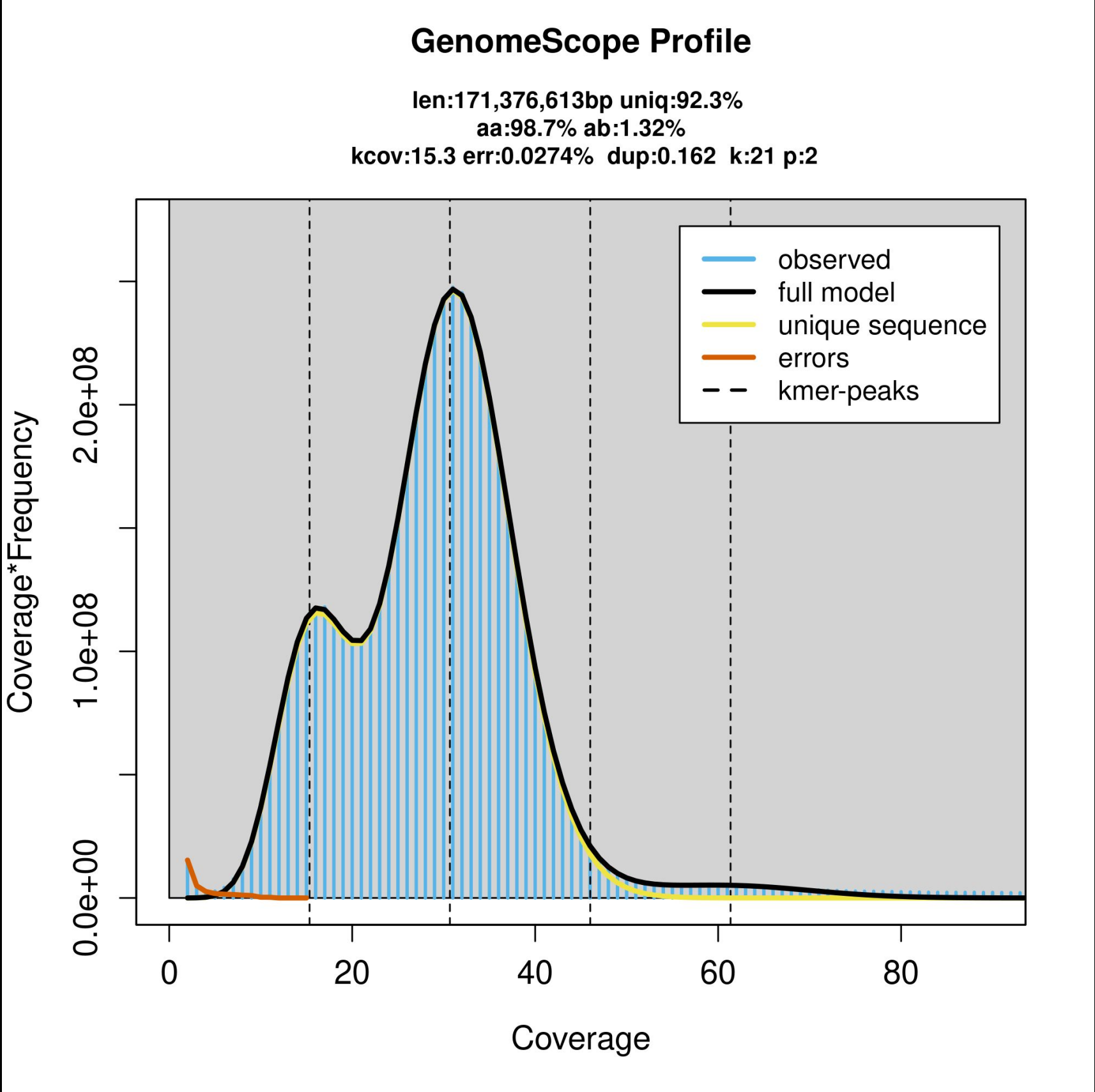
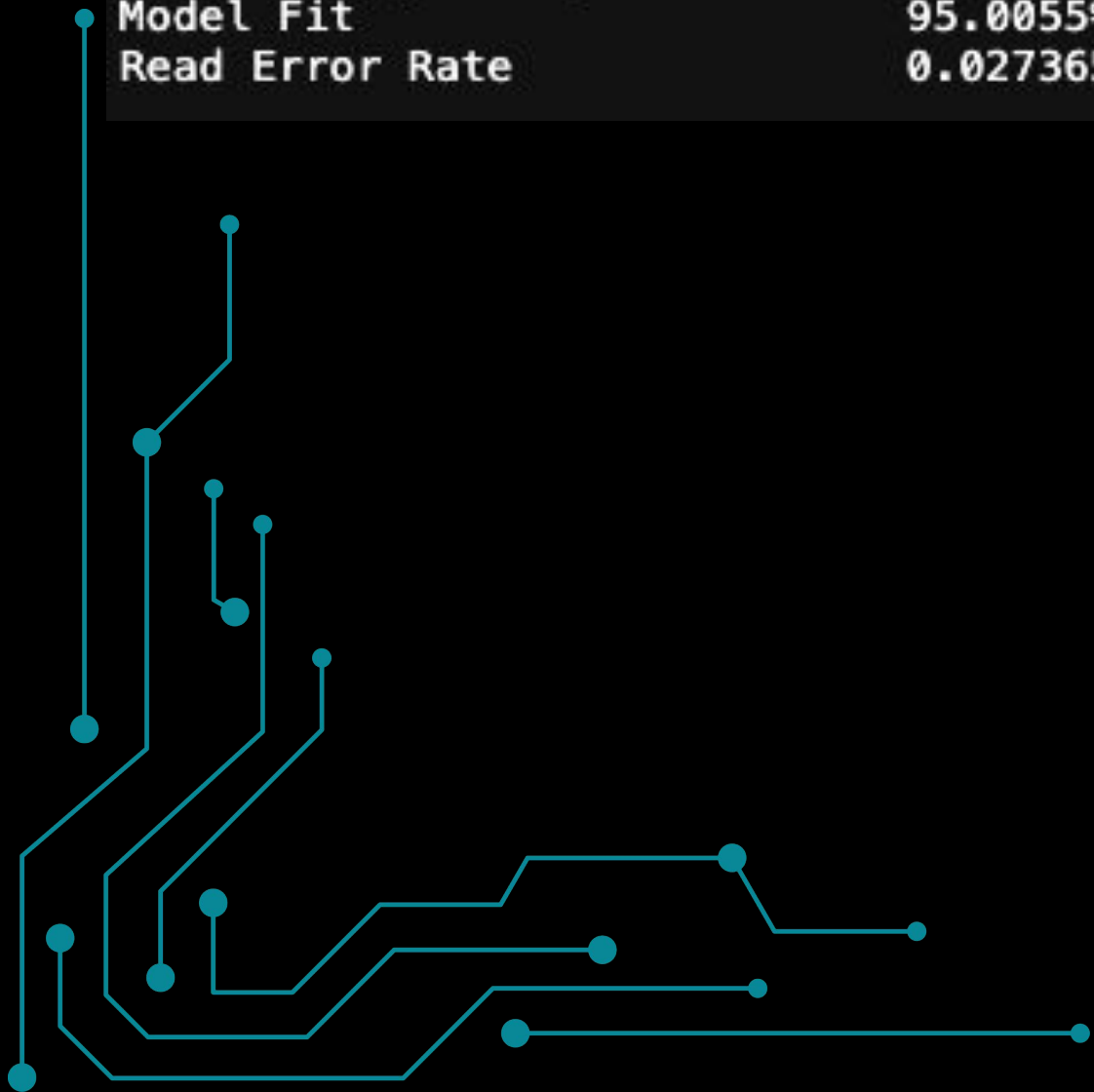
```
$ jellyfish histo -t 10 reads.jf > reads.histo
```



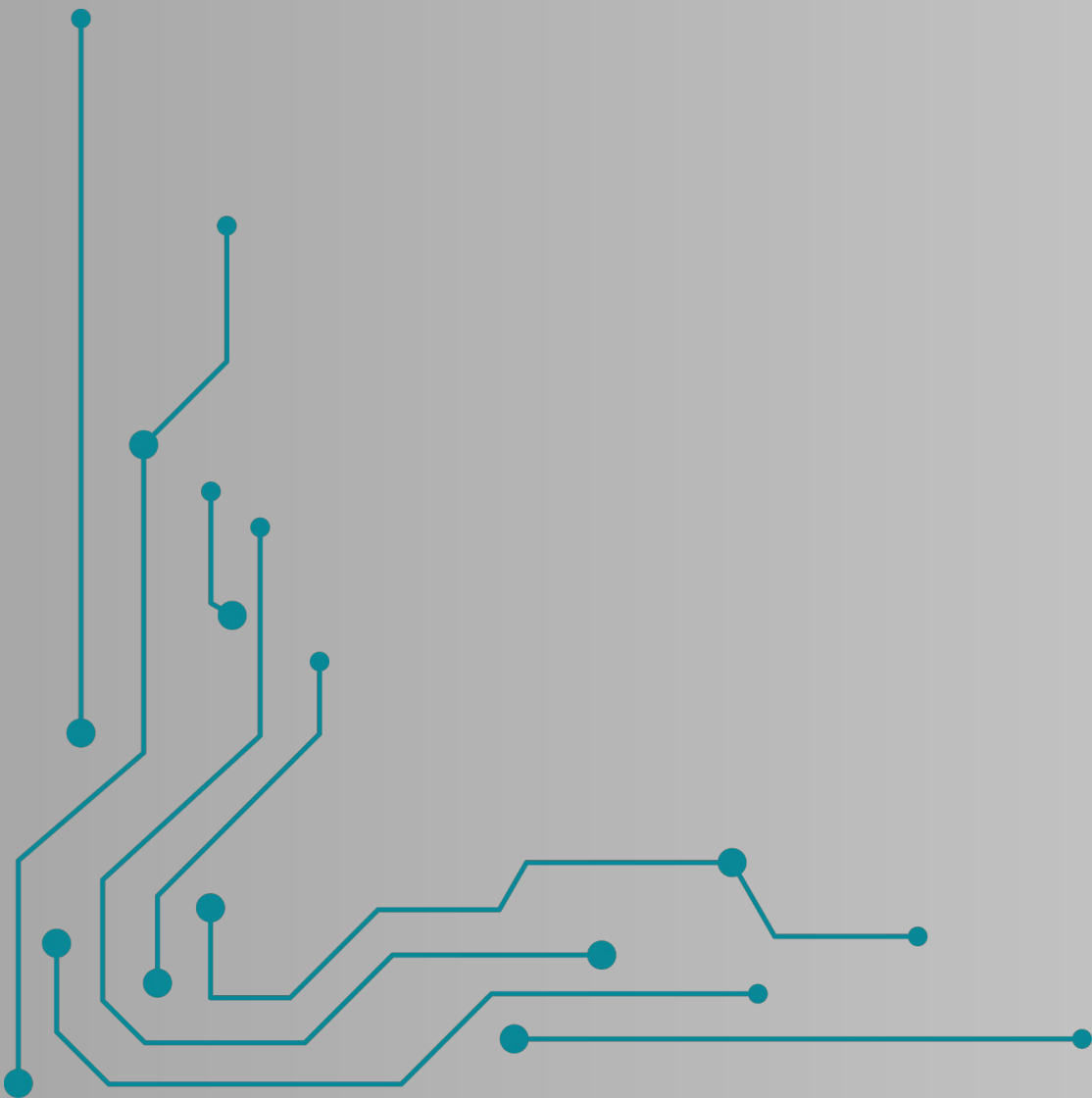
k-mer counter (kmc) output: kmers.histo

```
GenomeScope version 2.0
input file = user_uploads/yI0sZunedjAGcMsLrlCa
output directory = user_data/yI0sZunedjAGcMsLrlCa
p = 2
k = 21
```

property	min	max
Homozygous (aa)	98.6481%	98.7199%
Heterozygous (ab)	1.28014%	1.35188%
Genome Haploid Length	171,021,679 bp	171,376,613 bp
Genome Repeat Length	13,126,917 bp	13,154,161 bp
Genome Unique Length	157,894,761 bp	158,222,452 bp
Model Fit	95.0055%	99.4393%
Read Error Rate	0.0273657%	0.0273657%



Let's take a break!



*look over example submission
scripts & discuss outputs*

Our Assembly Pipeline

SRA toolkit

download publicly available
long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size,
heterozygosity

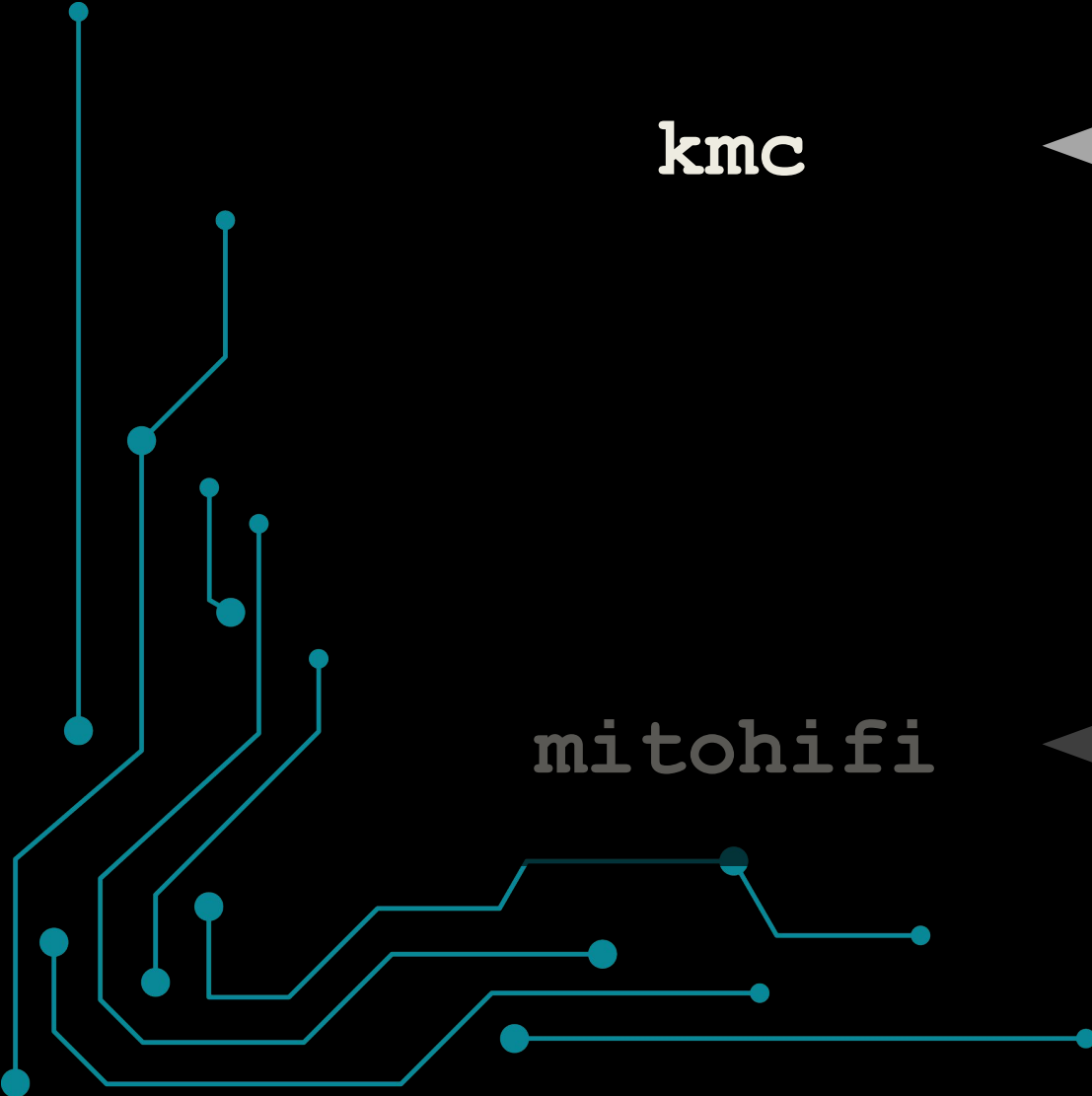
build assembly de novo using
long-read data

Hifiasm

mitohifi

pull mitochondrial genome
sequences from assembly

*build submission script, submit
job, and assess output*



Ploidy in genome assembly

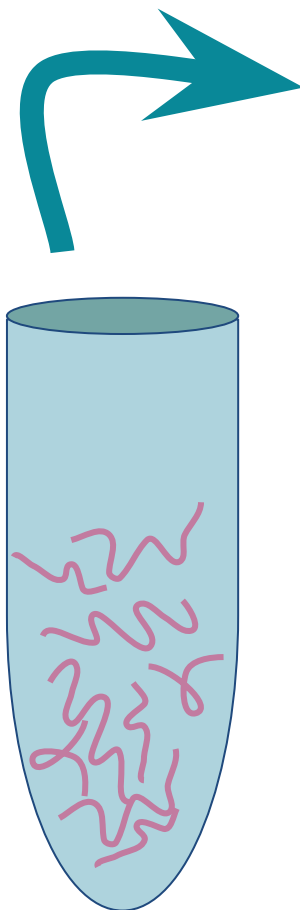
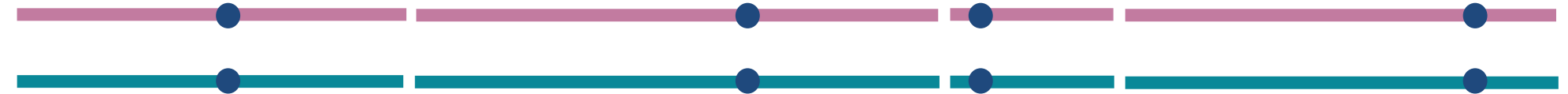
DNA instructions for the construction and maintenance
of a specific biological being



Most sexually reproducing organisms have two or more
slightly non-identical copies of their genome in each cell

Assembly structure vocabulary

DNA instructions for the construction and maintenance
of a specific biological being



Assembly structure vocabulary

DNA instructions for the construction and maintenance
of a specific biological being

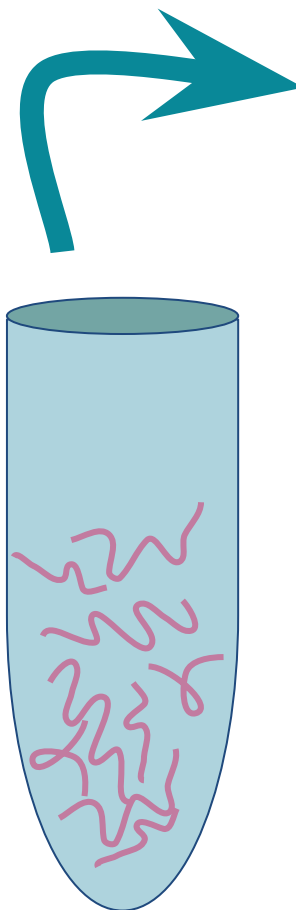
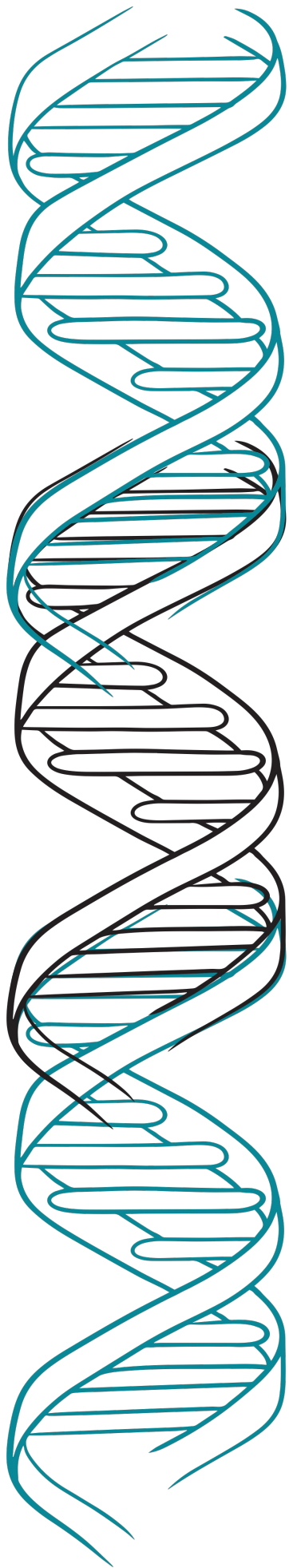
chromosomes



contigs

Chromosome (biological unit): sequences organized into distinct “packets” in the cell

Contig (assembly unit): a contiguous sequence without gaps.



Assembly structure vocabulary

DNA instructions for the construction and maintenance
of a specific biological being



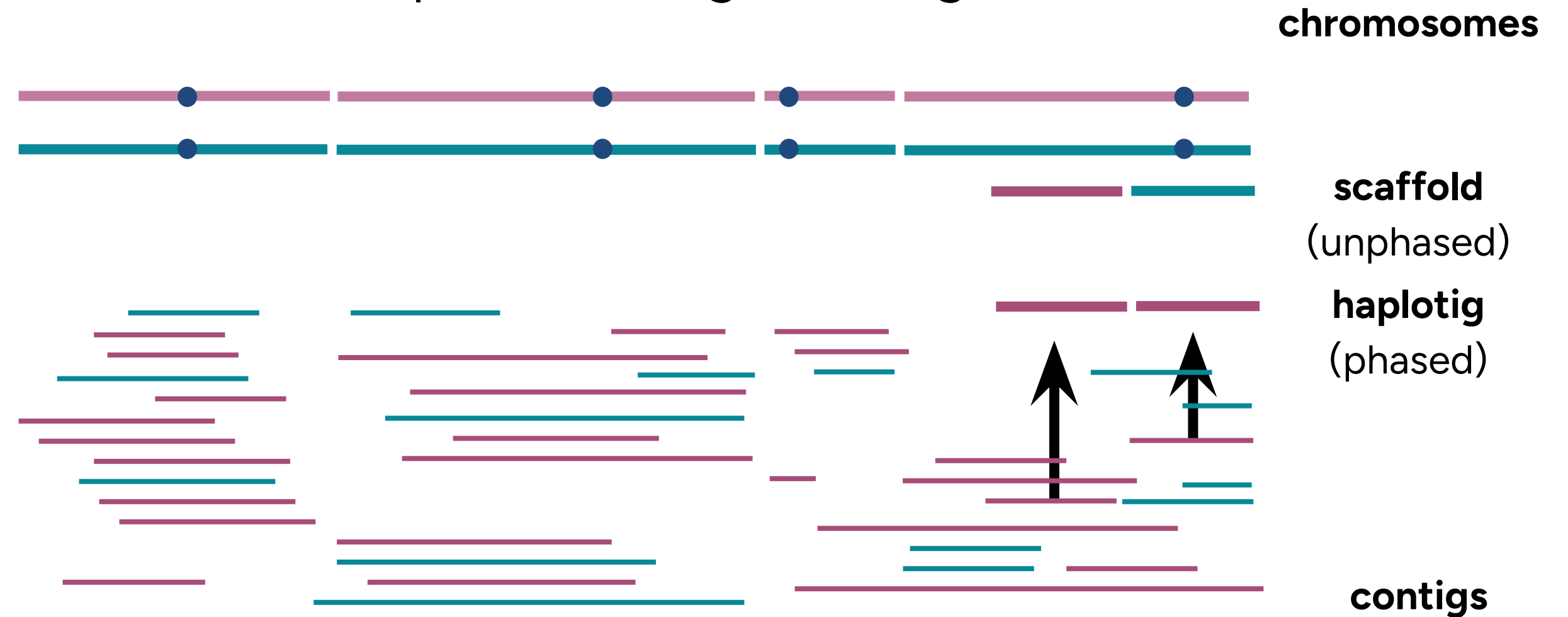
Chromosome (biological unit): sequences organized into distinct “packets” in the cell

Scaffold (assembly unit): a sequence consisting of one or multiple contigs connected by assembly gaps of various sizes.

Contig (assembly unit): a contiguous sequence without gaps.

Assembly structure vocabulary

DNA instructions for the construction and maintenance
of a specific biological being

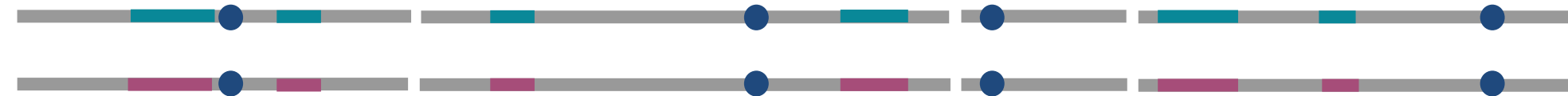


- Chromosome** (biological unit): sequences organized into distinct “packets” in the cell
- Haplotig** (type of scaffold): a contig that comes from the same haplotype. In an unphased assembly, a contig may join alleles from different parental haplotypes.
- Scaffold** (assembly unit): a sequence consisting of one or multiple contigs connected by assembly gaps of various sizes.
- Contig** (assembly unit): a contiguous sequence without gaps.



Assembly level vocabulary

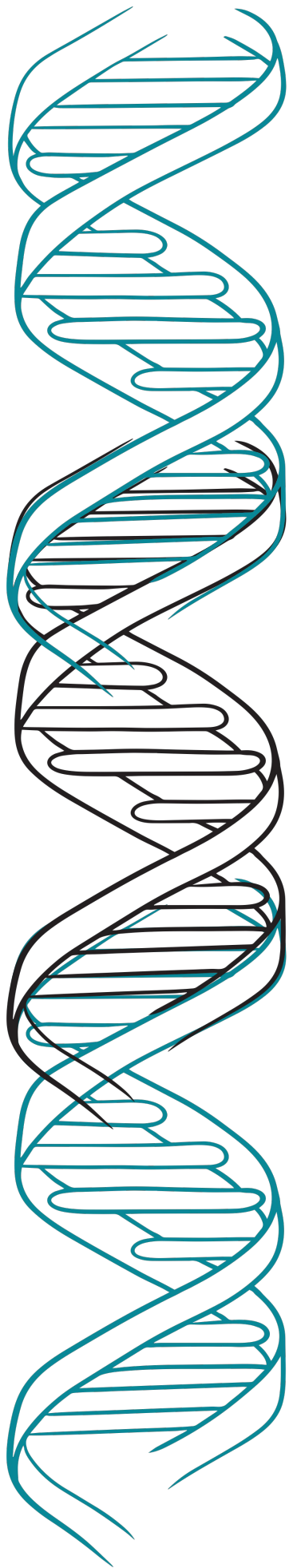
Chromosomes actually feature long stretches of the same sequences interspersed with allelic or parental haplotype differences



Traditionally, assemblers produced **collapsed assemblies** with parental alleles randomly switching in a contig

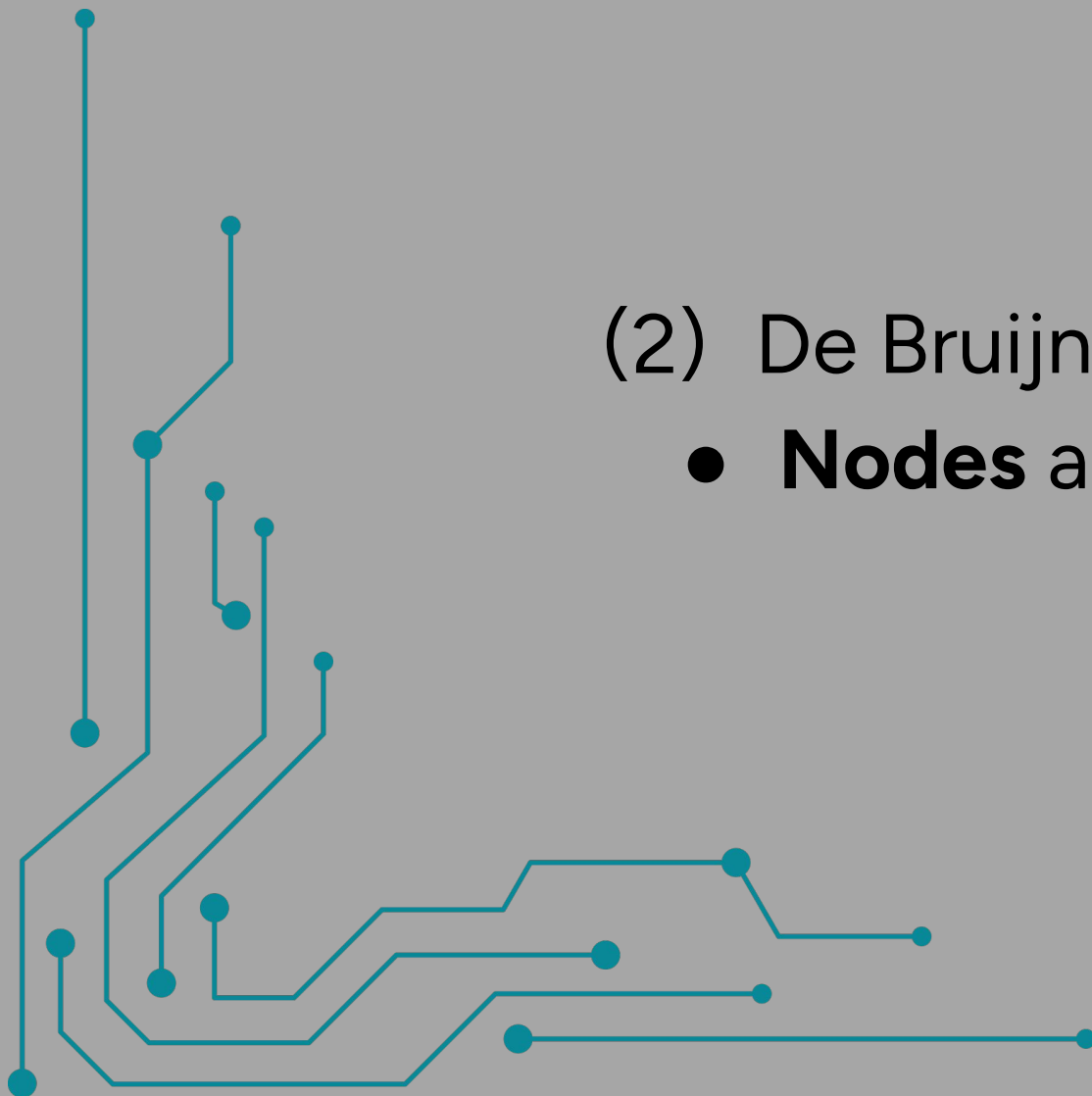
Newer assemblers produce **phased assemblies** that retain information from both (all) chromosomes and are represented as

- (1) Primary and Alternate assemblies
- (2) Partially phased assemblies
- (3) Haplotype-resolved assemblies



Two Main Approaches using Graph Theory models

- (1) Overlap Layout Consensus (OLC) graph
 - **Nodes** are reads; **Edges** are overlap between pairs of reads
- (2) De Bruijn Graph (DBG), a k-mer approach
 - **Nodes** are (k-1)-mers (prefixes and suffixes); **Edges** are kmers



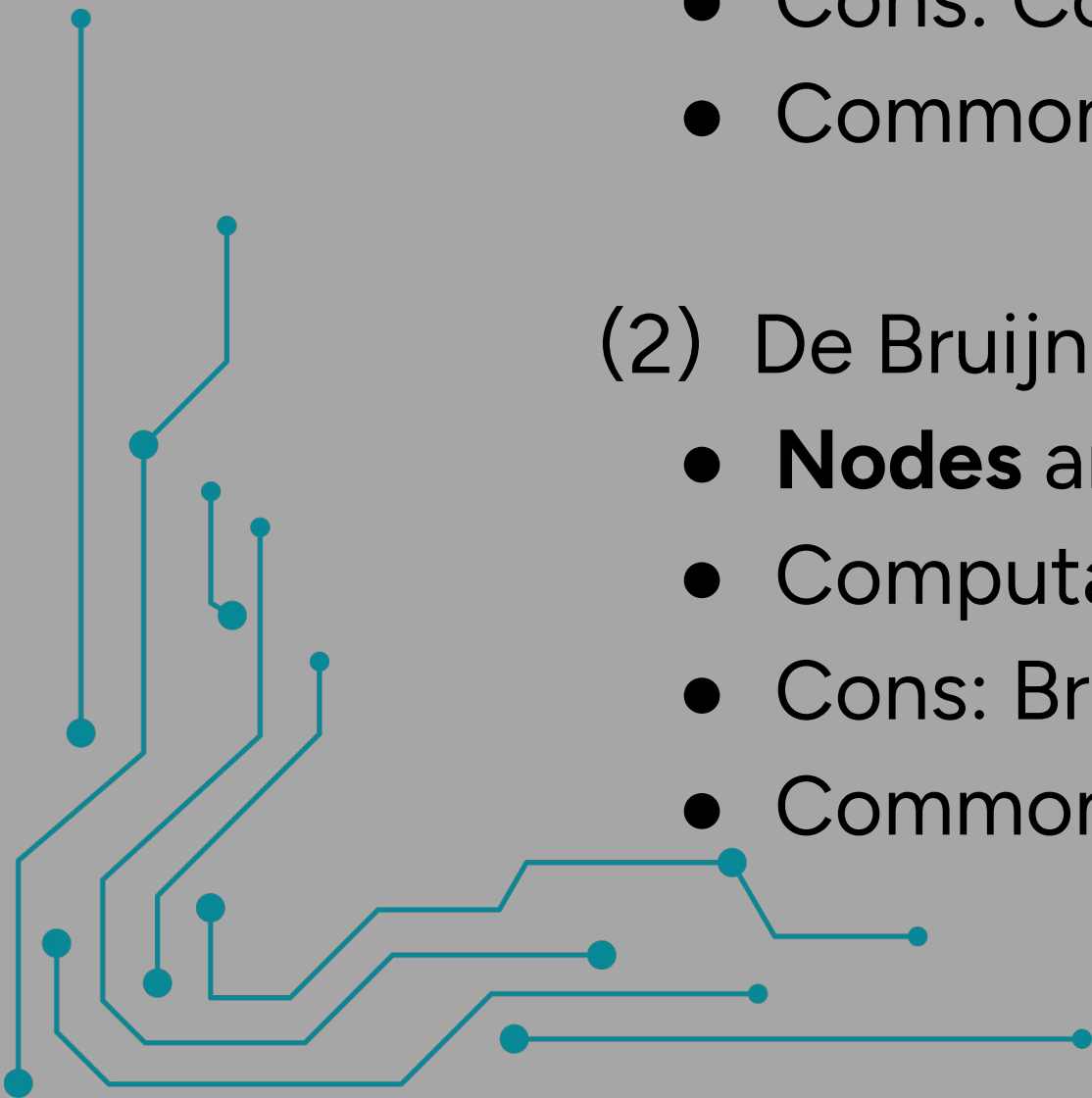
Two Main Approaches using Graph Theory models

(1) Overlap Layout Consensus (OLC) graph

- **Nodes** are reads; **Edges** are overlap between pairs of reads
- Intuitive and retains long sequences
- Cons: Computationally intensive; reconstructs repeat regions well
- Common for assembly of long-read sequences

(2) De Bruijn Graph (DBG), a k-mer approach

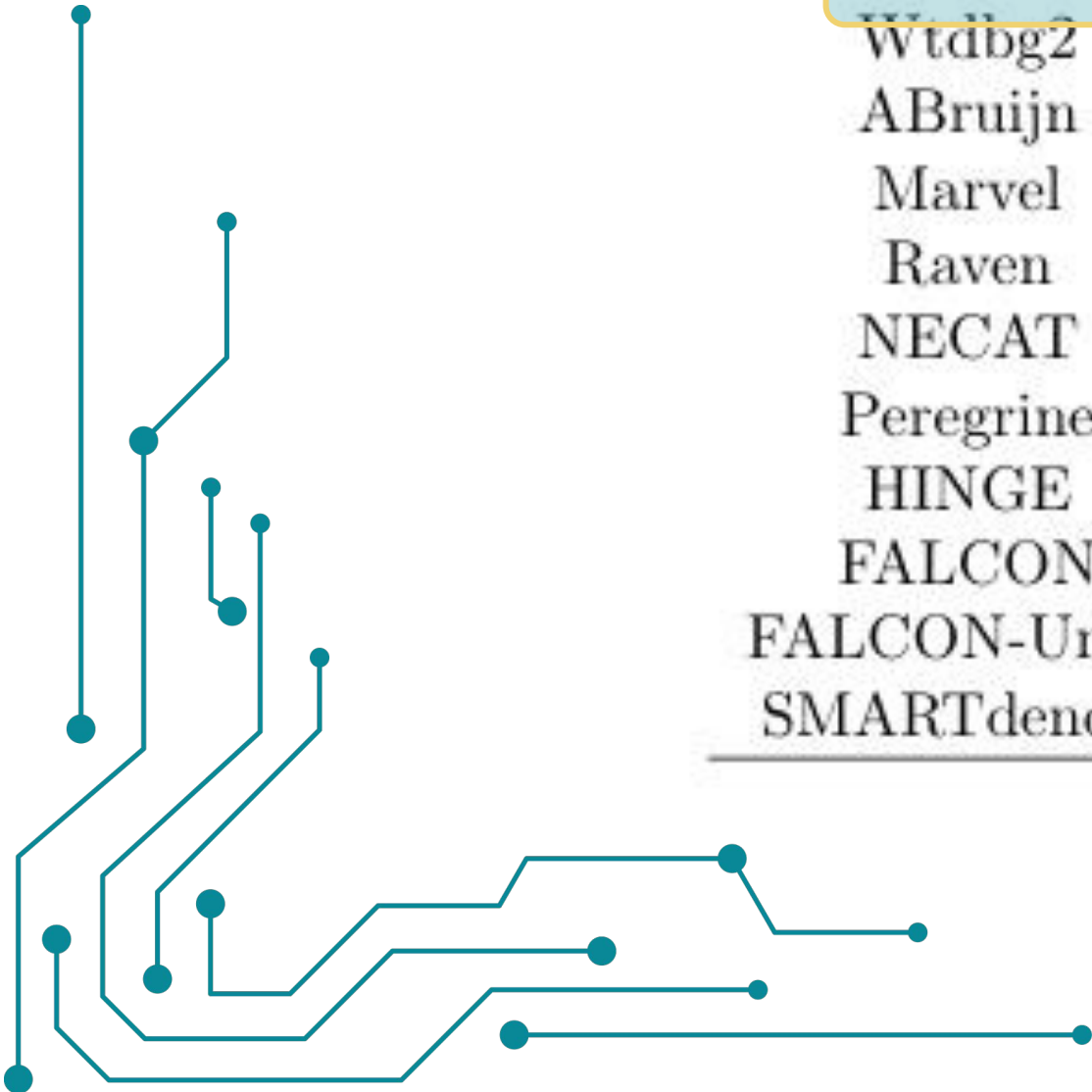
- **Nodes** are (k-1)-mers (prefixes and suffixes); **Edges** are kmers
- Computationally efficient and scalable
- Cons: Breaks reads into kmers; does not handle repeat regions well
- Common for assembly of short-read sequences

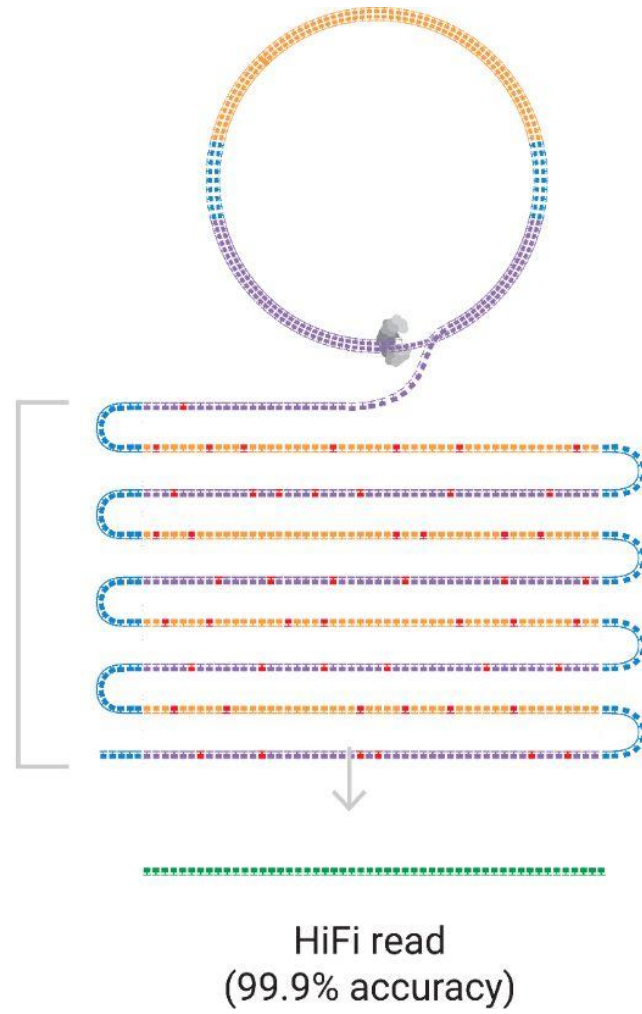
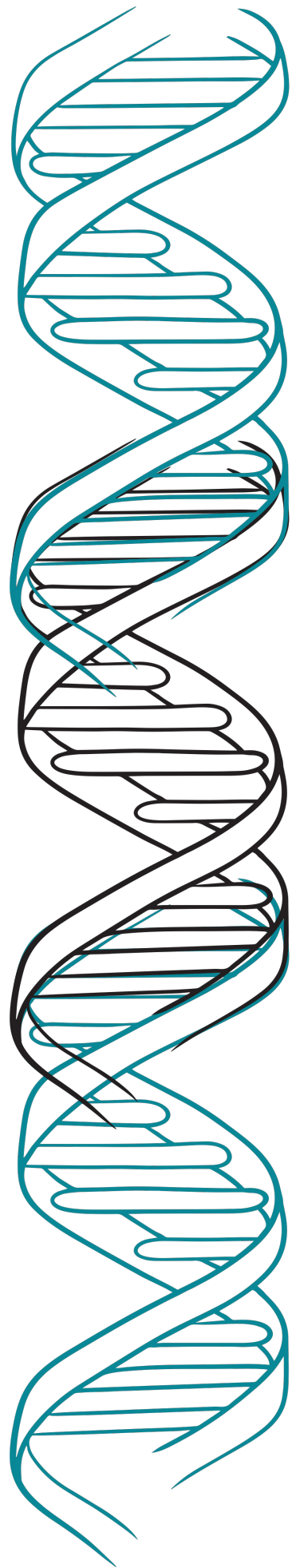


Genome Assembler Algorithms

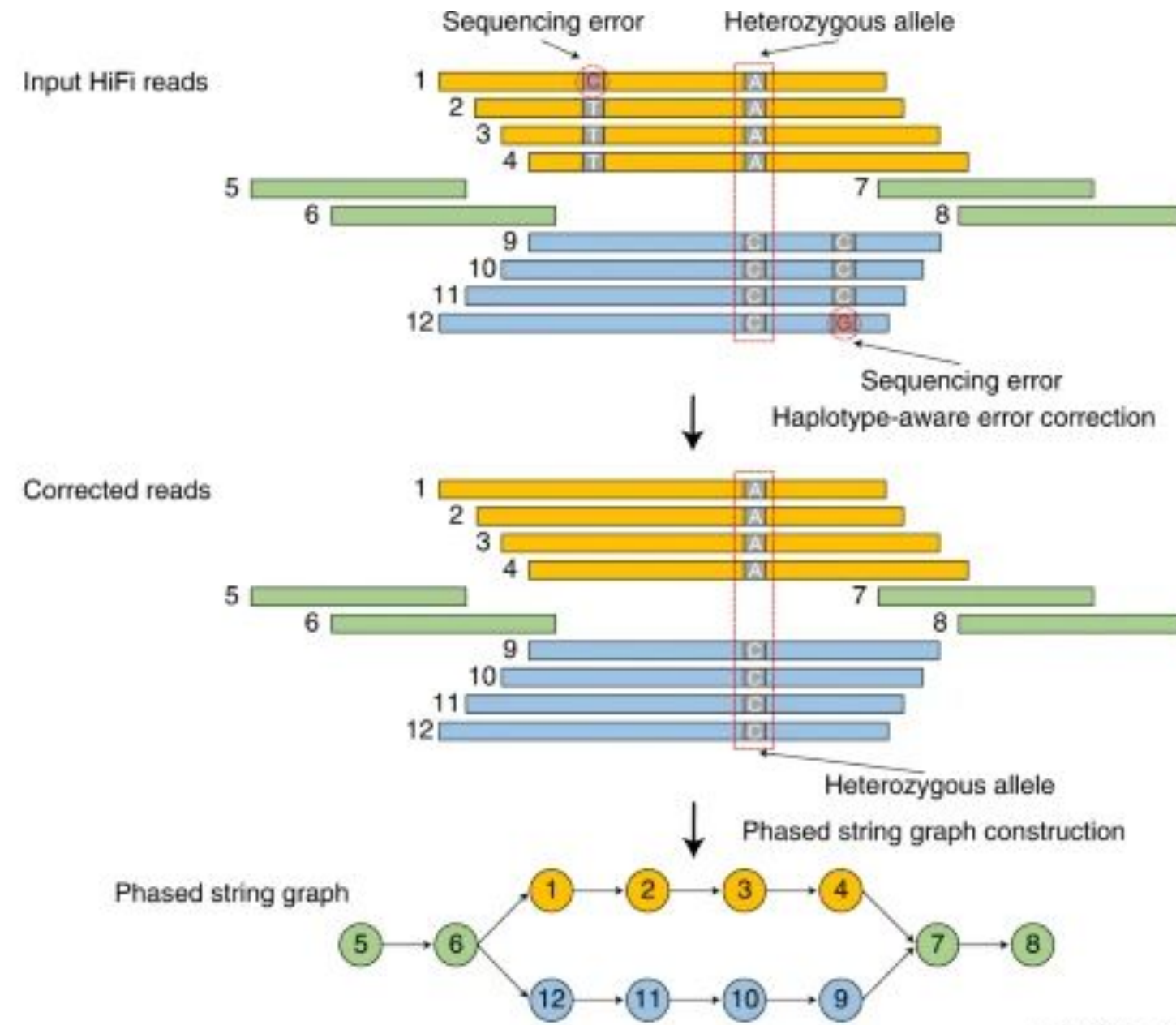
Table 2: Long-read de novo assemblers

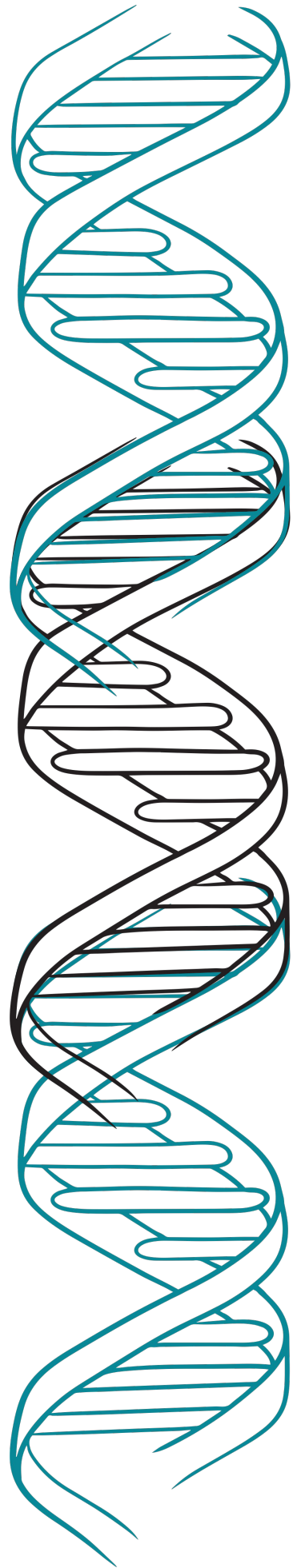
Assembler	Algorithm	Graph Structure	Sequencing data	Documentation
Miniasm	OLC-based	Overlap graph	ONT, PacBio	Miniasm
Canu	OLC-based	Overlap graph	ONT, PacBio	Canu
HiCanu	OLC-based	Overlap graph	PacBio	HiCanu
Flye	DBG-based	Repeat graph	ONT, PacBio	Flye
Shasta	DBG-based	Marker graph	ONT	Shasta
Hifiasm	OLC-based	String graph	PacBio	Hifiasm
Wtdbg2	DBG-based	Fuzzy Bruijn graph	ONT, PacBio	Wtdbg2
ABRuijn	DBG-based	A-Bruijn graph	ONT, PacBio	ABRuijn
Marvel	OLC-based	Overlap graph	PacBio	Marvel
Raven	OLC-based	Overlap graph	ONT, PacBio	Raven
NECAT	OLC-based	String graph	ONT	NECAT
Peregrine	OLC-based	Overlap graph	PacBio	Peregrine
HINGE	OLC-based	Overlap graph	PacBio	HINGE
FALCON	OLC-based	String graph	PacBio	FALCON
FALCON-Unzip	OLC-based	String graph	PacBio	FALCON-Unzip
SMARTdenovo	OLC-based	Overlap graph	ONT, PacBio	SMARTdenovo



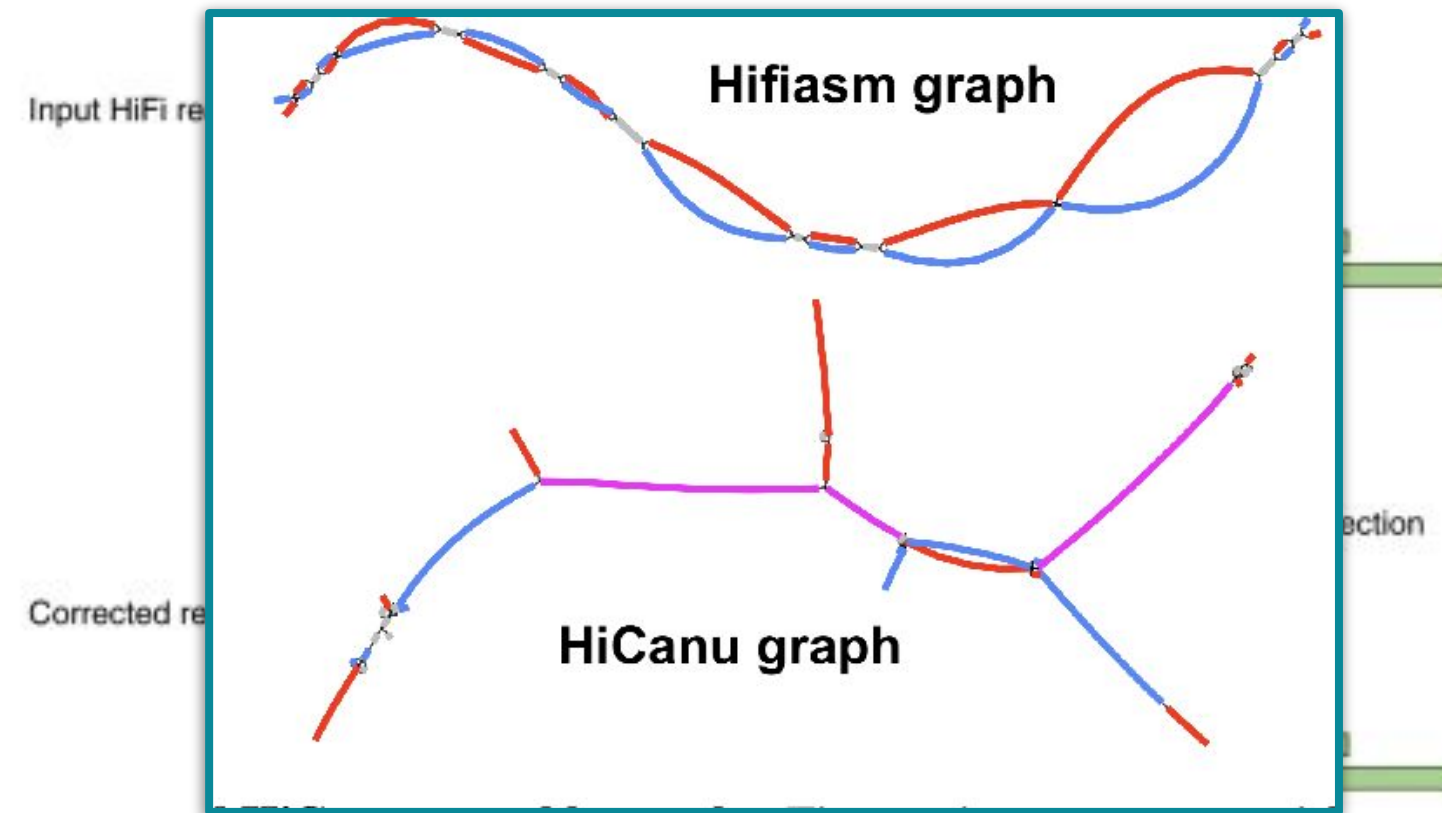


Producing phased assemblies with Hifiasm

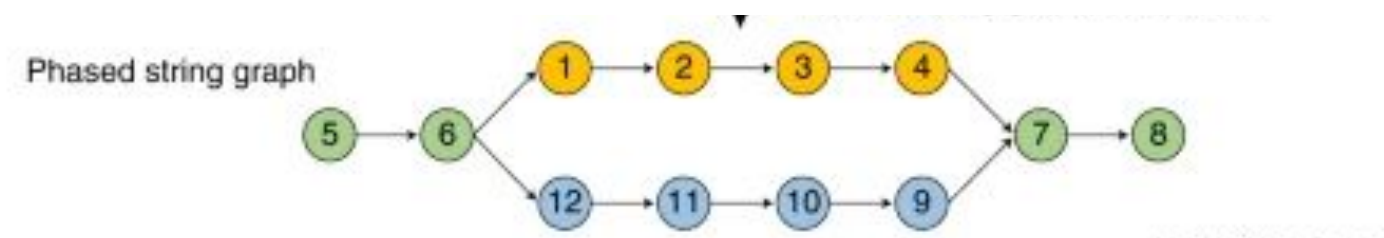




Producing phased assemblies with Hifiasm

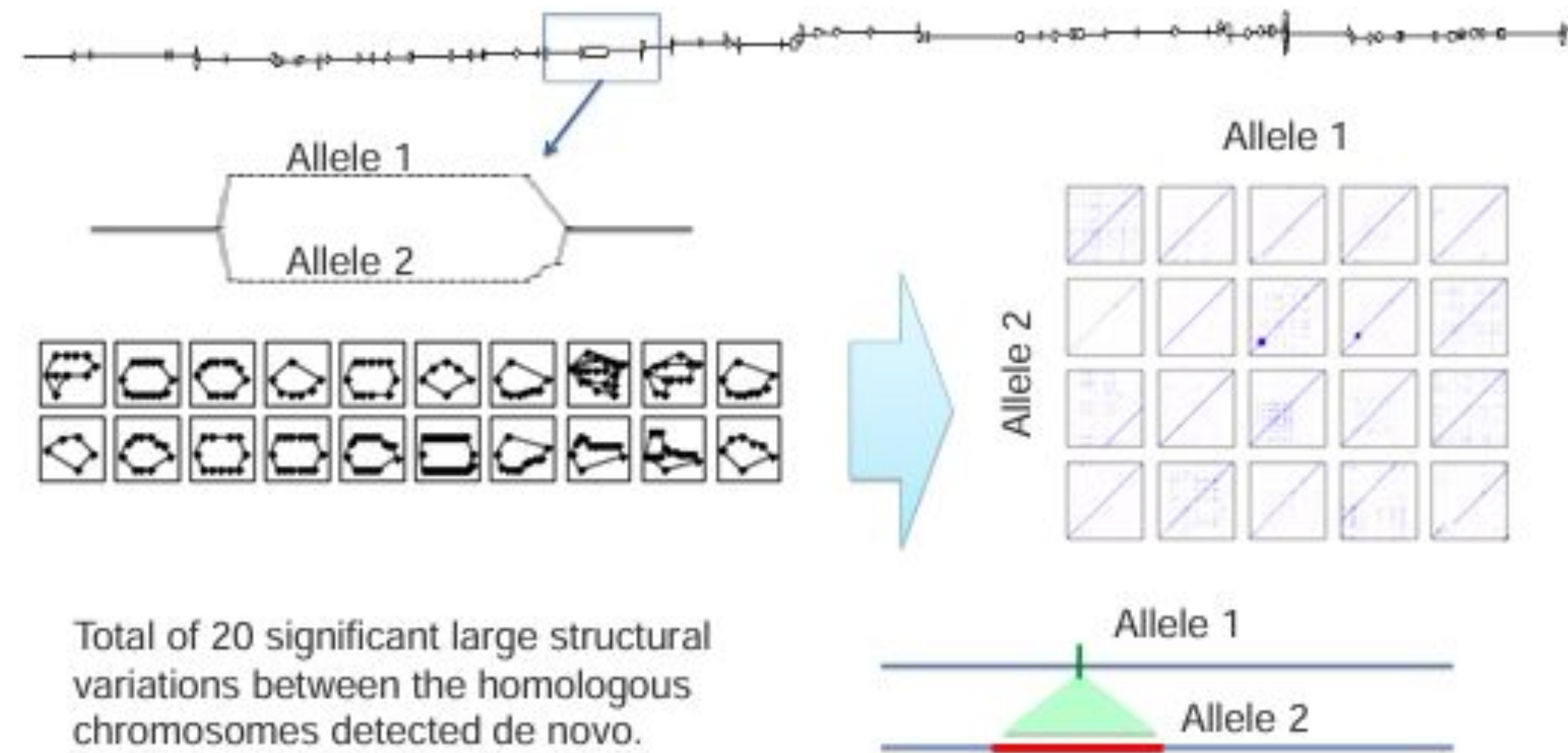


“Ideally, the phased assembly graph of hifiasm should be a chain of bubbles for diploid genomes” Cheng et al. (2020)



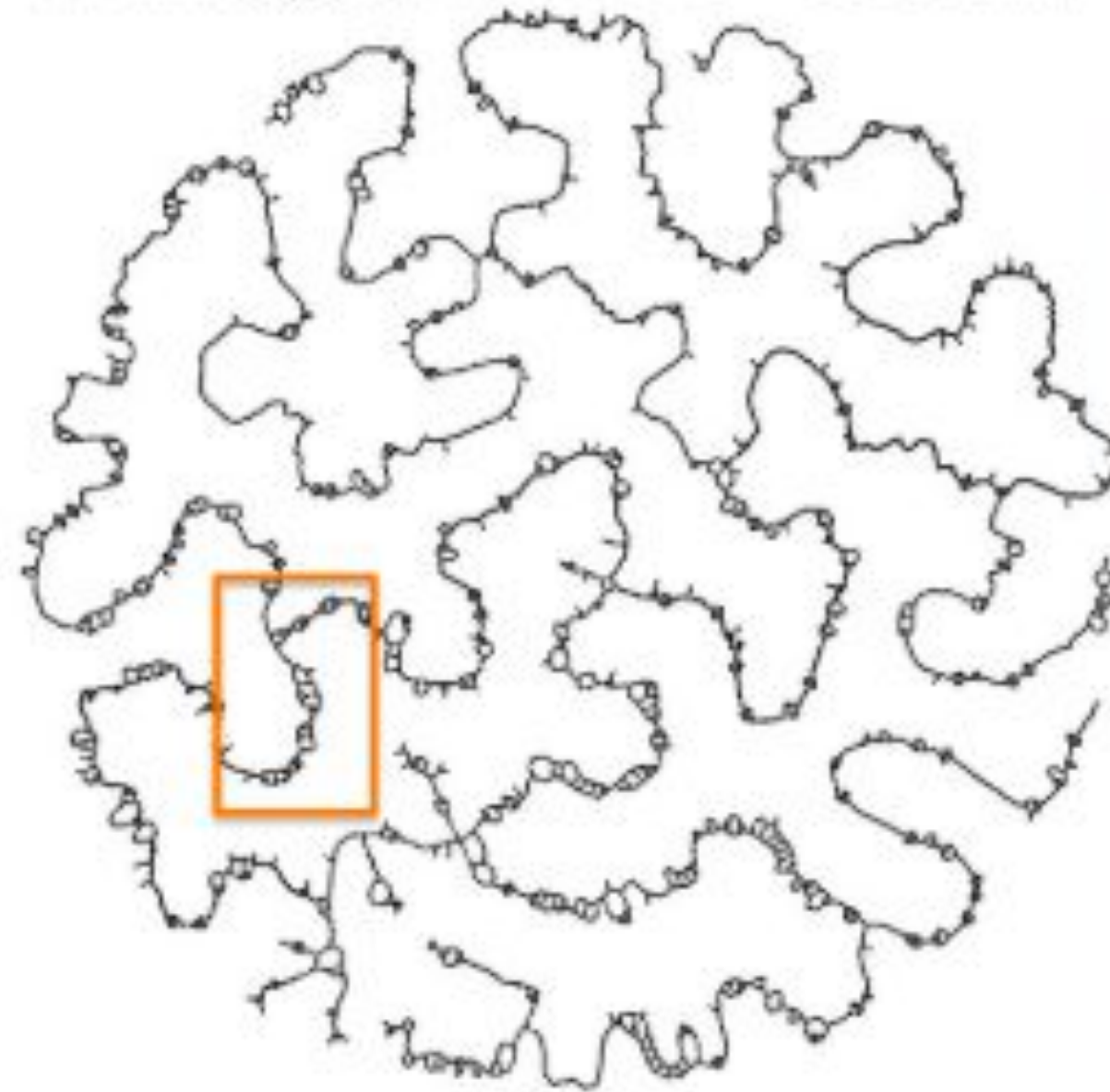
Assembly Graph of Diploid Human MHC Region

The assembly contig string graph contains many "bubbles"

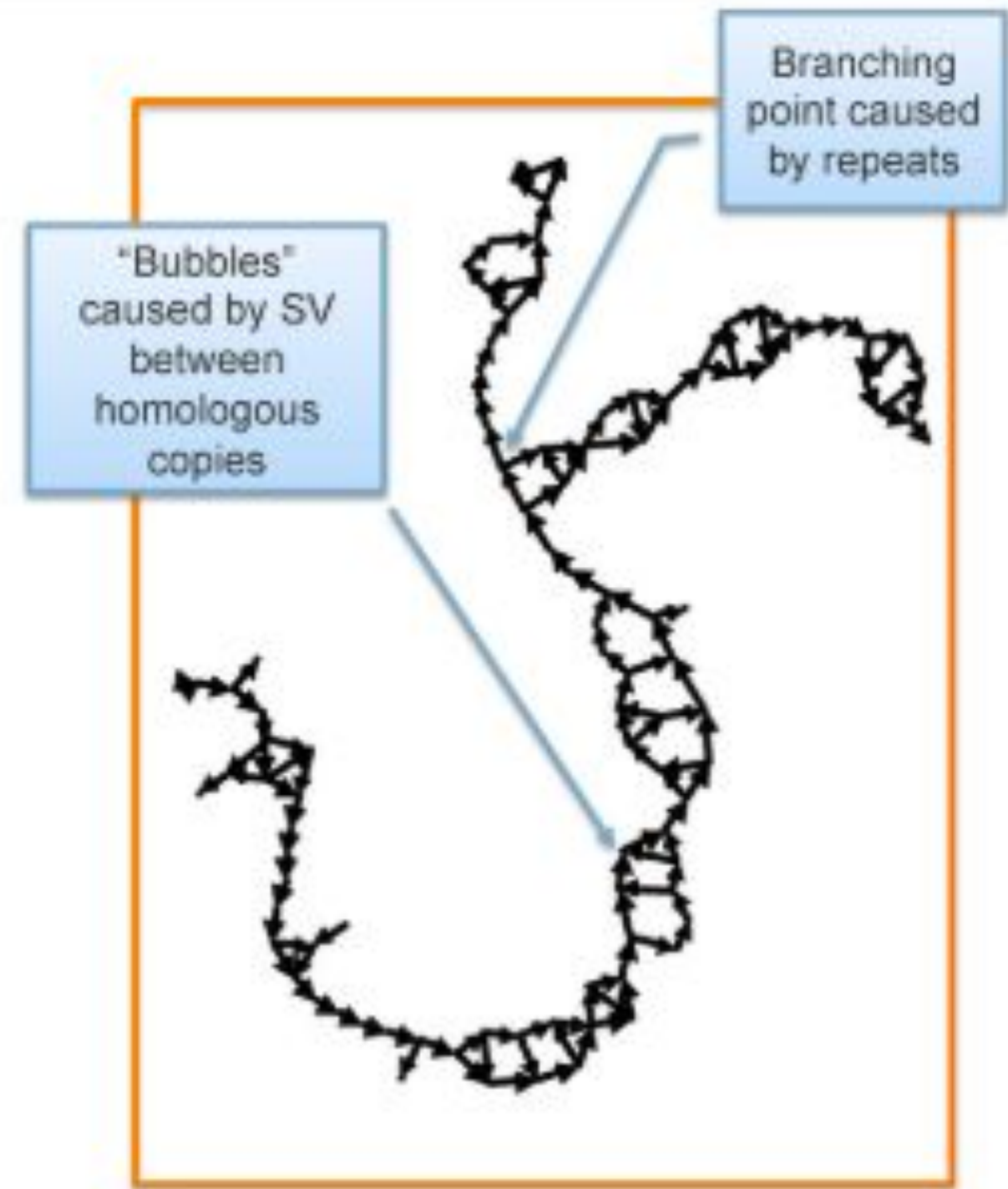


Genome Assembler Algorithms

An unitig graph from Ler-0 + Col-0 data

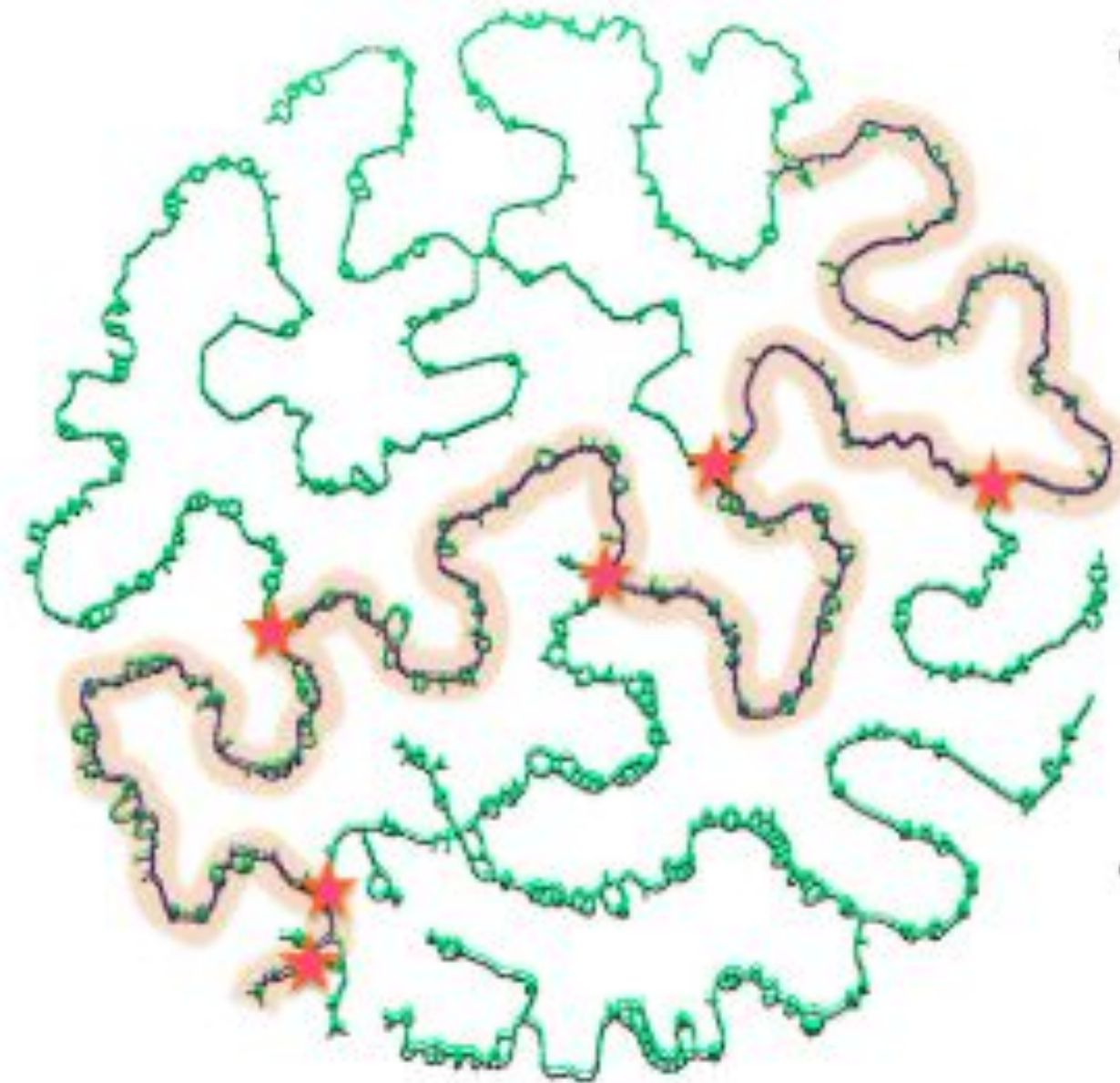


The graph "diameter" ~ 12 M bp
Mean edge size=17.4 k bp



12

Genome Assembler Algorithms

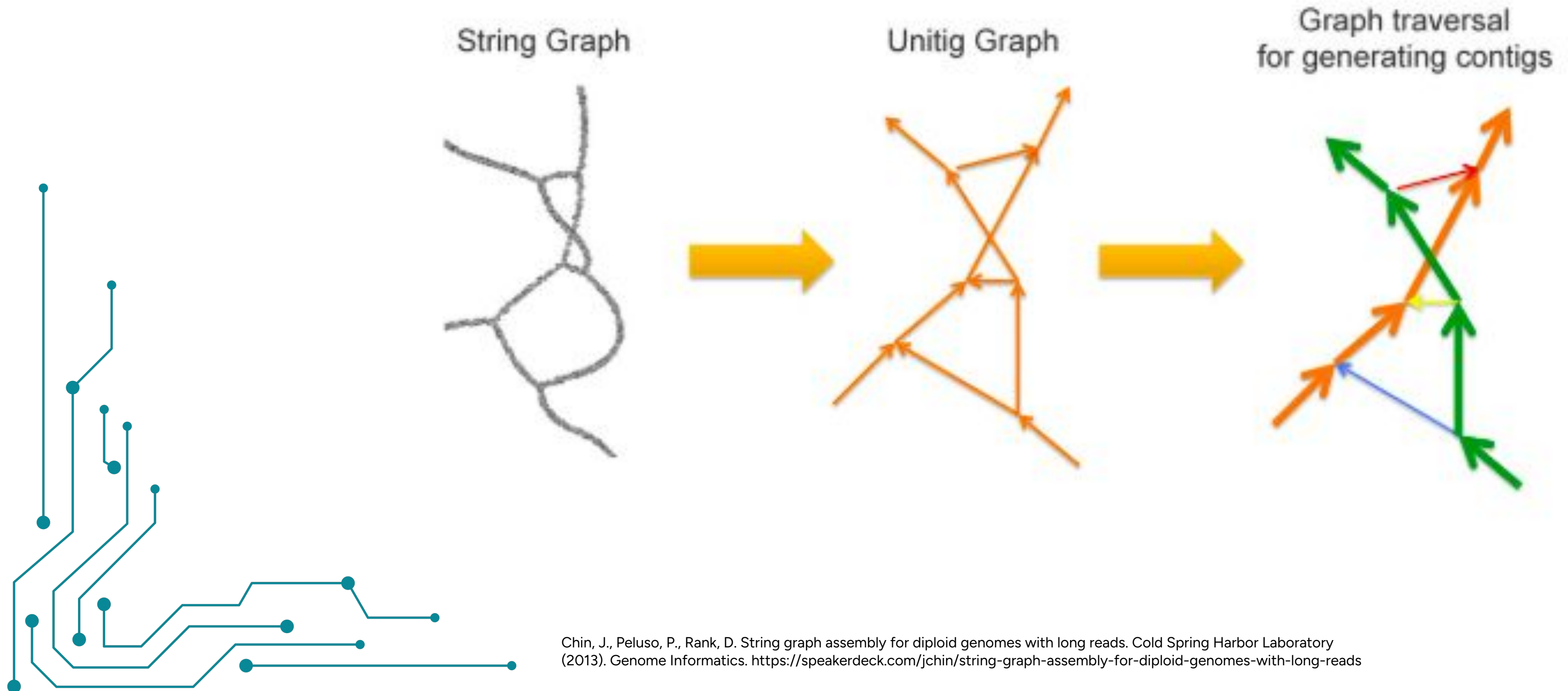


String bundle: compound paths that contain sequences from both haplotypes.

- Find an end-to-end path
⇒ A initial “primary contig”
- Build a “**string bundle**” along the path
⇒ the primary contig + locally “associated contigs”
- Repeat until no edge left
- Break the string bundle at branching point caused by repeats
⇒ Corrected primary contigs + locally associated contigs

Genome Assembler Algorithms

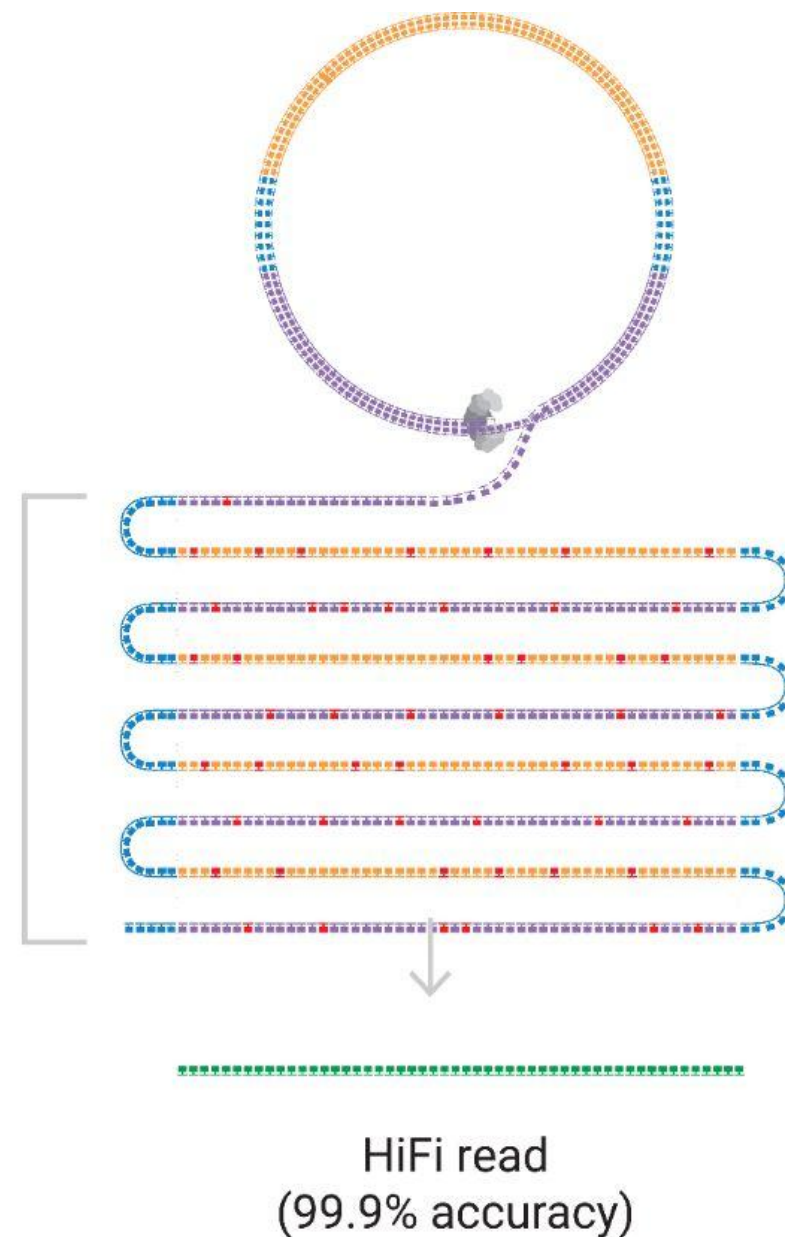
Building “unitigs” from the non-branching part of a string graph.



Chin, J., Peluso, P., Rank, D. String graph assembly for diploid genomes with long reads. Cold Spring Harbor Laboratory (2013). Genome Informatics. <https://speakerdeck.com/jchin/string-graph-assembly-for-diploid-genomes-with-long-reads>

De novo phased assemblies with **Hifiasm**

If you only have long-read data (no parental sequences or Hi-C),
you can produce:



- (1) **Primary and Alternate assemblies**
- (2) **Partially phased assemblies**
- (3) Haplotype-resolved assemblies

*look over example submission
scripts & discuss outputs*

Ok, now let's assemble a genome!!

SRA toolkit

download publicly available
long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size,
heterozygosity

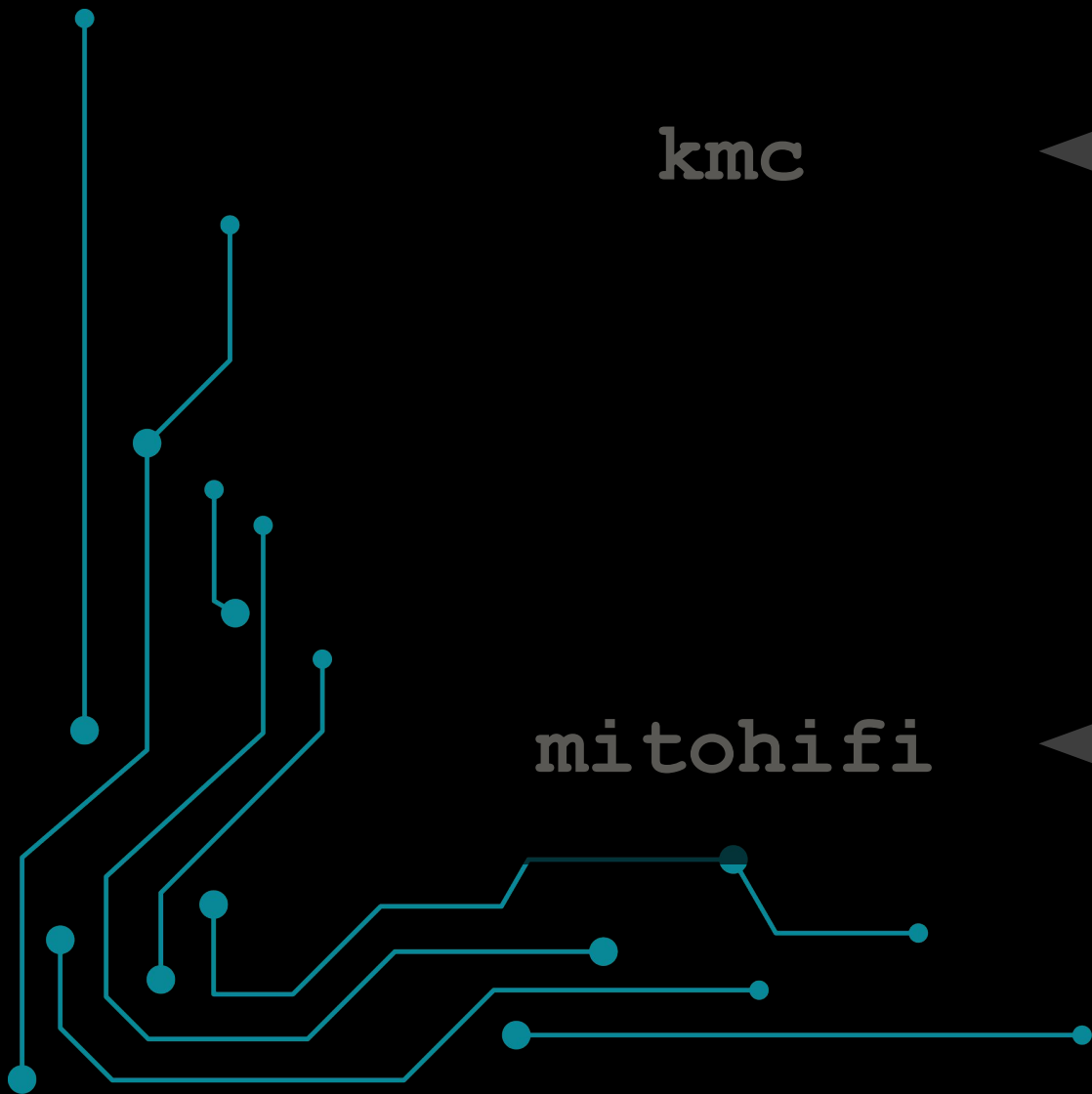
build assembly de novo using
long-read data

Hifiasm

mitohifi

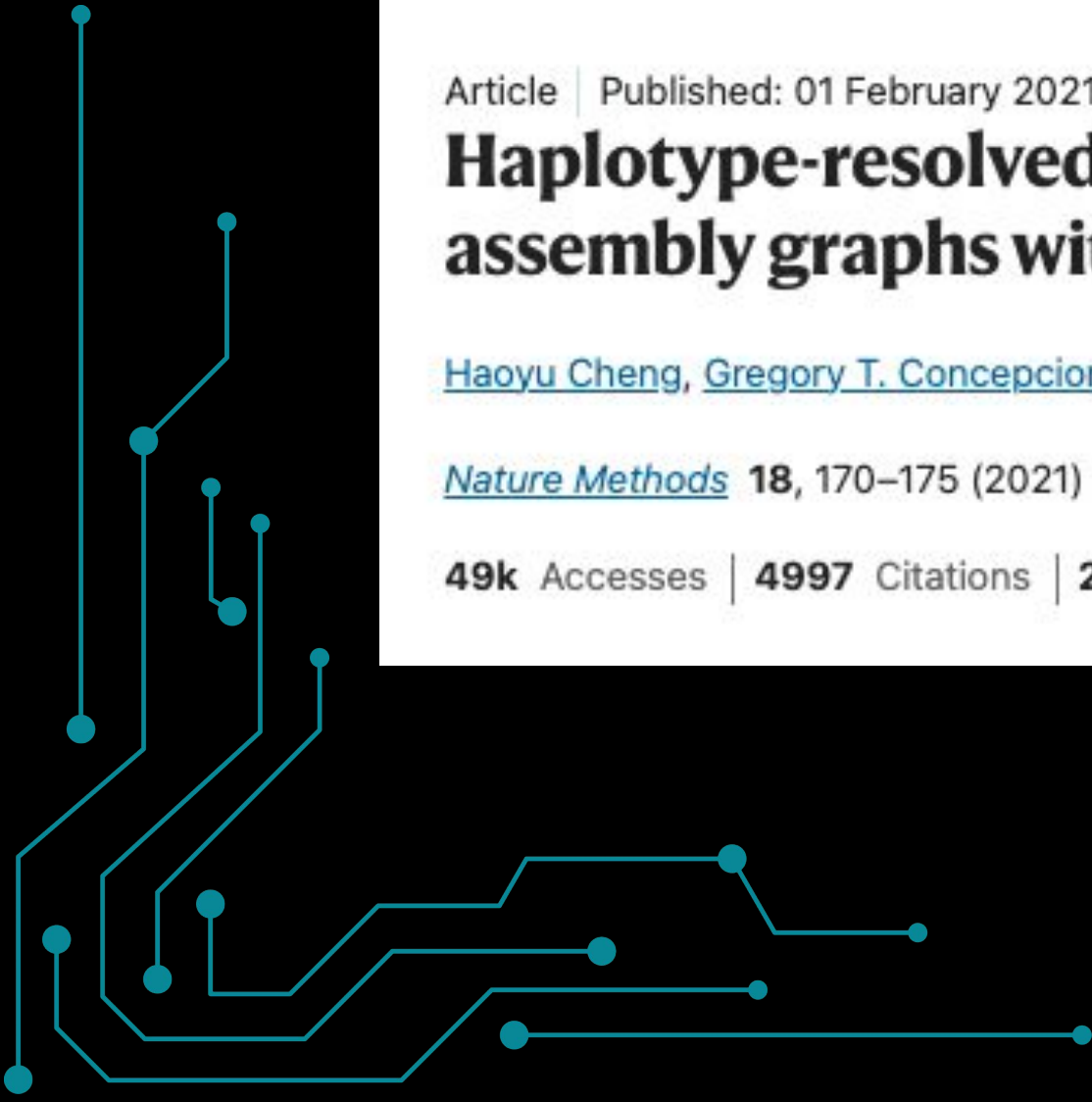
pull mitochondrial genome
sequences from assembly

*build submission script, submit
job, and assess output*



The Hifiasm assembler: how do we know what to do?

Read the paper!!!



nature methods

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Article | Published: 01 February 2021

Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm

[Haoyu Cheng](#), [Gregory T. Concepcion](#), [Xiaowen Feng](#), [Haowen Zhang](#) & [Heng Li](#) 

[Nature Methods](#) **18**, 170–175 (2021) | [Cite this article](#)

49k Accesses | **4997** Citations | **208** Altmetric | [Metrics](#)

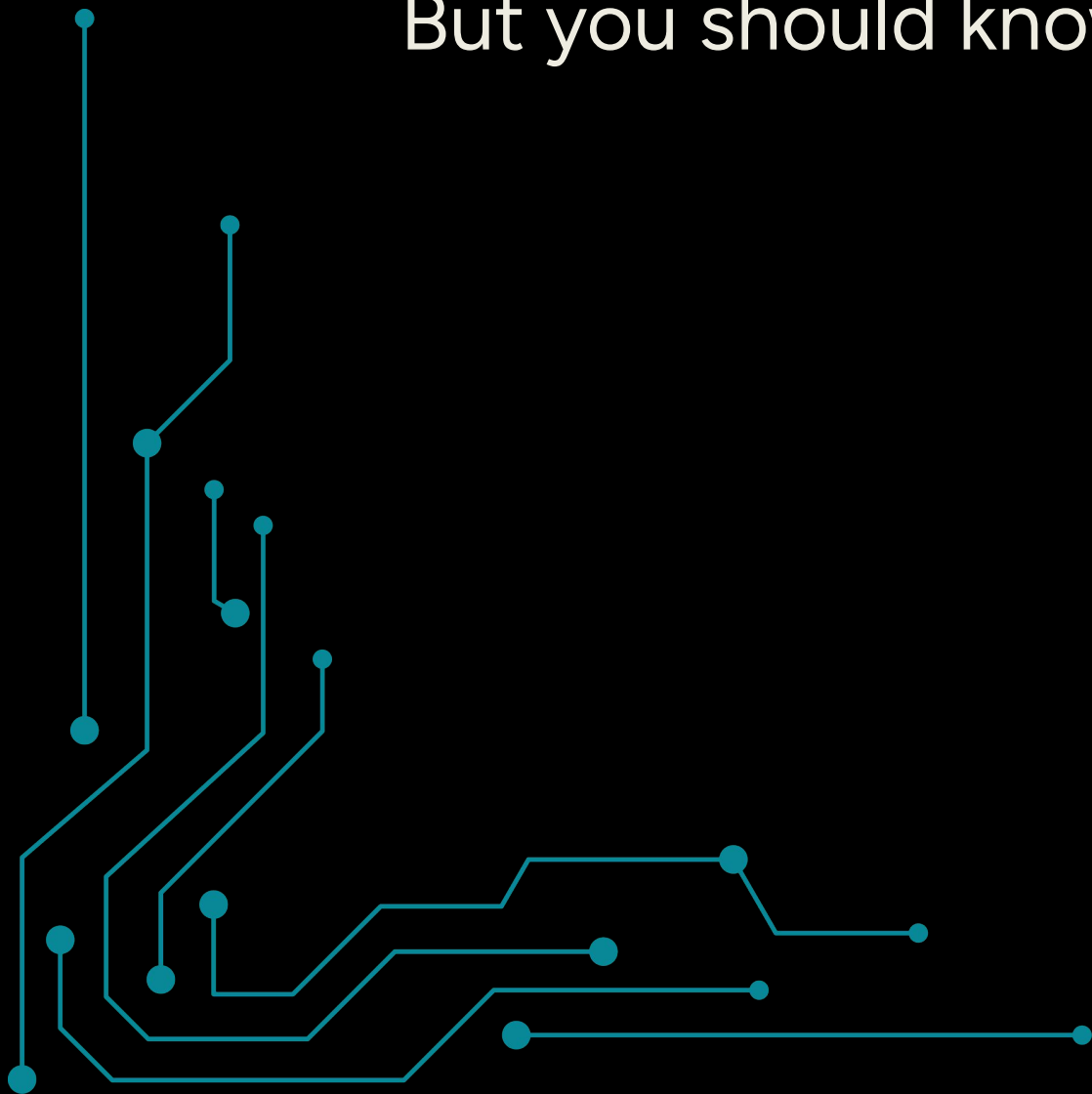
The Hifiasm assembler: how do we know what to do?

Just copy someone else's script!

Ask Chat! Or ask Claude!

Ok, sure. Realistically you will.

But you should know what you're doing – mistakes will be made!



The Hifiasm assembler: how do we know what to do?

Use the Hifiasm documentation

Hifi-reads only assembly

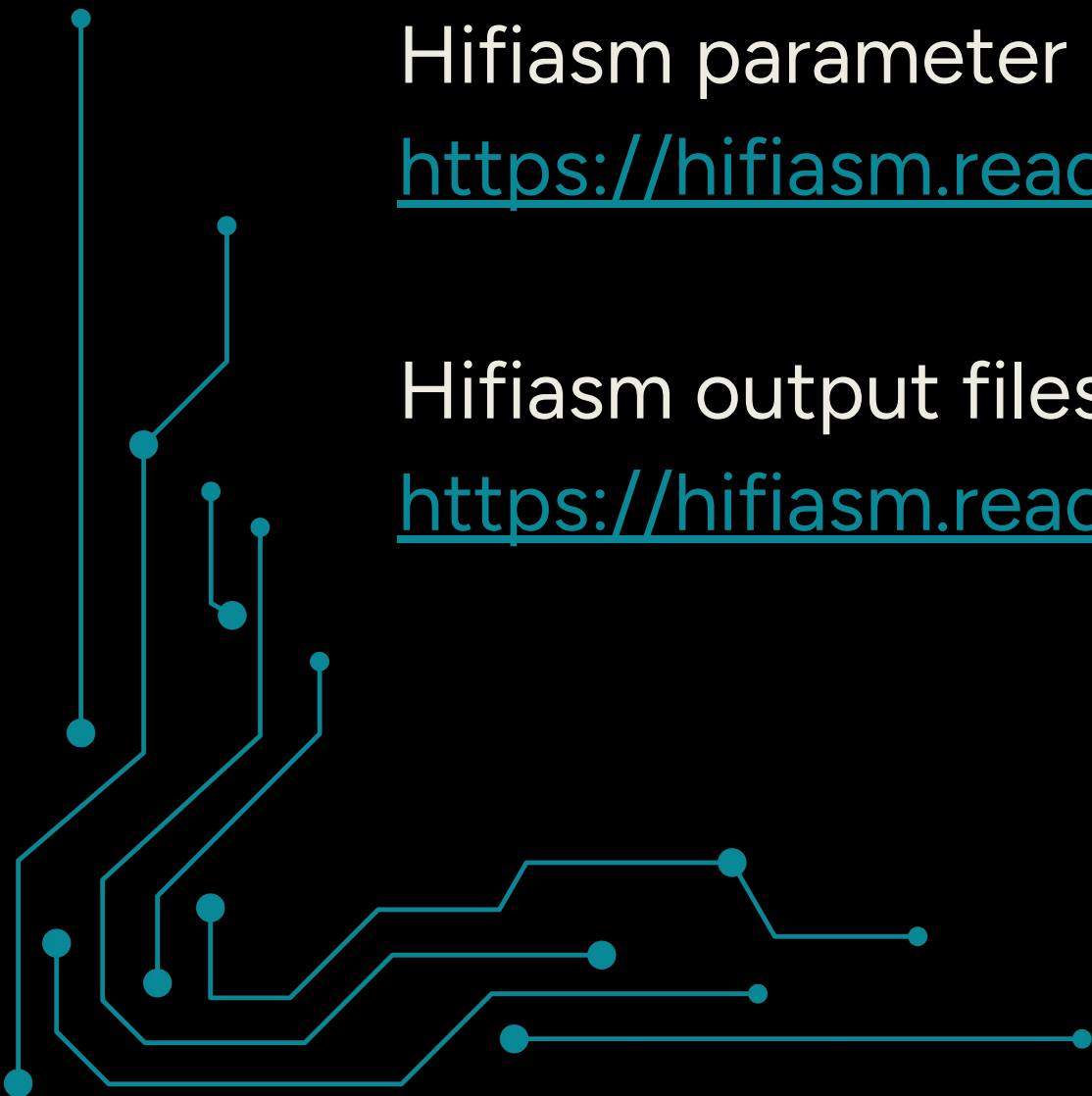
<https://hifiasm.readthedocs.io/en/latest/pa-assembly.html#pa-assembly>

Hifiasm parameter reference (the program settings)

<https://hifiasm.readthedocs.io/en/latest/parameter-reference.html>

Hifiasm output files

<https://hifiasm.readthedocs.io/en/latest/interpreting-output.html#interpreting-output>

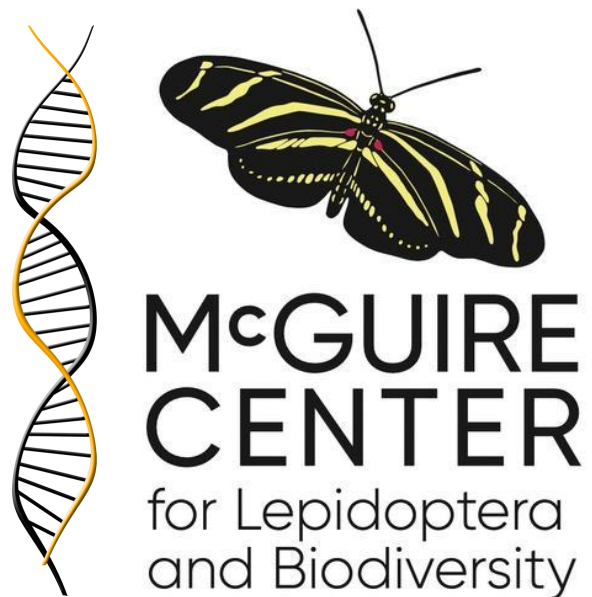


Module II: Genome Assembly

Sunday, January 18th 2026

R. Keating Godfrey, PhD
Assistant Professor
Neuroscience Program
Middlebury College

rkeatinggodfrey@middlebury.edu



*look over example submission
scripts & discuss outputs*

Ok, now let's assemble a genome!!

SRA toolkit

download publicly available
long-read sequencing data

longQC

Run quality check on data

kmc

assess genome size,
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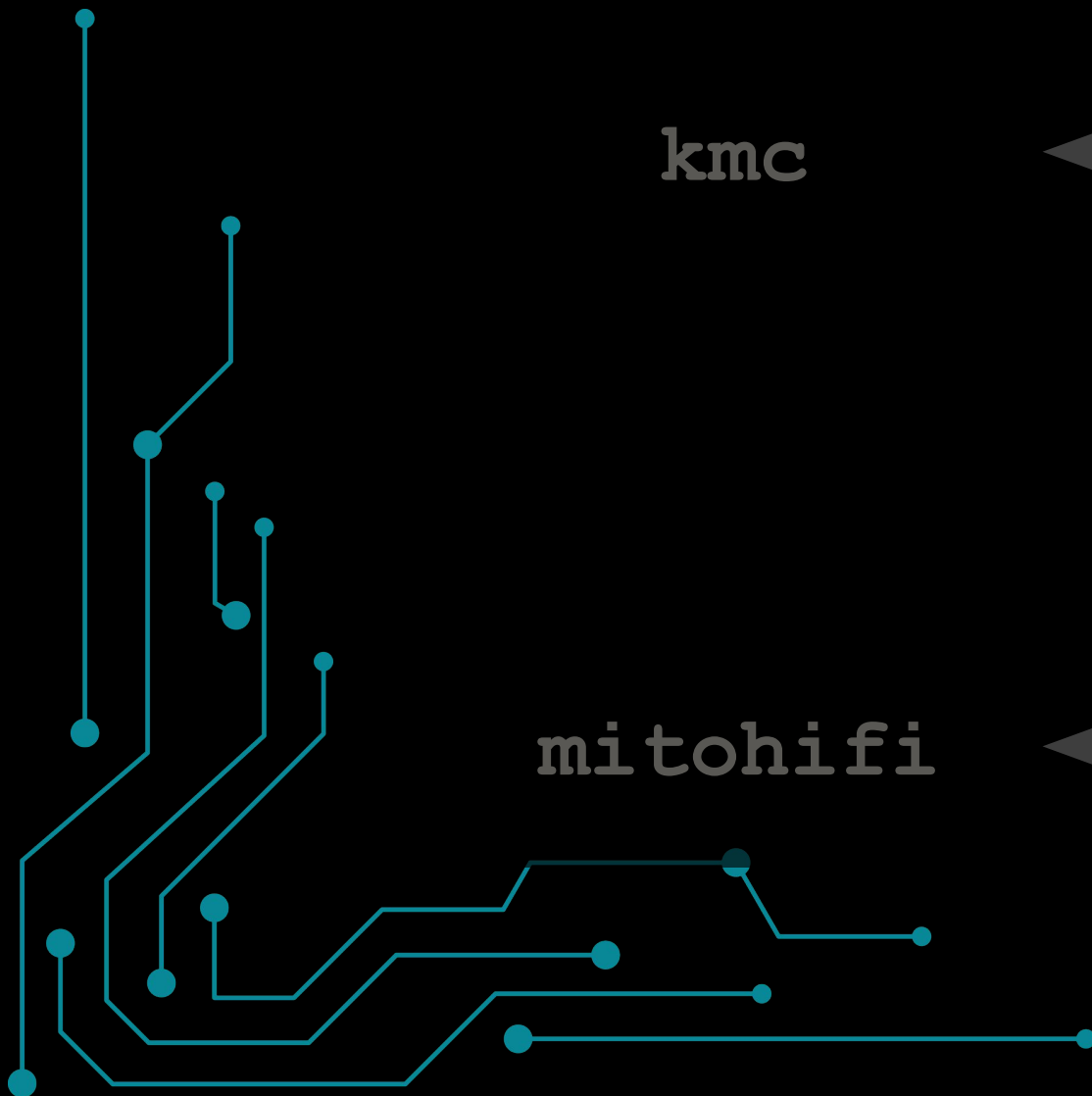
build assembly de novo using
long-read data

Hifiasm

mitohifi

pull mitochondrial genome
sequences from assembly

*build submission script, submit
job, and assess output*



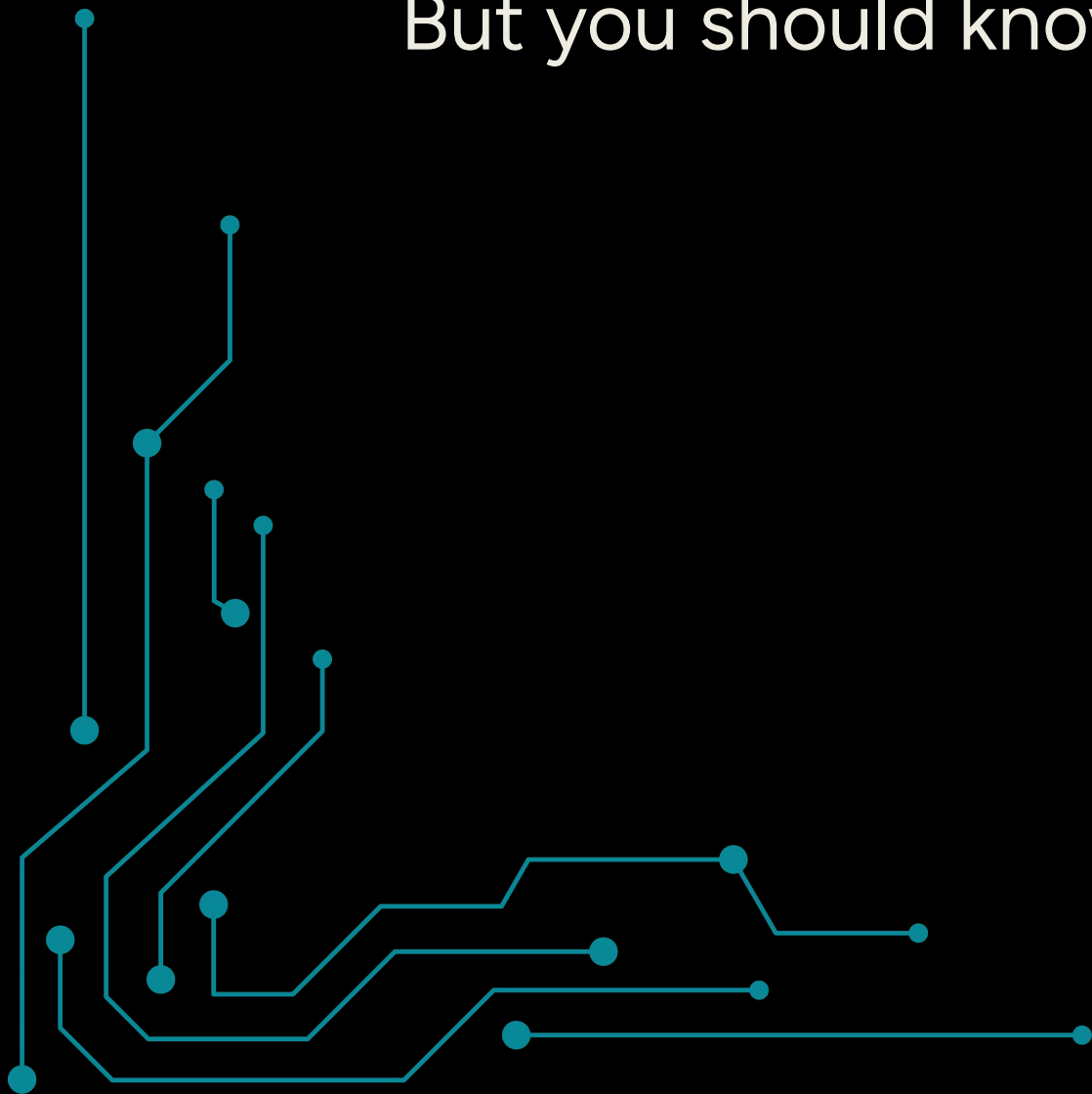
The HiFiasm assembler: what are the outputs?

Just copy someone else's script!

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But you should know what you're doing – mistakes will be made!



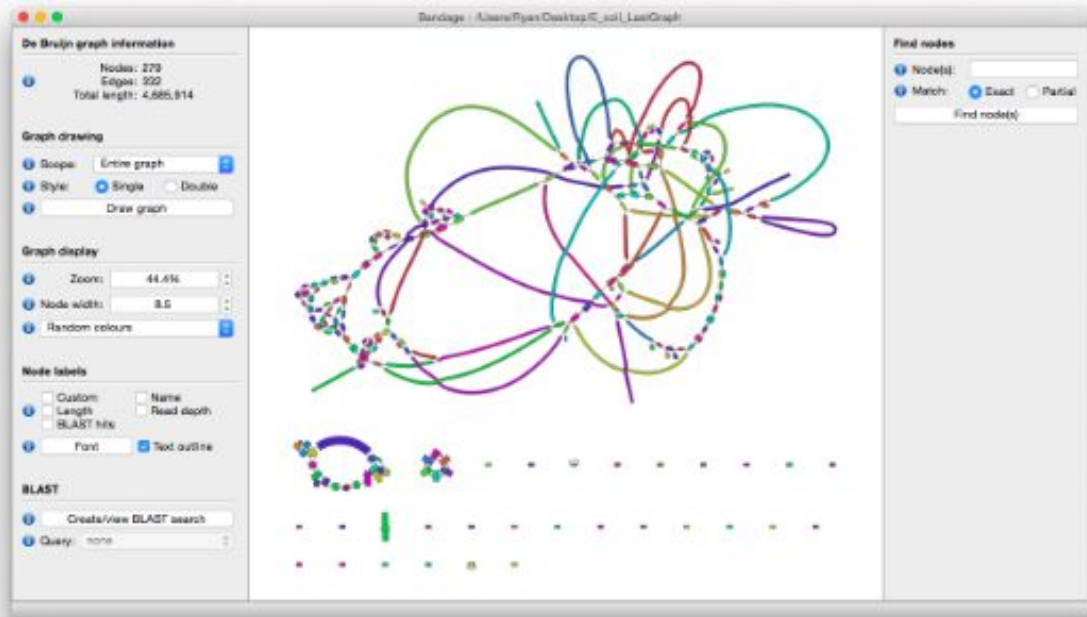
Visualizing GFA files with Bandage



Bandage

a Bioinformatics Application for Navigating De novo Assembly Graphs Easily

Bandage is a program for visualising *de novo* assembly graphs. By displaying connections which are not present in the contigs file, Bandage opens up new possibilities for analysing *de novo* assemblies.



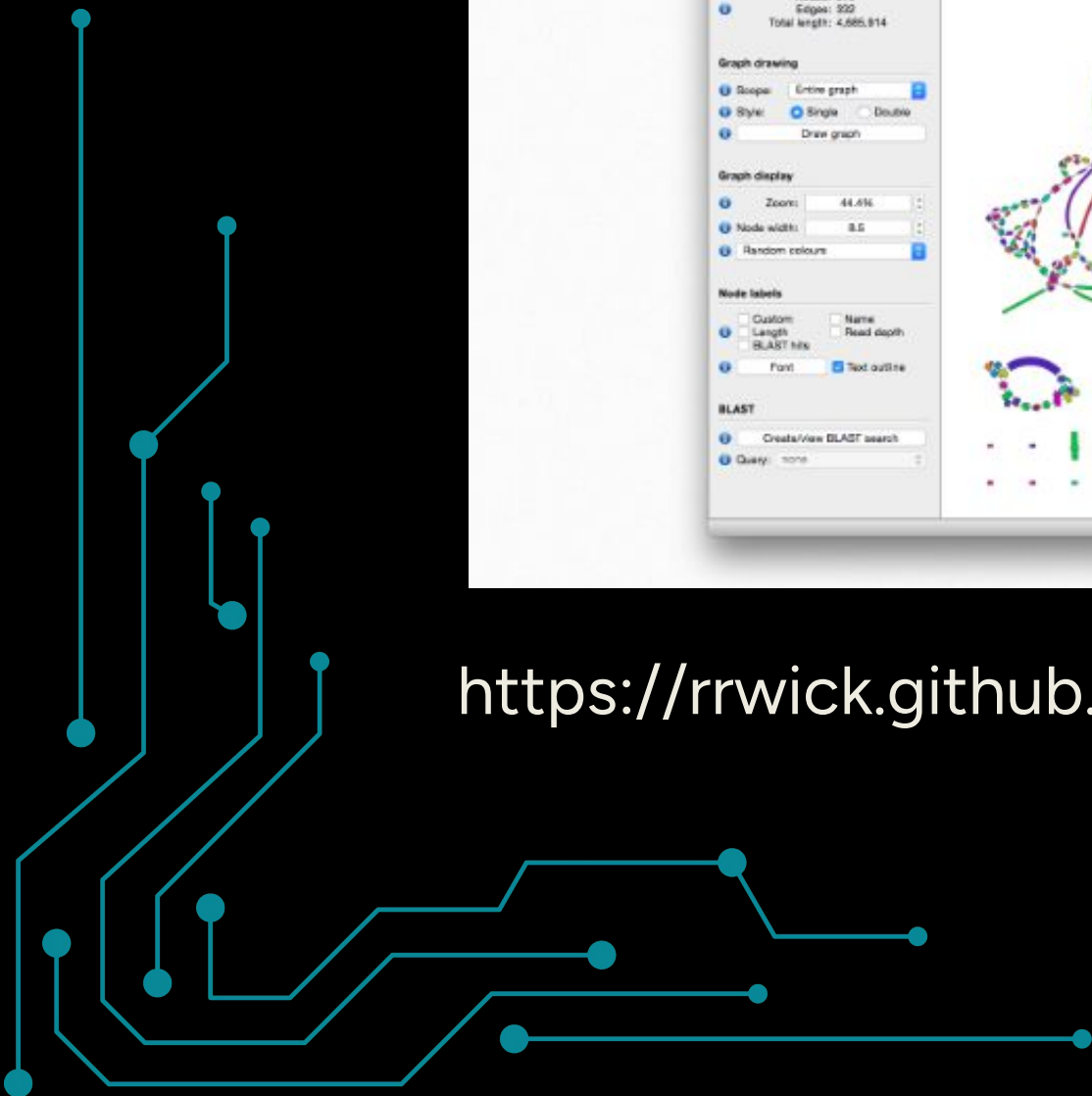
 Download
Mac

 Download
Linux

 Download
Windows

 View project on
GitHub

<https://rrwick.github.io/Bandage/>



Download your GFA locally

UF

Apps ▾Files ▾Jobs ▾Clusters ▾Interactive Apps ▾

?

>_ Open in Terminal

↻ Refresh

+ New File

New Directory

Upload

Download

Copy/Move

Delete

Home Directory

/orange/kawahara

/blue/kawahara

/blue/general_workshop

↑

/ blue / kawahara / rkeating.godfrey / genome_assembly / 05_hifiasm / Tbis_assembly_260116 /

Change directory

Copy path

☐ Show Owner/Mode

☐ Show Dotfiles

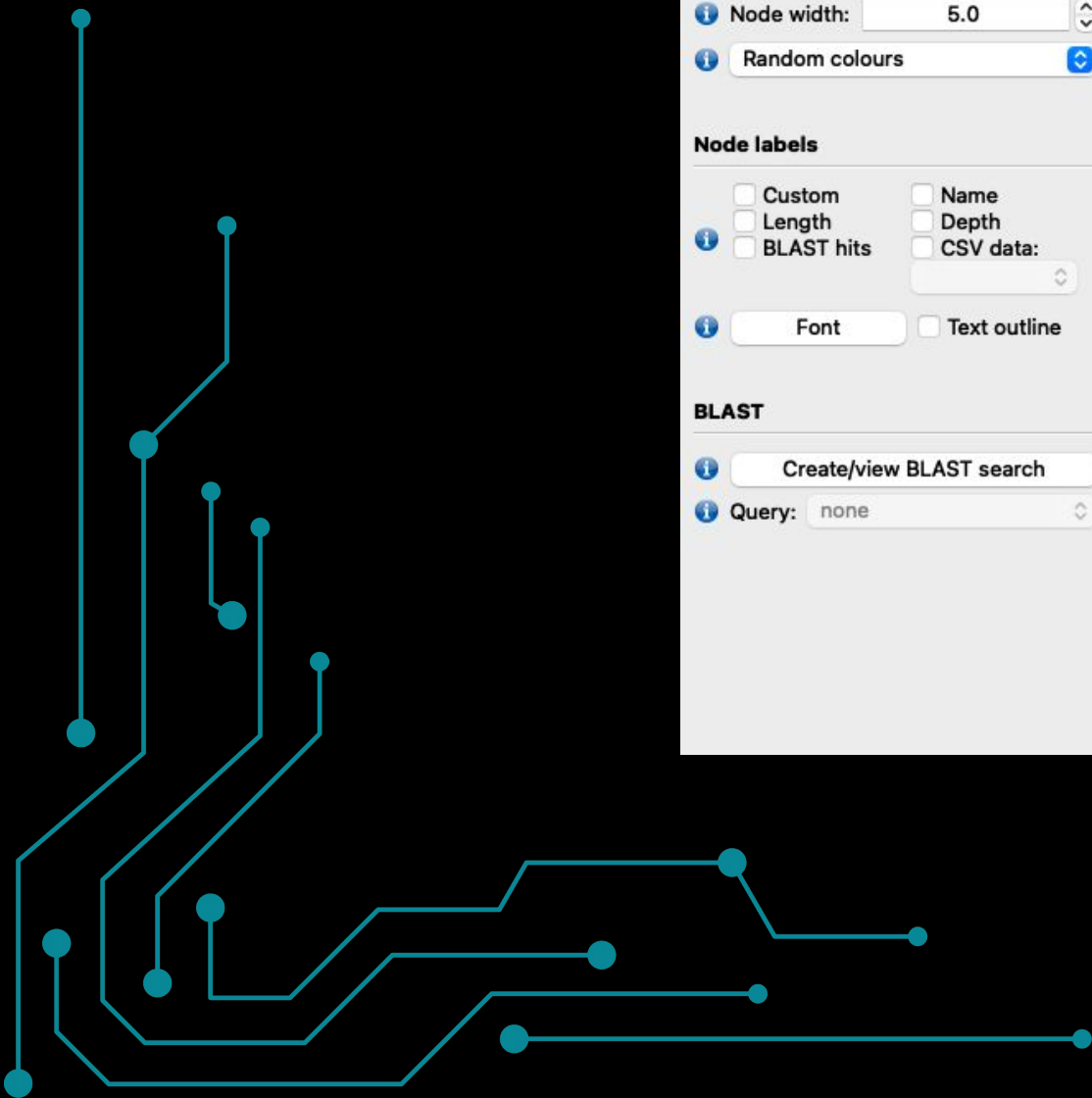
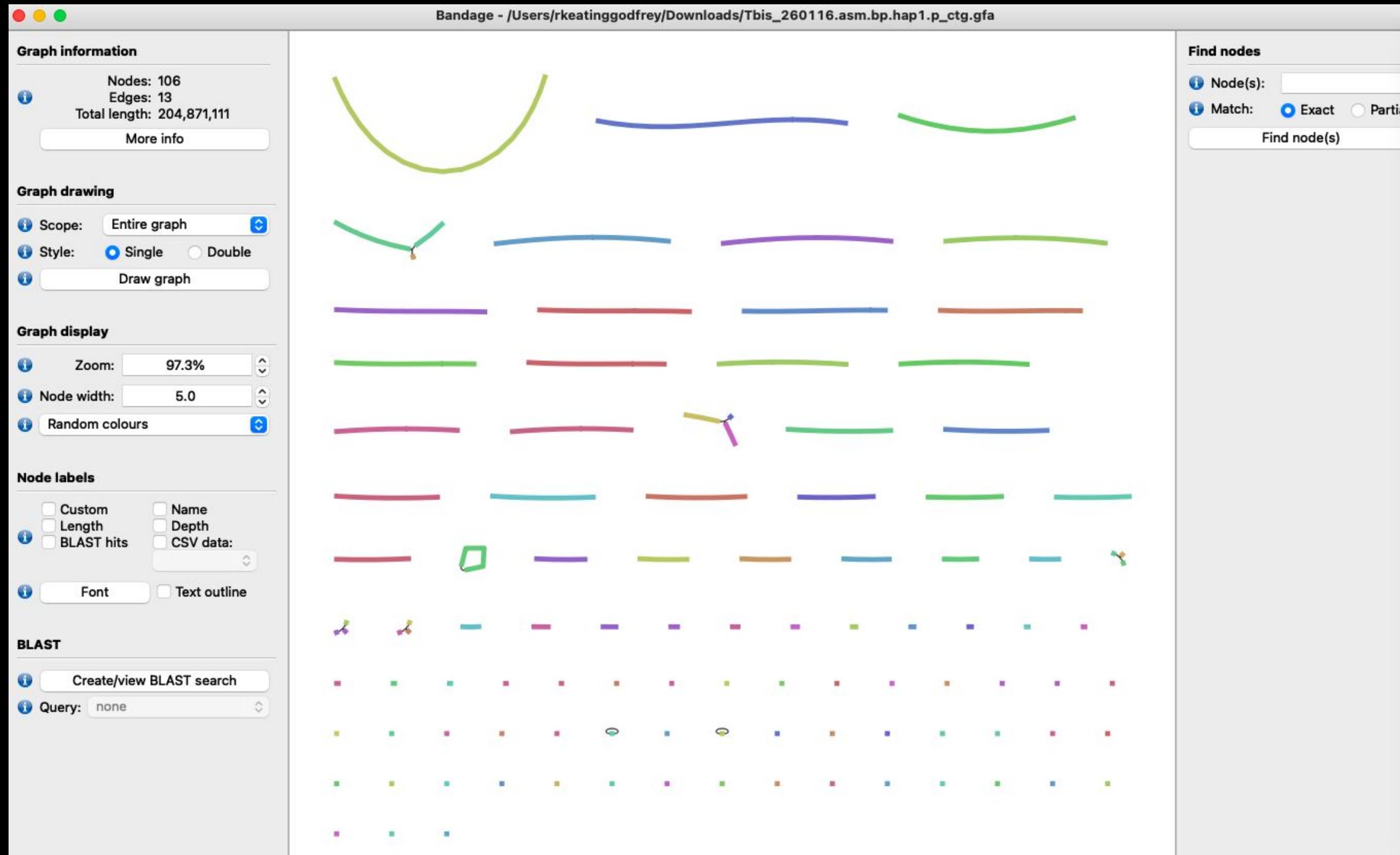
Filter:

Showing 19 rows - 1 rows selected

☐	Type	▲	Name		Size	Modified at
☒	File		Tbis_260116.asm.bp.hap1.p_ctg.gfa	⋮	210.16 MB	1/16/2026 4:12:22 PM
☐	File		Tbis_260116.asm.bp.hap1.p_ctg.lowQ.bed	⋮	75.69 kB	1/16/2026 4:12:22 PM



Visualize in Bandage

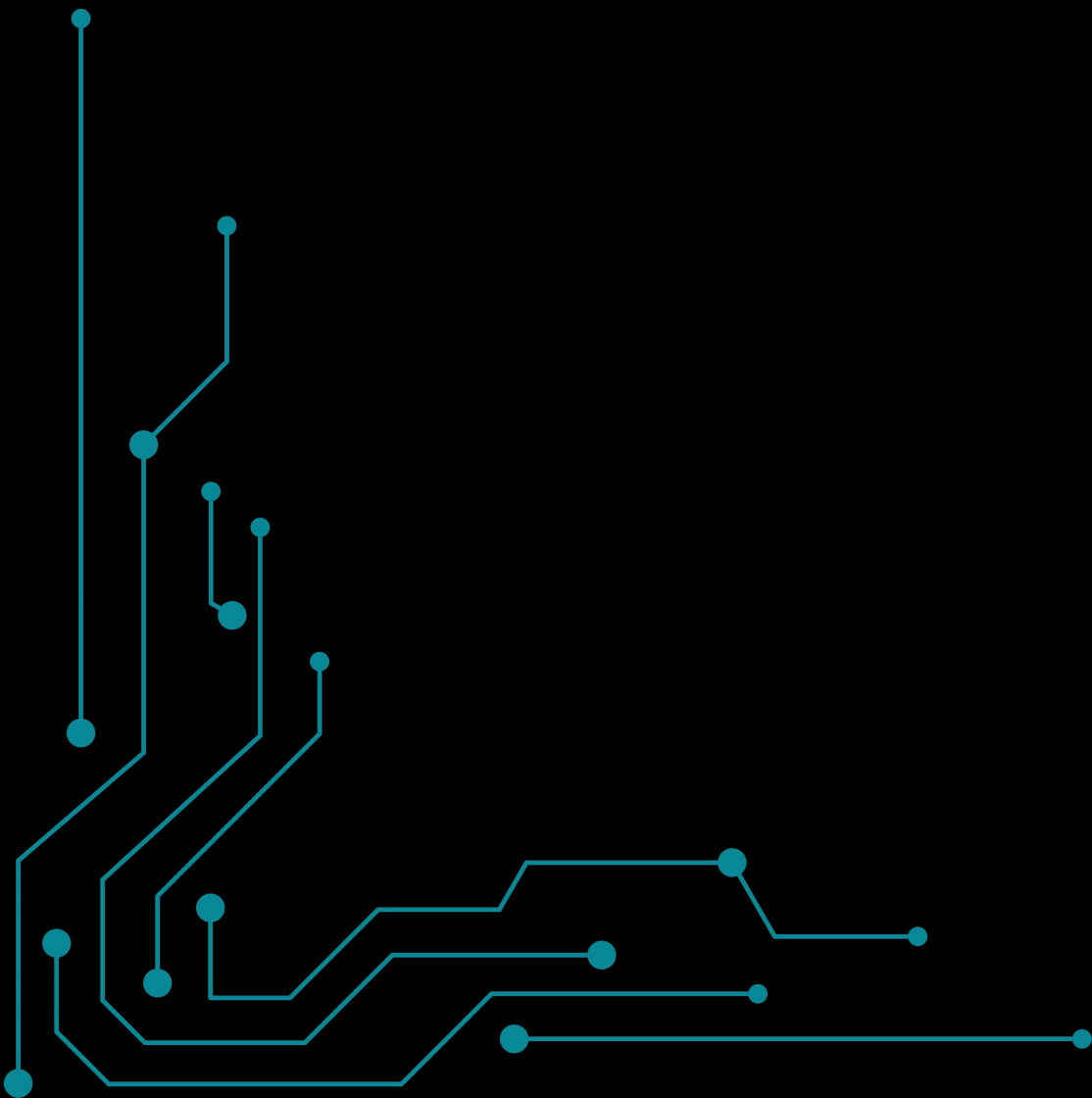


Getting a FASTA file for downstream analyses

How do I get contigs in FASTA?

The FASTA file can be produced from GFA as follows:

```
awk '/^S/{print ">"$2;print $3}' test.p_ctg.gfa > test.p_ctg.fa
```



*look over example submission
scripts & discuss outputs*

Assembling the mitochondrial genome

SRA toolkit

download publicly available
long-read sequencing data

longQC

Run quality check on data

kmc

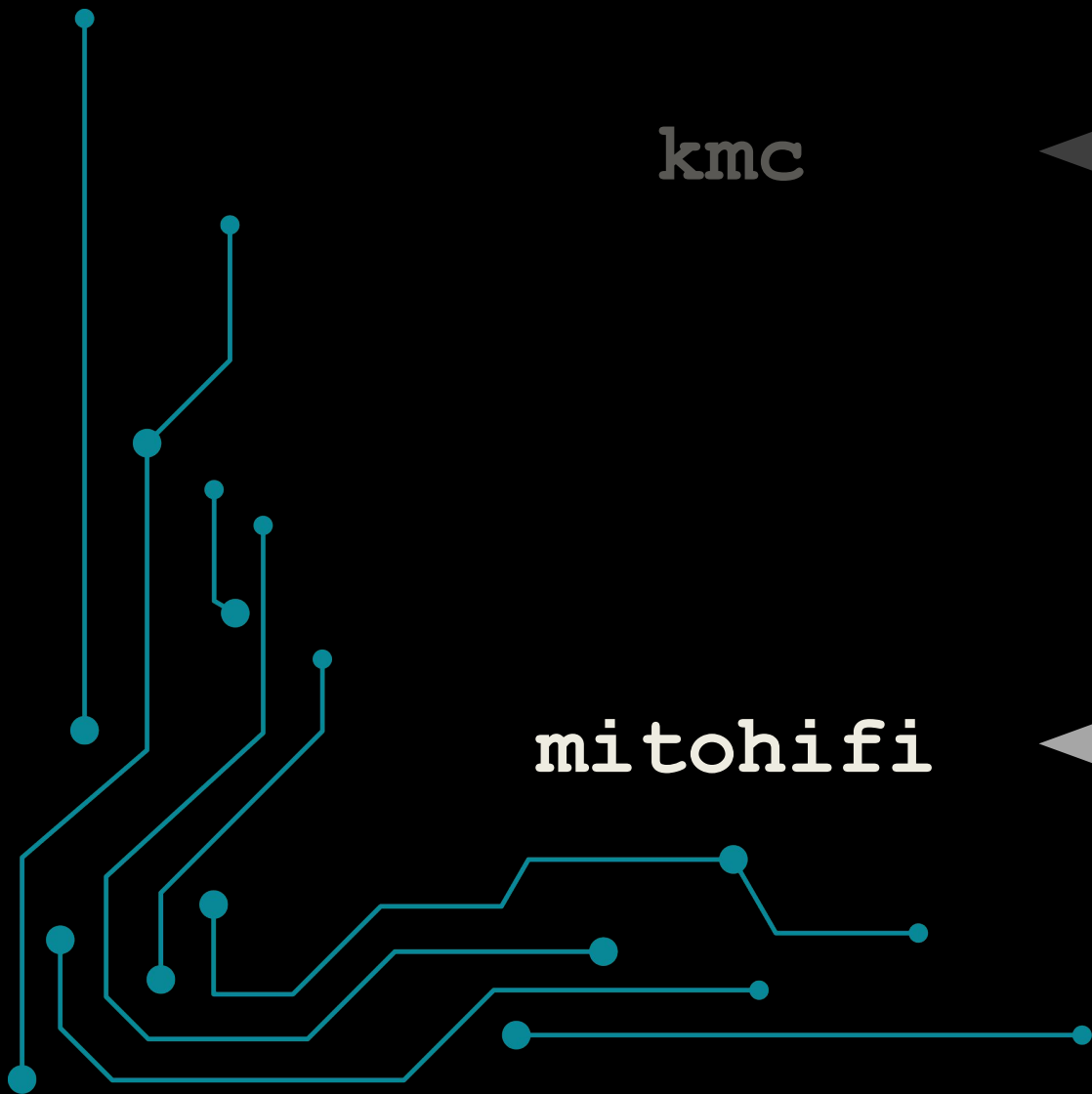
assess genome size,
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build assembly de novo using
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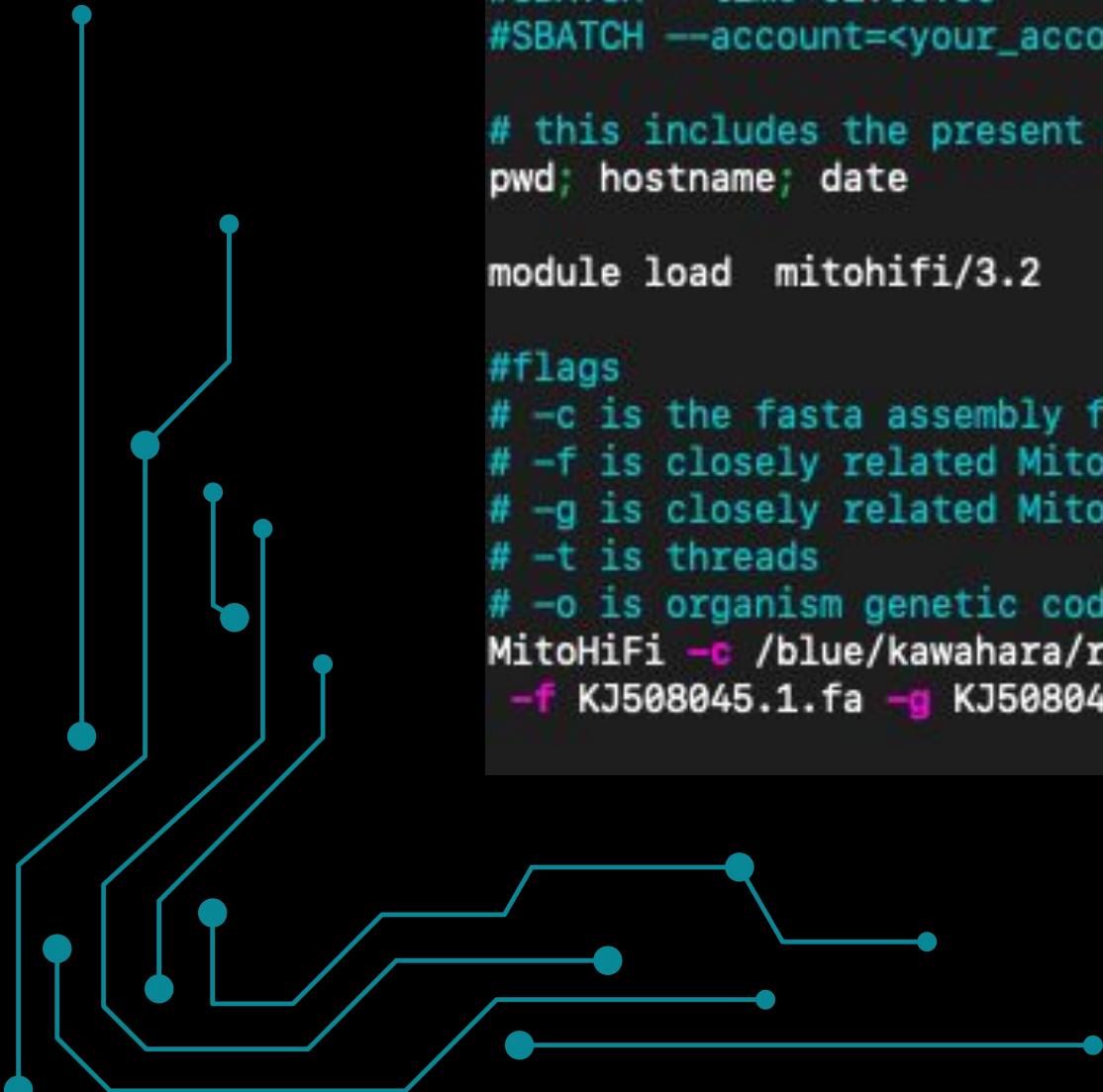
Hifiasm

mitohifi

pull mitochondrial genome
sequences from assembly



Let's take a look at the mitohifi github page: <https://github.com/marcelauliano/MitoHiFi>

A decorative circuit diagram in the bottom-left corner, featuring blue lines and dots representing electronic components and connections.

```
#!/bin/sh
#SBATCH --job-name=mitohifi
#SBATCH --output=mito_tbis_%j.out
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=<youremail>
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=2
#SBATCH --mem=4gb
#SBATCH --time=01:00:00
#SBATCH --account=<your_account>

# this includes the present working directory, host name, and date in output
pwd; hostname; date

module load mitohifi/3.2

#flags
# -c is the fasta assembly file from organism
# -f is closely related Mitogenome is fasta format
# -g is closely related Mitogenome is genbank format
# -t is threads
# -o is organism genetic code following NCBI table for mitochondrion annotation
MitoHiFi -c /blue/kawahara/rkeating.godfrey/genome_assembly/05_hifiasm/Tbis_assembly/Tbis_251224.asm.p_ctg.fa \
-f KJ508045.1.fa -g KJ508045.1.gb -t 4 -o 5
```

Module II: Genome Assembly

Saturday, January 17th 2026

R. Keating Godfrey, PhD
Assistant Professor,
Neuroscience Program
Middlebury College

rkeatinggodfrey@middlebury.edu

